

Mendel University in Brno
Faculty of Forestry and Wood Technology
Department of Wood Science and Wood Technology

Advanced Analysis of Acoustic Properties of Materials for Musical Instruments

Doctoral Thesis

Author: Ing. Patrik Nop
Supervisor: Ing. Jan Tippner, Ph.D.

2026

ZADÁNÍ DISERTAČNÍ PRÁCE

Autor práce: **Ing. Patrik Nop**

Studijní program: Materiálové inženýrství dřeva

Vedoucí práce: Ing. Jan Tippner, Ph.D.

Název práce: **Pokročilá analýza akustických vlastností materiálů pro hudební nástroje**

Rozsah práce: 111

Zásady pro vypracování:

1. Cílem disertační práce je studium kritérií, určujících vhodnost a kvalitu materiálu, jenž má být využit pro konstrukci daného hudebního nástroje. K popisu a vývoji těchto kritérií bude sloužit pokročilá analýza, založená na kombinaci současných přístupů, parametrů materiálu a metod, popisujících vibro-akustické vlastnosti materiálů.
2. Nalezené parametry a jejich korelace poslouží k posouzení vhodnosti některých druhů materiálů pro stavbu hudebních nástrojů, podle obecných principů vytváření zvuku a zvukové charakteristiky jednotlivých nástrojových skupin, a to rovněž s podporou FEM simulace dynamické odezvy ve frekvenční a časové oblasti.
3. Dokument práce (kompilace publikací) bude strukturou odpovídat standardům vědecké práce a respektovat platné předpisy o úpravě písemných prací.

Seznam odborné literatury:

1. BODIG, Jozsef; JAYNE, Benjamin A. 1993. *Mechanics of Wood and Wood Composites*. Malabar: Krieger Publish.Comp., 21 s. ISBN 0-89464-777-6.
2. BUCUR, Voichita. 1995. *Acoustics of Wood*. Boca Raton: CRC Press, 284 s. ISBN 0-8493-4801-3.
3. BUCUR, Voichita. 2003. *Nondestructive characterization and imaging of wood*. Berlin: Springer, 354 s. ISBN 978-3-642-07860-6.
4. *Urban forest acoustics*. Berlin: Springer-Verlag, 181 s. ISBN 978-3-540-30783-9.
5. MECHEL, F. 2002. *Formulas of Acoustics*. Berlin: Springer, 1175 s.
6. PELLERIN, Roy F.; ROSS, Robert J. 2002. *Nondestructive evaluation of wood*. Madison: Forest Products Society, 210 s. ISBN 1-892529-26-2.
7. ROSSING, Thomas. 2007. *Springer handbook of Acoustics*. USA: LE-TeX Gbr, Leipzig, 1167 s. ISBN 978-0-387-30446-5.
8. SYROVÝ, Václav. 2009. *Hudební zvuk: příspěvek k teorii zvukové tvorby*. 1. vyd. V Praze: Akademie múzických umění, 303 s. Akustická knihovna Zvukového studia Hudební fakulty AMU ;. ISBN 978-80-7331-161-2.
9. FLETCHER, Neville H.; ROSSING, Thomas D. *The physics of musical instruments*. 2. vyd. New York, N.Y.: Springer, 756 s. ISBN 978-1-4419-3120-7.
10. Brémaud, I. (2012). Acoustical properties of wood in string instruments soundboards and tuned idiophones: Biological and cultural diversity. *The Journal of the Acoustical Society of America*, 131(1), 807–818. <https://doi.org/10.1121/1.3651233>
11. Dunisch, O. (2019). Frequencies in vibrating wood – Does cell organization matter? *IAWA Journal*, 40(1), 124–142. <https://doi.org/10.1163/22941932-40190239>
12. Niemz, P., & Mannes, D. (2012). Non-destructive testing of wood and wood-based materials. *Journal of Cultural Heritage*, 13(3 SUPPL.), S26–S34. <https://doi.org/10.1016/j.culher.2012.04.001>
13. Obataya, E. (2017). Effects of natural and artificial ageing on the physical and acoustic properties of wood in musical instruments. *Journal of Cultural Heritage*, 27, S63–S69. <https://doi.org/10.1016/j.culher.2016.02.011>
14. Rojas, J. A. M., Alpuente, J., Postigo, D., Rojas, I. M., & Vignote, S. (2011). Wood species identification using stress-wave analysis in the audible range. *Applied Acoustics*, 72(12), 934–942. <https://doi.org/10.1016/j.apacoust.2011.05.016>
15. Rossing, T. D. (2010). The Science of String Instruments. https://doi.org/10.1007/978-1-4419-7110-4_2
16. Straže, A., Mitkovski, B., Tippner, J., Čufar, K., & Gorišek. (2015). Structural and acoustic properties of African padouk (*Pterocarpus soyauxii*) wood for xylophones. *European Journal of Wood and Wood Products*, 73(2), 235–243. <https://doi.org/10.1007/s00107-015-0878-0>
17. Wegst, U. G. K. (2006). Wood for sound. *American Journal of Botany*, 93(10), 1439–1448. <https://doi.org/10.3732/ajb.93.10.1439>

Datum zadání disertační práce:

září 2019

Termín odevzdání disertační práce:

červen 2023

L. S.

Ing. Patrik Nop

Autor práce

Ing. Jan Tippner, Ph.D.

Vedoucí práce

prof. Ing. Vladimír Gryc, Ph.D.

Vedoucí ústavu

prof. Ing. Vladimír Gryc, Ph.D.

Děkan LDF MENDELU

Declaration

I hereby declare that this thesis, entitled Advanced Analysis of Acoustic Properties of Materials for Musical Instruments was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final thesis.

I declare that the printed version of the thesis and the electronic version of the thesis published in the application of the Final Thesis in the University Information System are identical.

I am aware that my thesis is written in accordance with the Act. 121/2000 Coll., on Copyright and therefore Mendel University in Brno has the right to conclude licence agreements on the utilisation of the thesis as a school work in accordance with Article 60(1) of the Copyright Act.

Before concluding a licence agreement on the utilisation of the work by another person, I will request a written statement from the university that the licence agreement is not in contradiction to the legitimate interests of the university, and I will also pay a prospective fee to cover the cost incurred in creating the work to the full amount of such costs.

Brno, 30 April 2026

Ing. Patrik Nop

Preface and Acknowledgement

This doctoral thesis was prepared at the Department of Wood Science and Wood Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Czech Republic. The experimental work was carried out at the Josef Ressel Research Centre in Útěchov and at the University Forest Enterprise Masaryk Forest in Křtiny, in cooperation with the University of Primorska and InnoRenew CoE. in Slovenia.

This work was supported by the Ministry of Education, Youth and Sports of the Czech Republic (project ERC CZ no. LL1909), by the Internal Grant Agency of the Faculty of Forestry and Wood Technology at Mendel University in Brno (grant no. LDF_VP_2020013 and IGA-LDF-22-IP-016), and by the Internal Grant Schemes of Mendel University in Brno (registration no. CZ.02.2.69/0.0/0.0/19_073/0016670), funded by the European Social Fund. Open access publishing was supported by the institutions participating in the CzechELib Transformative Agreement.

I would like to thank my supervisor, Dr. Jan Tippner, for his guidance, advice, and for sharing his extensive experience and inspiring ideas. I would also like to thank my colleagues for creating a friendly and constructive environment during my whole study. Finally, I would like to thank my loved ones for their endless support.

Abstract

Nop, P. (2026), *Advanced Analysis of Acoustic Properties of Materials for Musical Instruments*, Doctoral Thesis, Mendel University in Brno.

Despite technological advances offering a wide range of materials, wood remains the most widely used material for the production of musical instruments. This is primarily due to its unique mechanical properties. These properties are related to the specific structure of wood, particularly selected species that have, over the course of the development of musical instruments, become established as reference materials in terms of sound quality. However, these high-quality woods are becoming increasingly valuable, which increases the need for their efficient use and at the same time emphasizes the importance of suitable alternatives.

The aim of this dissertation thesis is the investigation of key parameters for the qualitative evaluation of materials for musical instruments and of the influences that may affect their reliable determination. For the extension of existing methods, the following were assessed: 1) the influence of the dimensions of the tested samples; 2) the influence of material characteristics and of factors that affect them (e.g. moisture, degradation, modification); 3) the influence of the vibration mode from which the parameters are determined; 4) the influence of the measuring technique, including the instruments used as well as the measurement methodology; 5) the characteristic of the vibration spectrum.

The work consists of four publications in peer-reviewed journals and one conference proceeding paper. Various wood species (spruce, beech, ash, oak, lime, alder, larch, and maple) and wood-based composite materials (plywood made from pine, birch, and poplar) were tested. The main method used for determining vibro-acoustic parameters was the frequency resonance technique, which was complemented in individual studies by static, dynamic, and impact bending tests, as well as by numerical modal analysis using the finite element method.

The results show that when using the frequency resonance technique, it is necessary to consider specimen dimensions and to choose an appropriate measuring system. The investigated vibro-acoustic parameters allow for distinguishing between different wood species and wood-based composites and are sensitive to moisture-related and structural changes. Comparison with other methods revealed that parameters obtained under different loading conditions cannot be used interchangeably. It was found that the shape and order of vibration modes influence the determination of the evaluated parameters and that the spectral distribution of these modes provides information that can be used for the qualitative assessment of materials for musical instruments.

Keywords: frequency resonance technique, musical instruments, vibro-acoustic properties, dynamic response, modal analysis

Abstrakt

Nop, P. (2026), *Pokročilá analýza akustických vlastností materiálů pro hudební nástroje*, disertační práce, Mendelova univerzita v Brně.

Navzdory technologickému pokroku, který nabízí širokou škálu materiálů, zůstává dřevo nejvyužívanějším materiálem pro výrobu hudebních nástrojů. To je dáno především jeho jedinečnými mechanickými vlastnostmi. Tyto vlastnosti souvisejí se specifickou strukturou dřeva, zejména vybraných druhů, jež se v průběhu vývoje jednotlivých hudebních nástrojů etablovaly jako referenční z hlediska kvality zvuku. V současné době jsou však tato vysoce kvalitní dřeva stále cennějším artiklem, což zvyšuje potřebu jejich efektivního využití a zároveň akcentuje význam vhodných alternativ.

Cílem této disertační práce je studium klíčových parametrů pro kvalitativní hodnocení materiálů pro hudební nástroje a vlivů, které mohou ovlivnit jejich spolehlivé stanovení. Pro rozšíření stávajících metod byly posouzeny: 1) vliv rozměrů testovaných vzorků; 2) vliv materiálových charakteristik a faktorů, které je ovlivňují (např. vlhkost, degradace, modifikace); 3) vliv vibračního módu, ze kterého jsou parametry stanoveny; 4) vliv měřicí techniky, zahrnující použité přístroje i metodiku měření; 5) charakteristika vibračního spektra.

Práce je tvořena čtyřmi publikacemi v recenzovaných časopisech a jedním příspěvkem ve sborníku konference. Byly testovány různé druhy dřeva (smrk, buk, jasan, dub, lípa, olše, modřín a javor) a kompozitní materiály na bázi dřeva (překližka z borovice, břízy a topolu). Jako hlavní metoda pro stanovení vibro-akustických parametrů byla zvolena frekvenčně rezonanční technika, která byla v jednotlivých studiích doplněna o statickou, dynamickou a rázovou ohybovou zkoušku nebo o numerickou modální analýzu, využívající metodu konečných prvků.

Výsledky ukazují, že při použití frekvenčně rezonanční techniky je potřeba zohlednit rozměry testovaných vzorků a zvolit odpovídající měřicí systém. Zkoumané vibro-akustické parametry umožňují odlišit jednotlivé druhy dřeva i kompozity na bázi dřeva a jsou citlivé na vlhkostní či strukturální změny. Porovnání s dalšími metodami odhalilo, že nelze zaměňovat parametry získané při různých způsobech zatěžování. Bylo zjištěno, že tvar a pořadí vibračních módů ovlivňují stanovení sledovaných parametrů a že spektrální rozložení módů poskytuje informace, které mohou být využity při kvalitativním hodnocení materiálů pro hudební nástroje.

Klíčová slova: frekvenčně rezonanční technika, hudební nástroje, vibro-akustické vlastnosti, dynamická odezva, modální analýza

Content

1	Introduction.....	1
2	Literature Review	2
2.1	Physico-Mechanical Properties Related to Dynamic Response of Wood	2
2.2	Factors Influencing Vibro-Acoustical Properties	6
2.3	Materials for Wooden Musical Instruments	7
2.4	Methods for Vibro-Acoustic Properties Determination	9
3	References.....	13
4	Hypotheses and Objectives	23
I.	Effect of specimen dimensions	23
II.	Effect of material characteristics	23
III.	Effect of vibrational mode.....	24
IV.	Effect of the measurement technique	24
V.	Characteristics of the vibrational spectrum.....	24
5	Materials and Methods.....	25
5.1	Specimen Preparation.....	25
5.2	Material Modification	26
5.3	Numerical Modal Analysis.....	26
5.4	Frequency Resonance Technique	26
5.5	Three-point Bending Test Under Static Loading	29
5.6	Three-point Bending Test Under Oscillatory Loading.....	29
5.7	Three-point Bending Test Under Impact Loading	29
6	List of publications	30
	Paper I.....	31
	Paper II.....	45
	Paper III	51
	Paper IV	65
	Paper V.....	83
7	Summary	99
	List of Abbreviations	102

1 Introduction

Music and musical instruments are an integral part of human culture (Wegst, 2006). Despite technological progress, wood remains the preferred construction material across acoustic instrument groups (Wegst, 2006; Yoshikawa and Waltham, 2014; Zhou et al., 2022). There is an enormous diversity of musical instruments that evolved along different cultural pathways associated with local flora as well as musical and craft traditions (Carlier et al., 2019). Each instrument has a characteristic sound timbre, which is determined by a combination of sound production mechanism, material properties, geometry, and structural design (Fletcher and Rossing, 1991). However, across civilizations, woods selected for specific groups of musical instruments share certain common material characteristics (Brémaud, 2012a).

Traditional tonewood suitable for high-quality musical instruments is becoming increasingly scarce and valuable (Bennett, 2016; Liu et al., 2020). Currently, many tropical wood species are at risk of extinction and are listed under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) (Ahmed and Adamopoulos, 2018; Brémaud and Poidevin, 2018; Wegst, 2008). In addition, long-term studies indicate changes in wood properties, primarily associated with a decrease in wood density (largely driven by accelerated tree growth due to climate change), which may negatively affect future wood quality (González de Andr  es et al., 2025; Olivar et al., 2015; Torresan et al., 2024).

Consequently, there is a well-founded need to substitute endangered wood species with alternative, sustainable materials, as well as to enhance the properties of available lower-quality wood by modifications (Ahmed and Adamopoulos, 2018; Ahvenainen, 2019; Spro  mann et al., 2017). In addition to aesthetics—which also contributes to the demand for and valuation of materials used in musical instruments (Zhou et al., 2022)—the choice of wood or alternative materials should be guided by the properties of traditional tonewoods, particularly their distinctive mechanical performance and vibro-acoustic response (Bucur, 2006; Wegst, 2006).

2 Literature Review

2.1 Physico-Mechanical Properties Related to Dynamic Response of Wood

The dynamic response of a solid is governed by the generalised Hooke's law for viscoelastic materials, describing the relationship between stress (σ), strain (ε), and stiffness (C) (or equivalently its inverse, the compliance (S)) tensors (Bucur, 2006), as given in Equations 1 and 2.

$$[\sigma] = [C] \cdot [\varepsilon] \quad (1)$$

$$[\varepsilon] = [S] \cdot [\sigma] \quad (2)$$

When transitioning from the description of homogeneous, isotropic materials to the description of heterogeneous, anisotropic materials, the stiffness matrix is turned into a compliance matrix, which is expanded to include additional elastic constants—namely Young's moduli (E), shear moduli (G), and Poisson's ratios (ν) (Mechel, 2008). Wood exhibits independent properties along three mutually perpendicular directions: longitudinal (L), radial (R), and tangential (T), and as an orthotropic material, it can be described by three E , three G , and six ν (Wood Handbook, 2010). The distribution of elastic constants, as determined by the material's internal structure, governs the propagation of stress waves after dynamic loading, described by the equation of motion in continuum mechanics (Bucur, 2006). Along the principal symmetry directions or outside them, two types of waves can be distinguished: longitudinal waves (motion parallel to the propagation direction) and transverse (or shear) waves (motion perpendicular to the propagation direction) (Bucur, 2006). Additionally, surface waves (Rayleigh waves) are generated along the surface (Bucur, 2023). The speed of wave propagation (c) and wave attenuation represent the fundamental parameters describing the vibro-acoustical behavior of materials (Ross and Pellerin, 1994).

Speed of Sound

The speed of longitudinal (c_L) and transverse (c_T) waves in a material is given by the square root of the specific modulus of elasticity, i.e., the ratio of the corresponding modulus to the material's density (ρ), expressed in Equation 3 (Bucur and Archer, 1984; Fletcher and Rossing, 1991; Wegst, 2006).

$$c_L = \sqrt{\frac{E}{\rho}} ; c_T = \sqrt{\frac{G}{\rho}} \quad (3)$$

Due to the anatomical structure of wood and the associated distribution of stiffness, the c_L is generally highest in the L direction. The ratios of L/T and L/R velocities vary depending on the wood species (Bucur, 2006; Dackermann et al., 2016; Hasegawa et al., 2011). Bucur described the variation of wave velocity over a frequency range of 100 kHz to 1.5 MHz, where longitudinal waves in the L direction were strongly affected by frequency. In contrast, the velocities of transverse waves and longitudinal waves in transverse directions above 250kHz were insensitive to frequency change. Considering the dimensions of the measured samples and structural elements (with fibers vibrating in the 1–1.5 MHz range), it was concluded that once the wavelength (λ) becomes comparable to the dimensions of the samples, cells, or structural layers, the propagation velocity of wave becomes frequency-dependent (Bucur, 2006; Bucur and Feeney, 1992). Furthermore, it follows that by appropriately selecting the excitation frequency range, wood can be evaluated at the microstructural level; conversely, when a λ is much larger than the size of the inhomogeneities, it can be treated as a homogeneous solid of given dimensions (Bucur, 2006; Feeney et al., 1998).

In a solid body of finite dimensions, the propagation of an elastic wave leads to multiple reflections, subsequent interference of two or more waves moving in opposite directions, and the formation of standing waves (Hambric et al., 2016). When the excitation frequency coincides with the frequency of a standing wave—i.e., the natural resonance frequency of the body—this results in a high vibration amplitude and the formation of specific vibrational mode shapes (Gloth and Sinapius, 2004; Hambric et al., 2016). In addition to the geometry and material properties, the resulting mode shape and its frequency (f) are also influenced by the boundary conditions, such as the support of the body (Calim, 2016; Guo et al., 2025). For the n th longitudinal mode (standing longitudinal waves) or torsional mode (standing shear waves) of vibration, the f of a free-free body is given by:

$$f_n = n \frac{c}{2l} \quad (4)$$

since the λ is twice the body length (l) (Brancheriau and Bailleres, 2002; Hambric et al., 2016).

The calculation of the frequency of bending (or flexural) mode shape is more complex and must account for the dimensions of the investigated body. Based on this, the differential equation of motion proposed by Timoshenko (1921) is used, for which approximate solutions by Bernoulli–Euler (for very high length-to-depth ratios, with shear neglected), Rayleigh (for low length-to-depth ratios, with shear neglected), or Bordonné (for low length-to-depth ratios, with shear included), among others (Bork, 1995; Brancheriau and Bailleres, 2002; Calim, 2016; van Rensburg and van der Merwe, 2006). Based on the employed equation, the

intervals of higher modes (defined as frequency ratios) can be estimated, which, unlike longitudinal and torsional modes, are not in integer ratios (Bork, 1995; Bretos et al., 1999).

With knowledge of the locations of nodes and antinodes of standing waves, it is possible—through appropriate support of the specimen and proper placement of the excitation—to induce vibrational mode shapes using even impact excitation and, based on the calculated wave velocity, to determine additional vibro-acoustic properties (Ahmed and Adamopoulos, 2018; Bucur, 2006; Haines et al., 1996; Hambric et al., 2016; Hassan et al., 2013; Nop and Tippner, 2022).

Acoustic Impedance

The resistance of a material to the propagation of elastic waves is described by the characteristic impedance (Z), as given in Equation 5 (Wegst, 2006).

$$Z = c\rho = \sqrt{E\rho} \quad (5)$$

This property should be considered in the transmission of acoustic energy between media—for example, from the strings to the bridge and then to the soundboard—as a balance must be achieved between efficient energy transfer to the radiating element for audible sound production and preventing excessive energy loss from the oscillator (Fletcher and Rossing, 1991; Wegst, 2006). Low Z aids sound radiation into the surrounding air (Yoshikawa and Waltham, 2014).

Radiation Coefficient

The radiation coefficient (K) indicates how much energy a material is capable of radiating into the air. To achieve the highest possible sound output, it is necessary to obtain the highest possible value, according to Equation 6 (Wegst, 2006).

$$K = \frac{c}{\rho} = \sqrt{\frac{E}{\rho^3}} \quad (6)$$

Damping Coefficients

The amplitude of the wave over time is influenced, among other factors, by damping, which can be divided into hysteretic and viscous damping (Bucur, 2006; Merhar, 2024). Hysteretic damping can be regarded as internal friction arising from the material's internal structure, whereas viscous damping depends on the velocity of motion (Merhar, 2024). Although vibrational damping is a phenomenon resulting from various mechanisms, it has been accepted in materials science that it can be evaluated as internal friction (Alkadri et al., 2018; Baar et al., 2016; Bucur, 2006; Merhar, 2024; Traoré et al., 2010). There are many ways to express damping, such as the loss angle (δ) and loss tangent ($\tan \delta$), damping ratio (ζ), loss factor (η), quality factor (Q), logarithmic decrement of damping (LDD), and

temporal absorption coefficient (β), which are interrelated according to Equation 7 (Brémaud et al., 2012; Casiano, 2016; Magalas and Malinowski, 2003; Ono Sokki, n.d.).

$$\tan \delta = 2\zeta = \eta = \frac{1}{Q} = \frac{LDD}{\pi} = \frac{\beta}{\pi f} \quad (7)$$

The different expressions are mainly due to the distinction between the extraction methods (Casiano, 2016; Merhar, 2024). The most commonly used methods include determination from the amplitude decay in the time domain, from the frequency bandwidth of the resonance peak, or as the ratio of loss energy to storage energy (Ashaduzzaman et al., 2020; Brancheriau et al., 2010; Magalas and Malinowski, 2003; Mevada and Patel, 2016; Saxena and Gupta, 2018). Brémaud et al. compiled a large database of 445 wood species, which enabled them to confirm, on a broader scale, the relationship between the damping coefficient and the specific modulus of elasticity, as reported by Ono and Norimoto (Brémaud, 2012b; Brémaud et al., 2009; Ono and Norimoto, 1984, 1983). Therefore, the use of combined parameter (e.g., their ratio) is advantageous, as it enables a more efficient representation.

Acoustic Conversion Efficiency

The acoustic conversion efficiency (*ACE*) is given by the ratio of the radiation to the damping, according to Equation 8 (Straže et al., 2015).

$$ACE = \frac{K}{\tan \delta} = \frac{\sqrt{\frac{E}{\rho^3}}}{\tan \delta} \quad (8)$$

ACE has been accepted as an overall estimate of the suitability of materials for resonators—i.e., those parts of a musical instrument whose purpose is to produce the loudest possible response (Baar et al., 2016; Sedik et al., 2010). Straže et al. have found that relative acoustic conversion efficiency (*RACE*), in which the influence of density is reduced (see Equation 9), can be a more sensitive indicator of material quality, providing finer grading capability (Straže et al., 2015).

$$RACE = \frac{c}{\tan \delta} = \frac{\sqrt{\frac{E}{\rho}}}{\tan \delta} \quad (9)$$

2.2 Factors Influencing Vibro-Acoustical Properties

The physico-mechanical properties of wood are determined by its anatomical structure and extractive content, related to species and growth conditions (Baar et al., 2016; Bucur, 2006; Dunisch, 2019; Wood Handbook, 2010)

The main factor that significantly affects most physico-mechanical properties of wood is its moisture content (*MC*) (Gerhards, 1982; Ozyhar et al., 2016; Wang and Wang, 1999; Wood Handbook, 2010). In the green wood, *MC* can vary from 30% to 200%, depending on the species (Wood Handbook, 2010). In general, wood exhibits the highest mechanical performance at low *MC* (Babiak et al., 2018; Gerhards, 1982; Matsunaga et al., 2000). Above the fibre saturation point (FSP) ($MC \approx 30\%$), increases in *MC* do not cause significant changes, either dimensionally or in mechanical properties (Barrett and Hong, 2010; Nocetti et al., 2015; Wood Handbook, 2010). However, as *MC* decreases below the FSP, mechanical properties change rapidly, approximately linearly, with stiffness increasing and damping decreasing (Chan et al., 2011; Obataya et al., 2001). The strong correlation between the stiffness of green and dry wood enables property prediction and quality assessment already in the green state (Nocetti et al., 2017, 2015). *MC* also induces dimensional variations due to swelling and shrinkage (Vasques and Dias Rodrigues, 2011). These changes are almost negligible in the L direction ($\approx 0.1\text{--}0.2\%$), whereas they reach about 5% in the R direction and 10% in the T direction (Broda et al., 2024; Wood Handbook, 2010). If wood is dried under controlled conditions, its vibro-acoustical properties can be improved, as gradual seasoning (physical aging and stress relaxation processes) leads to increased sound propagation velocity and reduced damping, while minimising the formation of defects such as cracks caused by non-uniform stress distribution (Bucur, 2006; Noguchi et al., 2012; Obataya et al., 2020).

Another factor (largely associated with moisture) is temperature; an increase in temperature leads to a reduction in vibro-acoustical properties, with the rate of this change being approximately three times higher below the freezing point than above it (Bucur, 2006; Chan et al., 2011). When wood is exposed to temperatures above 100 °C for a certain period of time, irreversible changes (modifications or degradations) occur in the cell wall. These changes manifest as mass loss, decreased water absorption, and alterations in mechanical properties (such as stiffness and damping) (Danihelová et al., 2022; Hassan Vand et al., 2026; Wood Handbook, 2010). Thermal modification has been demonstrated to enhance dimensional stability and increase the *ACE* (Ahmed and Adamopoulos, 2018; Danihelová et al., 2022; Hassan Vand et al., 2026).

Dimensional stability and *ACE* can also be enhanced by chemical modification methods, including acetylation, furfurylation, extraction, formaldehyde treatment, and others (Ahmed and Adamopoulos, 2018; Bucur, 2006; Matsunaga et al., 2000; Roohnia et al., 2015;

Traoré et al., 2010). Chemical modifications within the cell wall can also modulate the anisotropic distribution of material properties (Brémaud et al., 2011).

Changes in wood properties caused by biological agents, such as wood-decaying or wood-staining fungi, are generally considered undesirable forms of degradation (Cristini et al., 2022). However, there are cases of controlled modification where, after a certain period of exposure, an increase K and damping has been observed (Danihelová et al., 2019; Schwarze et al., 2008; Schwarze and Morris, 2020).

Among the boundary conditions relevant to musical instruments, the influence of long-term mechanical loading on wood has been studied, showing an approximately 10% decrease in damping after just 5 hours of vibration. (Bucur, 2006). The effect of creep in wood (for example under long-term exposure to string tension) should also be taken into account (Wood Handbook, 2010).

2.3 Materials for Wooden Musical Instruments

The most widely used system for classifying musical instruments, published by E. Hornbostel and C. Sachs in 1913, is based on the physical principle of sound generation and distinguishes the following groups: aerophones – wind instruments, chordophones – string instruments, membranophones – membrane instruments, idiophones – self-sounding instruments, and electrophones, whose function depends on electrical energy (Bennett, 2016; Wegst, 2006). Although the oscillator is the primary vibrating element whose material properties, together with the boundary conditions, are crucial for the generated tone, in most instruments, the resulting sound characteristics depend on the combined interaction of other elements that are coupled in a complex system (Fletcher and Rossing, 1991).

Traditional Tonewoods

The sound characteristic of aerophones depends on the air column excitation (by a reed or at an edge), its length, its shape (cylindrical or conical), as well as the precise design and arrangement of the finger holes (Wegst, 2006). The main requirements for materials used in aerophones are dimensional stability and the ability to be drilled with high precision (Wegst, 2006). These are met by high-density woods ($\rho \approx 800 \text{ kg}\cdot\text{m}^{-3}$) with a fine structure (Wegst, 2006). Woodwind instruments are made from Bahia wood, Indian and Brazilian rosewood, Macassar ebony, grenadilla, cocobolo, sycamore, Norway maple, ma-had, cocuswood, chingchan, European, Castella, and West Indian boxwood, bubinga, red chacate, Indian kamba, olive wood, and fruitwoods such as sweet cherry, European crab apple, damson plum, and European pear (Danihelová et al., 2015; Fletcher and Rossing, 1991; Holz, 1996; Wegst, 2006).

The body of chordophones supports the strings and amplifies their oscillations, and must therefore have a high K ($\approx 12 \text{ m}^4\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$), low Z ($\approx 2.4 \text{ MPa}\cdot\text{s}\cdot\text{m}^{-1}$), but also resistance to long-term stress caused by string tension (Rossing, 2010; Waltham and Yoshikawa, 2018;

Wegst, 2006). The greatest attention is usually given to soundboards, as their contribution to the resulting timbre is crucial (Brémaud, 2012a; Carcagno et al., 2018; Roohnia et al., 2011; Vasques and Dias Rodrigues, 2011; Yano et al., 1994). For soundboards, a high axial-to-shear ratio (reflecting high material anisotropy) is recommended, as it leads to increased losses at high frequencies and provides a desirable softness to the sound (Brémaud, 2012a; Obataya et al., 2000; Yoshikawa and Waltham, 2014). Softwoods commonly used for soundboards in Western instruments exhibit lower ρ ($\approx 400 \text{ kg}\cdot\text{m}^{-3}$) and higher specific dynamic modulus (associated with a homogeneous anatomical structure—uniform annual ring width and a small proportion of latewood), while their damping ($\tan \delta \approx 0.006$) is approximately average across all species (Brémaud, 2012a; Bucur, 2016; Faktorová et al., 2024; Wegst, 2006). The value of K above which (mostly coniferous) wood is classified as “resonance wood” is $K \geq 12 \text{ m}^4\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$ in L direction and $K \geq 3.5 \text{ m}^4\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$ in R or T directions (Fedyukov et al., 2019). Woods used for the soundboards, bridges, ribs, sides, and backs of chordophones are most commonly Sitka and Norway spruce, silver and Douglas fir, sycamore, silver and Japanese maple, western red cedar, silver birch, Mediterranean cypress, American basswood, beech, paulownia, mulberry, Madagascar rosewood, and amboyna (Bennett, 2016; Bucur, 2016; Rossing, 2010; Wegst, 2006). Hardwoods such as Honduran and West Indian mahogany, Indian and Brazilian rosewood, African blackwood, Ceylon, Gaboon, and Macassar ebony, and black walnut are used for necks and fretboards (Bennett, 2016; Liu et al., 2020). Bows are typically made from pernambuco, Brazilwood, Sappan wood, redwood, snakewood, plum tree, and lower-quality bows were historically made from ironwood and beech (Bucur, 2016; Fletcher and Rossing, 1991; Waltham and Yoshikawa, 2018).

The sound of membranophones is primarily produced by membrane vibrations, which depend on the applied tension and membrane diameter (Bucur, 2022). The dimensions of the drum shell (the volume of the air cavity) are designed to support membrane oscillations at the desired frequencies, so a dimensionally stable material that also maintains constant membrane tension is required (Bucur, 2022; Ibáñez-Arnal et al., 2020). The shells are made from Honduras rosewood, mahogany, bubinga, jackfruit, cordia, maple, walnut, cherry, birch, beech, oak, poplar, and tree of heaven (Aiyeloja et al., 2015; Bucur, 2022; Damodaran et al., 2015; Fanxizi et al., 2021). Tropical hardwoods used for membranophones and idiophones across different regions all share a high density with high moduli ($E_L \approx 30 \text{ GPa}$) – and thus average specific moduli – and low damping ($\tan \delta \approx 0.004$) (Brémaud, 2012a; Waltham and Yoshikawa, 2018).

Because idiophones are usually excited impulsively, low damping is essential, leading to a brighter sound, as higher-frequency modes can vibrate for a longer time (Bork, 1995; Wegst, 2006). At the same time, the highest possible K is required, and due to repeated mallet impacts, high hardness is necessary (Wegst, 2006). For idiophones (represented by xylophones and marimbas, woodblocks, castanetes), hardwoods such as Amazon and

Honduran rosewood, African padauk, grenadilla, merbau, wenge, cristobal, ching-chan, mahad, and pau-rosa are used (Bucur, 2022; Čulík and Danihelová, 2025; Rujinirun et al., 2005; Straže et al., 2015; Wegst, 2006). The sound of percussion instruments is also influenced by the mallets (both the head and the stick), whose softness—related to rebound speed—determines how many higher modes can develop without being damped (Bucur, 2022; Wegst, 2006). Drumsticks and mallets are made from rosewood, lancewood, hickory, elm, ash, beech, birch, maple, walnut, hornbeam, and boxwood (Bucur, 2022, 2006; Mania et al., 2020; Waltham and Yoshikawa, 2018).

Material Innovations in Musical Instruments

The concept of substituting wood is not new. A report from the turn of the millennium states that, after two decades of development, musical instruments—particularly chordophones—constructed from wood-based composites such as plywood can compete with their traditional wooden counterparts (Besnainou, 1998). Bucur notes that already Baroque violins had fingerboards constructed from plywood—ebony or maple veneers with a willow or spruce core—and that wood-based composites are used across a range of musical instruments, including upright pianos, harps, guitars, drum shells, and drumsticks (Bucur, 2022, 2016). With the advancement of technology, a wide range of materials has emerged as alternatives, including bio-based fibre (flax, hemp, ramie, and jute), or man-made fibre (fiberglass, carbon, and graphite) reinforced composites, as well as 3D-printed materials and nanocomposites (Brezas et al., 2024; Bucur, 2022; Fanxizi et al., 2021). These alternatives, with the possibility of fine-tuning their properties through engineering, have the potential to compete with traditional reference woods, further develop established musical instruments, or enable the creation of new ones (Brezas et al., 2024). However, not all traditional selection methods can be applied to new materials or modified woods, and as emphasized by some instrument makers, there is a need for reliable material assessment based on vibro-acoustic properties (Buksnowitz et al., 2007; Plath et al., 2022).

2.4 Methods for Vibro-Acoustic Properties Determination

The selection of suitable wood for a musical instrument has been (and in most cases still is) based on the maker's empirical approach (Spycher et al., 2008). This is based, among other things, on the visual assessment of the macroscopic characteristics of the wood (annual ring width, regularity, absence of defects, color, etc.) (Buksnowitz et al., 2007; Dinulică et al., 2021; Spycher et al., 2008). These parameters are related to ρ , and thus also to the moduli of elasticity and damping, as well as to hardness or dimensional stability. (Ahvenainen, 2019; Dinulică et al., 2021; Dündar et al., 2016; Quintavalla et al., 2022; Spycher et al., 2008). However, the strength of this correlation varies with species and is generally insufficient for reliable prediction (Brémaud, 2012a; Dündar et al., 2016). Thus, a multi-criteria approach integrating different disciplines is required; alongside visual grading, mechanical methods

should be included, as they allow for the detection of hidden defects and contribute to more precise material classification (Brémaud, 2012a; Carlier et al., 2019; Yang et al., 2003). The maker's evaluation of sound characteristics is performed by sequentially holding the sample at several points (mostly corresponding to the nodal points of individual vibration modes), tapping it, and listening to the response, with the resulting grading being purely subjective (Buksnowitz et al., 2007; Danihelova and Culik, 2005; Straže et al., 2015). Moreover, most instrument makers specialize in a particular type of instrument and consistently source similar species of wood with comparable dimensions and condition from specialized suppliers, whom they trust for wood preselection (Carlier et al., 2019). To select suitable material across different alternatives, shapes, and conditions, more advanced methods must be employed.

Static tests are commonly used to determine the mechanical properties of materials, including all elastic constants (Bucur, 2006; Merhar, 2020). However, these methods are often time-consuming and experimentally demanding, as they require a set of specimens with prescribed dimensions and geometry for each test (Aira et al., 2014; Milch et al., 2016). In contrast, properties determined by non-destructive dynamic methods show a very high correlation with static properties and therefore offer an efficient alternative, particularly for valuable or limited materials (Barrett and Hong, 2010; Bucur, 2006).

Time of Flight Method

The determination of the speed of wave propagation based on the direct measurement of the travel time between two probes or sensors is carried out using the time-of-flight method (ToF) (Bakar et al., 2023; Gil-Martín et al., 2025). The emitted wave can be generated either by an impulse induced by a hammer impact or by means of a high-frequency transmitter (Gonçalves et al., 2019; Grabianowski et al., 2006). This method is commonly used across a range of scales—from small specimens, through timber samples and felled logs, to standing trees (Bakar et al., 2023; El-Hadad et al., 2018; Faktorová et al., 2024; Grabianowski et al., 2006). With an appropriately selected specimen geometry, the ToF method can be used to completely characterize the elastic properties (Bucur, 2006; Grabianowski et al., 2006).

Dynamic Mechanical Analysis

Dynamic mechanical analysis (DMA) is an established technique for determining the glass transition temperature of polymers and composite materials under cyclic loading (Ashaduzzaman et al., 2020; Birkinshaw et al., 1986). During DMA, the specimen is clamped (using clamps and test setups similar to static testing) and subjected to a constant sinusoidal or frequency-sweep load (typically 0.1–100 Hz) (Aslan and Aktürk, 2024; Saxena and Gupta, 2018). This allows the measurement of the storage modulus (E') and loss modulus (E''), as well as $\tan \delta$, defined as the phase angle between the applied stress and the observed strain (Ashaduzzaman et al., 2020; Corn et al., 2013). Nowadays, in addition to temperature-

controlled chambers, DMA also offers humidity-controlled chambers, which enable new possibilities for studying the dynamic behavior of wood (Goken et al., 2018; Kutnar et al., 2021). Smaller specimens (30–60 mm × 5–13 mm × 1–3 mm), determined by the chamber size and test configuration, are ideal for testing various wood modifications or limited amounts materials (Hamdan et al., 2010; Kaboorani and Blanchet, 2014; Kutnar et al., 2021, 2013; Saxena and Gupta, 2018).

Experimental Modal Analysis

Experimental modal analysis (EMA) is the process of determining the mode shapes along with their frequencies and corresponding damping of a linear time-invariant system (Kawrza et al., 2022; Vasques and Dias Rodrigues, 2011). It is predominantly used for systems with complex geometry or demanding boundary conditions (Corradi et al., 2017; Kawrza et al., 2022). To perform EMA, appropriate instrumentation is required, including multi-point measurements of the system (using accelerometers or a scanning vibrometer) and an excitation source (e.g., an impact hammer or a mechanical shaker with continuous excitation), together with the measurement of the input force or excitation signal (Kawrza et al., 2022; Longo et al., 2018). From the recorded signals of the system's dynamic response, by applying a fast Fourier transform (FFT) and others numerical techniques, the frequency spectra with corresponding mode shapes can be determined (Corradi et al., 2017; Longo et al., 2018; Vasques and Dias Rodrigues, 2011). After proper identification of the mode shapes and considering the boundary conditions, the corresponding material properties, such as dynamic moduli, ν , and damping coefficients, can be derived from the frequency response function (Longo et al., 2018; McIntyre and Woodhouse, 1988; Vasques and Dias Rodrigues, 2011).

Frequency Resonance Technique

A simplified form of EMA applied to systems with simple geometry and boundary conditions is called the frequency resonance technique (FRT) (Nop et al., 2024, 2023; Nop and Tippner, 2022). The method is based on the same principle used by instrument makers in auditory evaluation, but with objectively defined parameters obtained using measurement devices (e.g., a microphone) (Danilova and Culik, 2005; Straže et al., 2015). With appropriate placement of the specimen supports (at nodal points), location of excitation and response sensing (at antinodes), the vibrations of specific modes can be measured using only a single sensor (Alkadri et al., 2018; ASTM E1876 – 15, 2013; Merhar, 2024). Typically, a free–free supported specimen is used, in the form of a slender prismatic beam or strip (Gerhard et al., 2020; Gilani et al., 2016; Sedik et al., 2010). It is a highly efficient method for determining elastic constants, applicable across a wide range of specimen scales—from small samples suitable for modifications, through specimens of standardized dimensions for static bending tests, up to freshly cut logs (Cristini et al., 2022; Gerhard et al., 2020; Nop et

al., 2024, 2023; Straže and Novak, 2022). Using FRT on green logs, Straže and Novak (2022) found that lower wood quality (as determined visually and by *RACE*) exhibits deviations from the theoretical intervals of longitudinal modes, i.e., integer multiples of the fundamental mode frequency, indicating that acoustic response alone—without knowledge of additional material properties—can be used, for example, for pre-grading (Straže and Novak, 2022). For more complex geometries or boundary conditions, the absence of visualization makes correct identification of mode shapes and their corresponding frequencies and damping more challenging; therefore, in addition to the examiner's experience, multi-point excitation together with a larger number of differently positioned sensors, or a theoretical background based on predictive numerical models, is recommended (Merhar, 2024; Nop et al., n.d.; Qi and Kaiyin, 2024; Tippner et al., 2023).

Numerical Modelling

Finite-element methods (FEM) have been developed to describe the behavior of complex structures under specific conditions of interest (Viala et al., 2016). Numerical simulations are based on the definition of the material model, geometry, and boundary conditions (Hambric et al., 2016; Ray et al., 2021). Experimentally determined material properties are often supplemented with values from the literature, which are subsequently adjusted to match the measured response (Gonzalez et al., 2022; Kabała and Barczewski, 2020; Merhar, 2020; Merhar and Humar, 2020). Once the numerical model is validated against experimental data, a high level of accuracy can be expected in predicting the system response for modified geometries or varying loading conditions (Merhar and Humar, 2020). The combination of simultaneous non-destructive determination of multiple vibro-acoustic properties with numerical simulations provides a powerful tool for improving material grading and designing musical instruments, enabling the planning of their sound characteristics (Ray et al., 2021; Viala et al., 2020).

3 References

- Ahmed, S.A., Adamopoulos, S., 2018. Acoustic properties of modified wood under different humid conditions and their relevance for musical instruments. *Applied Acoustics* 140, 92–99. <https://doi.org/10.1016/j.apacoust.2018.05.017>
- Ahvenainen, P., 2019. Anatomy and mechanical properties of woods used in electric guitars. *IWA Journal* 40, 106–123. <https://doi.org/10.1163/22941932-40190218>
- Aira, J.R., Arriaga, F., Íñiguez-González, G., 2014. Determination of the elastic constants of Scots pine (*Pinus sylvestris* L.) wood by means of compression tests. *Biosystems Engineering* 126, 12–22. <https://doi.org/10.1016/j.biosystemseng.2014.07.008>
- Alkadri, A., Carlier, C., Wahyudi, I., Gril, J., Langbour, P., Brémaud, I., 2018. Relationships between anatomical and vibrational properties of wavy sycamore maple. *IWA Journal* 39, 63–86. <https://doi.org/10.1163/22941932-20170185>
- Ashaduzzaman, M., Hale, M.D., Ormondroyd, G.A., Spear, M.J., 2020. Dynamic mechanical analysis of Scots pine and three tropical hardwoods. *International Wood Products Journal* 11, 189–203. <https://doi.org/10.1080/20426445.2020.1799910>
- Aslan, G., Aktürk, N., 2024. Experimental and theoretical analysis of frequency- and temperature-dependent characteristics in viscoelastic materials using Prony series. *Applied Mechanics* 5, 786–803. <https://doi.org/10.3390/applmech5040044>
- ASTM International, 2015. Standard test method for dynamic Young's modulus, shear modulus, and Poisson's ratio by impulse excitation of vibration. ASTM E1876-15. *ASTM International*, West Conshohocken, PA. <https://doi.org/10.1520/E1876-15>
- Baar, J., Tippner, J., Gryc, V., 2016. Wood anatomy and acoustic properties of selected tropical hardwoods. *IWA Journal* 37, 69–83. <https://doi.org/10.1163/22941932-20160121>
- Babiak, M., Gaff, M., Sikora, A., Hysek, Š., 2018. Modulus of elasticity in three- and four-point bending of wood. *Composite Structures* 204, 544–551. <https://doi.org/10.1016/j.compstruct.2018.07.113>
- Bakar, A.H.A., Legg, M., Konings, D., Alam, F., 2023. Estimation of the rod velocity in wood using multi-frequency guided wave measurements. *Applied Acoustics* 202, 109108. <https://doi.org/10.1016/j.apacoust.2022.109108>
- Barrett, J.D., Hong, J.P., 2010. Moisture content adjustments for dynamic modulus of elasticity of wood members. *Wood Science and Technology* 44, 485–495. <https://doi.org/10.1007/s00226-009-0292-z>
- Bennett, B.C., 2016. The sound of trees: Wood selection in guitars and other chordophones. *Economic Botany* 70, 49–63. <https://doi.org/10.1007/s12231-016-9336-0>
- Besnainou, C., 1998. Composite materials for musical instruments: The maturity. *Journal of the Acoustical Society of America* 103, 2872–2873. <https://doi.org/10.1121/1.421525>
- Birkinshaw, C., Buggy, M., Henn, G.G., 1986. Dynamic mechanical analysis of wood. *Journal of Materials Science Letters* 5, 898–900. <https://doi.org/10.1007/BF01729266>
- Bork, I., 1995. Practical tuning of xylophone bars and resonators. *Applied Acoustics* 46 (1), 103–127. [https://doi.org/10.1016/0003-682X\(95\)93953-F](https://doi.org/10.1016/0003-682X(95)93953-F)

- Brancheriau, L., Baillères, H., 2002. Natural vibration analysis of clear wooden beams: A theoretical review. *Wood Science and Technology* 36, 347–365. <https://doi.org/10.1007/s00226-002-0143-7>
- Brancheriau, L., Kouchade, C., Brémaud, I., 2010. Internal friction measurement of tropical species by various acoustic methods. *Journal of Wood Science* 56, 371–379. <https://doi.org/10.1007/s10086-010-1111-8>
- Brémaud, I., 2012a. Acoustical properties of wood in string instruments soundboards and tuned idiophones: Biological and cultural diversity. *Journal of the Acoustical Society of America* 131, 807–818. <https://doi.org/10.1121/1.3651233>
- Brémaud, I., 2012b. What do we know on “resonance wood” properties? Selective review and ongoing research, in: *Proceedings of the Acoustics 2012 Nantes Conference*, Nantes, France, 23–27 April 2012, pp. 2759–2764.
- Brémaud, I., El Kaïm, Y., Guibal, D., Minato, K., Thibaut, B., Gril, J., 2012. Characterisation and categorisation of the diversity in viscoelastic vibrational properties between 98 wood types. *Annals of Forest Science* 69, 373–386. <https://doi.org/10.1007/s13595-011-0166-z>
- Brémaud, I., Gril, J., Thibaut, B., 2011. Anisotropy of wood vibrational properties: Dependence on grain angle and review of literature data. *Wood Science and Technology* 45, 735–754. <https://doi.org/10.1007/s00226-010-0393-8>
- Brémaud, I., Minato, K., Thibaut, B., 2009. Mechanical damping of wood as related to species classification: A preliminary survey, in: *Proceedings of the 6th Plant Biomechanics Conference*, Cayenne, French Guiana, pp. 1–7.
- Brémaud, I., Poidevin, N., 2013. Cultural and mechanical approaches in the choice of woods in instrument making: The case of early bows, in: Castellengo, M., Genevoix, H. (eds.), *La musique et ses instruments / Music and its instruments*. Éditions Delatour France, pp. 1–27. ISBN 978-2-7521-0159-4.
- Bretos, J., Santamaría, C., Alonso Moral, J., 1999. Finite element analysis and experimental measurements of natural eigenmodes and random responses of wooden bars used in musical instruments. *Applied Acoustics* 56, 141–156. [https://doi.org/10.1016/S0003-682X\(98\)00030-9](https://doi.org/10.1016/S0003-682X(98)00030-9)
- Brezas, S., Katsipis, M., Kaleris, K., Papadaki, H., Katerelos, D.T.G., Papadogiannis, N.A., Bakarezos, M., Dimitriou, V., Kaselouris, E., 2024. Review of manufacturing processes and vibro-acoustic assessments of composite and alternative materials for musical instruments. *Applied Sciences* 14, 2293. <https://doi.org/10.3390/app14062293>
- Broda, M., Popescu, C.M., Poszwa, K., Roszyk, E., 2024. How thermal treatment affects the chemical composition and the physical, mechanical and swelling properties of Scots pine juvenile and mature wood. *Wood Science and Technology* 58, 1153–1180. <https://doi.org/10.1007/s00226-024-01561-2>
- Bucur, V., 2023. A review on acoustics of wood as a tool for quality assessment. *Forests* 14, 1545. <https://doi.org/10.3390/f14081545>
- Bucur, V., 2022. Handbook of Materials for Percussion Musical Instruments. *Springer Nature*. <https://doi.org/10.1007/978-3-030-98650-6>
- Bucur, V., 2016. Handbook of Materials for String Musical Instruments. *Springer*. <https://doi.org/10.1007/978-3-319-32080-9>

- Bucur, V., 2006. Acoustics of wood. *Springer*, Berlin. ISBN 978-3-540-30594-7
- Bucur, V., Archer, R.R., 1984. Elastic constants for wood by an ultrasonic method. *Wood Science and Technology* 18, 255–265. <https://doi.org/10.1007/BF00353361>
- Bucur, V., Feeney, F., 1992. Attenuation of ultrasound in solid wood. *Ultrasonics* 30, 76–81. [https://doi.org/10.1016/0041-624X\(92\)90037-M](https://doi.org/10.1016/0041-624X(92)90037-M)
- Buksnowitz, C., Teischinger, A., Müller, U., Pahler, A., Evans, R., 2007. Resonance wood (Picea abies (L.) Karst.): Evaluation and prediction of violin makers' quality grading. *Journal of the Acoustical Society of America* 121, 2384–2395. <https://doi.org/10.1121/1.2434756>
- Calim, F.F., 2016. Transient analysis of axially functionally graded Timoshenko beams with variable cross-section. *Composites Part B: Engineering* 98, 472–483. <https://doi.org/10.1016/j.compositesb.2016.05.040>
- Carcagno, S., Bucknall, R., Woodhouse, J., Fritz, C., Plack, C.J., 2018. Effect of back wood choice on the perceived quality of steel-string acoustic guitars. *Journal of the Acoustical Society of America* 144, 3533–3547. <https://doi.org/10.1121/1.5084735>
- Carlier, C., Alkadri, A., Gril, J., Brémaud, I., 2018. Revisiting the notion of “resonance wood” choice: A decompartmentalised approach from violin makers' opinion and perception to characterization of material properties' variability, in: *Wooden Musical Instruments – Different Forms of Knowledge. Book of end of WoodMusICK COST Action FP1302*, pp. 119–132.
- Casiano, M.J., 2016. Extracting damping ratio from dynamic data and numerical solutions. NASA/TM-2016-218227. *NASA Technical Reports Server (NTRS)*, 1–29.
- Chan, J.M., Walker, J.C.F., Raymond, C.A., 2011. Effects of moisture content and temperature on acoustic velocity and dynamic MOE of radiata pine sapwood boards. *Wood Science and Technology* 45, 609–626. <https://doi.org/10.1007/s00226-010-0350-6>
- Corn, S., Dupuy, J.S., Ienny, P., Daridon, L., 2013. Vibration analysis techniques for detecting filler–matrix decohesion in composites. *HAL (hal-00797606)*. <https://hal.archives-ouvertes.fr/hal-00797606>
- Corradi, R., Miccoli, S., Squicciarini, G., Fazioli, P., 2017. Modal analysis of a grand piano soundboard at successive manufacturing stages. *Applied Acoustics* 125, 113–127. <https://doi.org/10.1016/j.apacoust.2017.04.010>
- Cristini, V., Tippner, J., Nop, P., Zlámál, J., Hassan Vand, M., Šeda, V., 2022. Degradation of beech wood by Kretzschmaria deusta: Its heterogeneity and influence on dynamic and static bending properties. *Holzforschung* 76, 813–824. <https://doi.org/10.1515/hf-2022-0039>
- Čulík, M., Danihelová, A., 2025. European mountain ash as a possible substitute for Honduran rosewood in the production of xylophones. *Akustika* 48, 82–88. <https://doi.org/10.36336/akustika20254882>
- Dackermann, U., Elsener, R., Li, J., Crews, K., 2016. A comparative study of using static and ultrasonic material testing methods to determine the anisotropic material properties of wood. *Construction and Building Materials* 102, 963–976. <https://doi.org/10.1016/j.conbuildmat.2015.07.195>

- Danihelová, A., Vidholdová, Z., Gergeľ, T., Spišiaková Kružlicová, L., Pástor, M., 2022. Thermal modification of spruce and maple wood for special wood products. *Polymers* 14, 2813. <https://doi.org/10.3390/polym14142813>
- Danihelová, A., Čulík, M., 2005. Influence of ageing process on sound quality of wood for stringed musical instrument making, in: *Proceedings of Forum Acusticum 2005*, Budapest, Hungary.
- Danihelová, A., Spišiak, D., Halachan, P., Gergeľ, T., Kružlicová, L., 2015. Physico-acoustical characteristics of usual and unusual wood species. *Akustika* 23, 22–27.
- Danihelová, A., Spišiak, D., Reinprecht, L., Gergeľ, T., Vidholdov, Z., Ondrejka, V., 2019. Acoustic properties of Norway spruce wood modified with staining fungus (*Sydowia polyspora*). *BioResources* 14, 3432–3444. <https://doi.org/10.15376/biores.14.2.3432-3444>
- Dinulică, F., Stanciu, M.D., Savin, A., 2021. Correlation between anatomical grading and acoustic–elastic properties of resonant spruce wood used for musical instruments. *Forests* 12, 1122. <https://doi.org/10.3390/f12081122>
- Dündar, T., Wang, X., As, N., Avci, E., 2016. Potential of ultrasonic pulse velocity for evaluating the dimensional stability of oak and chestnut wood. *Ultrasonics* 66, 86–90. <https://doi.org/10.1016/j.ultras.2015.11.007>
- Dünisch, O., 2019. Frequencies in vibrating wood: Does cell organization matter? *IAWA Journal* 40, 124–142. <https://doi.org/10.1163/22941932-40190239>
- El-Hadad, A., Brodie, G.I., Ahmed, B.S., 2018. The effect of wood condition on sound wave propagation. *Open Journal of Acoustics* 8, 37–51. <https://doi.org/10.4236/oja.2018.83004>
- Faktorová, D., Stanciu, M.D., Krbata, M., Savin, A., Kohutiar, M., Chlada, M., Năstac, S.M., 2024. Analysis of the anisotropy of sound propagation velocity in thin wooden plates using Lamb waves. *Polymers* 16, 753. <https://doi.org/10.3390/polym16060753>
- Fanxizi, L., Keqin, W., Chenhong, L., Fuwang, G., Jinhua, J., Yiping, Q., 2021. Mechanical and acoustic emission properties of vegetable fiber-reinforced epoxy composites for percussion instrument drums. *Polymer Composites* 42, 2864–2871. <https://doi.org/10.1002/pc.26020>
- Fedyukov, V.I., Saldaeva, E.Y., Chernova, M.S., Chernov, V.Y., 2019. Research into dendro-acoustic properties of introduced clones' wood as material for manufacturing musical instruments. *South-east European Forestry* 10 (2), 173–179. <https://doi.org/10.15177/see-for.19-18>
- Feeney, F.E., Chivers, R.C., Evertsen, J.A., Keating, J., 1998. The influence of inhomogeneity on the propagation of ultrasound in wood. *Ultrasonics* 36, 449–453. [https://doi.org/10.1016/S0041-624X\(97\)00066-8](https://doi.org/10.1016/S0041-624X(97)00066-8)
- Fletcher, N.H., Rossing, T.D., 1991. *The Physics of Musical Instruments*. Springer, New York. <https://doi.org/10.1007/978-1-4612-2980-3>
- Gerhard, S., Loidl, D., Stanzl-Tschegg, S., Maierhofer, J., Puchegger, S., 2020. Elastic constants of six wood species measured with the resonant beam technique. *Wood Research* 65, 347–352. <https://doi.org/10.37763/wr.1336-4561/65.2.347352>
- Gerhards, C.C., 1982. Effect of moisture content and temperature on the mechanical properties of wood: An analysis of immediate effects. *Wood and Fiber* 14, 4–36.
- Gil-Martín, I., Suarez, E., Roldán, A., 2025. Calibration strategy for a nondestructive wood characterization tool using optimized time-of-flight determination and material propagation analysis. *Measurement Science and Technology* 36. <https://doi.org/10.1088/1361-6501/adc75c>

- Gilani, M.S., Neuenschwander, J., Heeb, M., Furrer, R., Sanabria, S.J., Stoel, B.C., Schwarze, F.W.M.R., 2016. Influence of incubation time on the vibration and mechanical properties of mycowood. *Holzforschung* 70, 557–565. <https://doi.org/10.1515/hf-2015-0128>
- Gloth, G., Sinapius, M., 2004. Analysis of swept-sine runs during modal identification. *Mechanical Systems and Signal Processing* 18, 1421–1441. [https://doi.org/10.1016/S0888-3270\(03\)00087-6](https://doi.org/10.1016/S0888-3270(03)00087-6)
- Goken, J., Fayed, S., Schafer, H., Enzenauer, J., 2018. A study on the correlation between wood moisture and the damping behaviour of the tonewood spruce. *Acta Physica Polonica A* 133, 1241–1260. <https://doi.org/10.12693/APhysPolA.133.1241>
- Gonçalves, R., Garcia, G.H.L., Brazolin, S., Bertoldo, C., Ruy, M., 2019. Methodology for the characterization of elastic constants of wood from tree branches. *BioResources* 14, 8439–8454. <https://doi.org/10.15376/biores.14.4.8439-8454>
- González de Andr  es, E., Gazol, A., Colangelo, M., Valeriano, C., Cantero, A., Camarero, J.J., 2025. Recent declines in radial growth and wood density characterize dieback in European beech and pedunculate oak. *Dendrochronologia* 90, 126300. <https://doi.org/10.1016/j.dendro.2025.126300>
- Gonzalez, S., Chacra, E., Carre  o, C., Espinoza, C., 2022. Wooden mechanical metamaterials: Towards tunable wood plates. *Materials & Design* 221, 110952. <https://doi.org/10.1016/j.matdes.2022.110952>
- Grabianowski, M., Manley, B., Walker, J.C.F., 2006. Acoustic measurements on standing trees, logs and green lumber. *Wood Science and Technology* 40, 205–216. <https://doi.org/10.1007/s00226-005-0038-5>
- Guo, Z., Zhang, Y., Sheng, M., Liu, L., Li, Y., 2025. Vibration characteristics of a beam with elastic time-varying stiffness boundaries. *Applied Sciences* 15, 11365. <https://doi.org/10.3390/app152111365>
- Haines, D.W., Leban, J.M., Herb  , C., 1996. Determination of Young’s modulus for spruce, fir and isotropic materials by the resonance flexure method with comparisons to static flexure and other dynamic methods. *Wood Science and Technology* 30, 253–263. <https://doi.org/10.1007/BF00229348>
- Hambric, S.A., Sung, S.H., Nefske, D.J. (Eds.), 2016. Engineering vibroacoustic analysis: Methods and applications. *John Wiley & Sons*, Chichester. ISBN 978-1-119-95344-9
- Hamdan, S., Talib, Z.A., Rahman, M.R., Ahmed, A.S., Islam, M.S., 2010. Dynamic Young’s modulus measurement of treated and post-treated tropical wood polymer composites (WPC). *BioResources* 5, 324–342.
- Hasegawa, M., Takata, M., Matsumura, J., Oda, K., 2011. Effect of wood properties on within-tree variation in ultrasonic wave velocity in softwood. *Ultrasonics* 51, 296–302. <https://doi.org/10.1016/j.ultras.2010.10.001>
- Hassan, K.T.S., Hor   ek, P., Tippner, J., 2013. Evaluation of stiffness and strength of Scots pine wood using resonance frequency and ultrasonic techniques. *BioResources* 8, 1634–1645. <https://doi.org/10.15376/biores.8.2.1634-1645>
- Hassan Vand, M., Nop, P., Tippner, J., 2026. Assessment of the impact strength properties of thermally modified wood by non-destructive testing. *Journal of Nondestructive Evaluation* 45. <https://doi.org/10.1007/s10921-025-01313-7>

- Holz, D., 1996. Tropical hardwoods used in musical instruments: Can we substitute them by temperate zone species? *Holzforschung* 50, 121–129. <https://doi.org/10.1515/hfsg.1996.50.2.121>
- Ibáñez-Arnal, M., Doménech-Ballester, L., Sánchez-López, F., 2020. Design methodology and material evaluation criteria for membranophone shells. *Applied Acoustics* 169, 107449. <https://doi.org/10.1016/j.apacoust.2020.107449>
- Kabała, A., Barczewski, R., 2020. Shell-solid FEM model of a violin resonance body. *Vibrations in Physical Systems* 31, 1–8. <https://doi.org/10.21008/j.0860-6897.2020.3.08>
- Kaboorani, A., Blanchet, P., 2014. Determining the linear viscoelastic region of sugar maple wood by dynamic mechanical analysis. *BioResources* 9, 4392–4409. <https://doi.org/10.15376/biores.9.3.4392-4409>
- Kawrza, M., Furtmüller, T., Adam, C., 2022. Experimental and numerical modal analysis of a cross laminated timber floor system in different construction states. *Construction and Building Materials* 344, 128032. <https://doi.org/10.1016/j.conbuildmat.2022.128032>
- Kutnar, A., O'Dell, J., Hunt, C., Frihart, C., Kamke, F., Schwarzkopf, M., 2021. Viscoelastic properties of thermo-hydro-mechanically treated beech (*Fagus sylvatica* L.) determined using dynamic mechanical analysis. *European Journal of Wood and Wood Products* 79, 263–271. <https://doi.org/10.1007/s00107-020-01629-3>
- Kutnar, A., Widmann, R., Brémaud, I., 2013. Preliminary studies for use of dynamic mechanical analysis (DMA) to verify intensity of thermal wood modifications. *International Wood Products Journal* 4, 158–165. <https://doi.org/10.1179/2042645313Y.0000000044>
- Liu, M., Peng, L., Lyu, S., Lyu, J., 2020. Properties of common tropical hardwoods for fretboard of string instruments. *Journal of Wood Science* 66, 62. <https://doi.org/10.1186/s10086-020-01862-7>
- Longo, R., Laux, D., Pagano, S., Delaunay, T., Le Clézio, E., Arnould, O., 2018. Elastic characterization of wood by resonant ultrasound spectroscopy (RUS): A comprehensive study. *Wood Science and Technology* 52, 383–402. <https://doi.org/10.1007/s00226-017-0980-z>
- Magalas, L.B., Malinowski, T., 2003. Measurement techniques of the logarithmic decrement. *Solid State Phenomena* 89, 247–260. <https://doi.org/10.4028/www.scientific.net/ssp.89.247>
- Mania, P., Siuda, F., Roszyk, E., 2020. Effect of slope grain on mechanical properties of different wood species. *Materials* 13, 1503. <https://doi.org/10.3390/ma13071503>
- Matsunaga, M., Obataya, E., Minato, K., Nakatsubo, F., 2000. Working mechanism of adsorbed water on the vibrational properties of wood impregnated with extractives of pernambuco (*Guilandina echinata* Spreng.). *Journal of Wood Science* 46, 122–129. <https://doi.org/10.1007/BF00777358>
- McIntyre, M.E., Woodhouse, J., 1988. On measuring the elastic and damping constants of orthotropic sheet materials. *Acta Metallurgica* 36, 1397–1416. [https://doi.org/10.1016/0001-6160\(88\)90209-X](https://doi.org/10.1016/0001-6160(88)90209-X)
- Mechel, F.P., 2008. Elastic acoustics, in: Mechel, F.P. (Ed.), *Formulas of acoustics*, 2nd ed. Springer, Berlin, pp. 1133–1196.

- Merhar, M., 2024. Identification of damping of spruce wood (*Picea abies*) under various levels of moisture content using time-scale decomposition. *Polymers* 16, 1313. <https://doi.org/10.3390/polym16101313>
- Merhar, M., 2020. Determination of elastic properties of beech plywood by analytical, experimental and numerical methods. *Forests* 11, 1221. <https://doi.org/10.3390/f11111221>
- Merhar, M., Humar, M., 2020. The influence of wood modification on transfer function of a violin bridge. *Drvna Industrija* 71, 163–169. <https://doi.org/10.5552/drvid.2020.1966>
- Mevada, H., Patel, D., 2016. Experimental determination of structural damping of different materials. *Procedia Engineering* 144, 110–115. <https://doi.org/10.1016/j.proeng.2016.05.013>
- Milch, J., Tippner, J., Sebera, V., Brabec, M., 2016. Determination of the elasto-plastic material characteristics of Norway spruce and European beech wood by experimental and numerical analyses. *Holzforschung* 70, 1081–1092. <https://doi.org/10.1515/hf-2015-0267>
- Nocetti, M., Brunetti, M., Bacher, M., 2015. Effect of moisture content on the flexural properties and dynamic modulus of elasticity of dimension chestnut timber. *European Journal of Wood and Wood Products* 73, 51–60. <https://doi.org/10.1007/s00107-014-0861-1>
- Nocetti, M., Pröller, M., Brunetti, M., Dowse, G.P., Wessels, C.B., 2017. Investigating the potential of strength grading green *Eucalyptus grandis* lumber using multi-sensor technology. *BioResources* 12, 9273–9286. <https://doi.org/10.15376/biores.12.4.9273-9286>
- Noguchi, T., Obataya, E., Ando, K., 2012. Effects of aging on the vibrational properties of wood. *Journal of Cultural Heritage* 13, S21–S26. <https://doi.org/10.1016/j.culher.2012.02.008>
- Nop, P., Cristini, V., Zlámál, J., Hassan Vand, M., Šeda, V., Tippner, J., 2023. Dynamic properties of wood obtained by frequency resonance technique and dynamic mechanical analysis. *Wood and Fiber Science* 55, 131–142. <https://doi.org/10.22382/wfs-2023-12>
- Nop, P., Cristini, V., Zlámál, J., Vand, M.H., Šeda, V., Tippner, J., 2024. Determination of the static bending properties of green beech and oak wood by the frequency resonance technique. *Forests* 15, 150. <https://doi.org/10.3390/f15010150>
- Nop, P., Loula, K., Tippner, J., submitted. A multimodal vibro-acoustic evaluation of wood and wood-based materials for musical instrument applications. **Manuscript submitted for publication.**
- Nop, P., Tippner, J., 2022. Influence of dimensions of wooden samples for determination of acoustic parameters and sound timbre. *Applied Acoustics* 196, 108895. <https://doi.org/10.1016/j.apacoust.2022.108895>
- Obataya, E., Minato, K., Tomita, B., 2001. Influence of moisture content on the vibrational properties of hematoxylin-impregnated wood. *Journal of Wood Science* 47, 317–321. <https://doi.org/10.1007/BF00766720>
- Obataya, E., Ono, T., Norimoto, M., 2000. Vibrational properties of wood along the grain. *Journal of Materials Science* 35, 2993–3001. <https://doi.org/10.1023/A:1004782827844>
- Obataya, E., Zeniya, N., Endo-Ujiie, K., 2020. Effects of seasoning on the vibrational properties of wood for the soundboards of string instruments. *Journal of the Acoustical Society of America* 147, 998–1005. <https://doi.org/10.1121/10.0000723> <https://doi.org/10.1121/10.0000723>

- Olivar, J., Rathgeber, C., Bravo, F., 2015. Climate change, tree-ring width and wood density of pines in Mediterranean environments. *IAWA Journal* 36, 257–269. <https://doi.org/10.1163/22941932-20150098>
- Ono Sokki Co., Ltd., n.d. Damping materials and their performance measurement. Technical report. https://www.onosokki.co.jp/English/hp_e/whats_new/SV_rpt/damp/index.htm
- Ono, T., Norimoto, M., 1984. On physical criteria for the selection of wood for soundboards of musical instruments. *Rheologica Acta* 23, 652–656. <https://doi.org/10.1007/BF01438805>
- Ono, T., Norimoto, M., 1983. Study on Young's modulus and internal friction of wood in relation to the evaluation of wood for musical instruments. *Japanese Journal of Applied Physics* 22, 611–614. <https://doi.org/10.1143/JJAP.22.611>
- Ozyhar, T., Mohl, L., Hering, S., Hass, P., Zeindler, L., Ackermann, R., Niemz, P., 2016. Orthotropic hygric and mechanical material properties of oak wood. *Wood Material Science and Engineering* 11, 36–45. <https://doi.org/10.1080/17480272.2014.941930>
- Plath, N., Linke, S., Mores, R., 2022. On the angle-dependent vibrational behavior of fiber composite plates and its implications for musical instrument making. *Journal of the Acoustical Society of America* 151, 1956–1970. <https://doi.org/10.1121/10.0009799>
- Qi, S., Kaiyin, Z., 2024. Study on resonance of multi-degree-of-freedom structure based on modal orthogonal basis. *J. Low Freq. Noise Vib. Act. Control* 43, 755–764. <https://doi.org/10.1177/14613484231219575>
- Qi, S., Kaiyin, Z., 2024. Study on resonance of multi-degree-of-freedom structure based on modal orthogonal basis. *Journal of Low Frequency Noise, Vibration and Active Control* 43, 755–764. <https://doi.org/10.1177/14613484231219575>
- Quintavalla, M., Gabrielli, F., Canevari, C., 2022. Grading materials for stringed instruments soundboards: An approach considering the orthotropic elastic and damping properties. *Applied Acoustics* 187, 108521. <https://doi.org/10.1016/j.apacoust.2021.108521>
- Ray, T., Kaljun, J., Dolšak, B., 2021. Numerical model application to predict the sound quality of an instrument. *International Journal of Simulation Modelling* 20, 696–706. <https://doi.org/10.2507/IJSIMM20-4-580>
- Roohnia, M., Kohantorabi, M., Tajdini, A., 2015. Maple wood extraction for a better acoustical performance. *European Journal of Wood and Wood Products* 73, 139–142. <https://doi.org/10.1007/s00107-014-0871-z>
- Roohnia, M., Tajdini, A., Manouchehri, N., 2011. Assessing wood in sounding boards considering the ratio of acoustical anisotropy. *NDT & E International* 44, 13–20. <https://doi.org/10.1016/j.ndteint.2010.09.001>
- Ross, R.J. (ed.), 2010. Wood handbook: Wood as an engineering material. General Technical Report FPL–GTR–190. U.S. Department of Agriculture, Forest Service, Forest Products Laboratory, Madison, WI. <https://doi.org/10.2737/FPL-GTR-190>
- Ross, R.J., Pellerin, R.F., 1994. Nondestructive evaluation of wood, in: *Proceedings of the Materials Engineering Conference*, ASCE, pp. 1233–1241.
- Rossing, T.D., 2010. Plucked strings, in: *The Science of String Instruments*. Springer, pp. 63–79. https://doi.org/10.1007/978-1-4419-7110-4_2

- Rujinirun, C., Phinyocheep, P., Prachyabrued, W., Laemsak, N., 2005. Chemical treatment of wood for musical instruments. Part I: Acoustically important properties of wood for the Ranad (Thai traditional xylophone). *Wood Science and Technology* 39, 77–85. <https://doi.org/10.1007/s00226-004-0275-z>
- Saxena, M., Gupta, M.K., 2018. Hybrid mango/shorea robusta wood reinforced epoxy composite: Crystalline behavior and dynamic mechanical analysis. *Materials Today: Proceedings* 5, 19359–19366. <https://doi.org/10.1016/j.matpr.2018.06.295>
- Schwarze, F.W.M.R., Morris, H., 2020. Banishing the myths and dogmas surrounding the biotech Stradivarius. *Plants, People, Planet* 2, 237–243. <https://doi.org/10.1002/ppp3.10097>
- Schwarze, F.W.M.R., Spycher, M., Fink, S., 2008. Superior wood for violins: Wood decay fungi as a substitute for cold climate. *New Phytologist* 179, 1095–1104. <https://doi.org/10.1111/j.1469-8137.2008.02524.x>
- Sedik, Y., Hamdan, S., Jusoh, I., Hasan, M., 2010. Acoustic properties of selected tropical wood species. *Journal of Nondestructive Evaluation* 29, 38–42. <https://doi.org/10.1007/s10921-010-0063-7>
- Sproßmann, R., Zauer, M., Wagenführ, A., 2017. Characterization of acoustic and mechanical properties of common tropical woods used in classical guitars. *Results in Physics* 7, 1737–1742. <https://doi.org/10.1016/j.rinp.2017.05.006>
- Spycher, M., Schwarze, F.W.M.R., Steiger, R., 2008. Assessment of resonance wood quality by comparing its physical and histological properties. *Wood Science and Technology* 42, 325–342. <https://doi.org/10.1007/s00226-007-0170-5>
- Straže, A., Mitkovski, B., Tippner, J., Čufar, K., Gorišek, Ž., 2015. Structural and acoustic properties of African padouk (*Pterocarpus soyauxii*) wood for xylophones. *European Journal of Wood and Wood Products* 73, 235–243. <https://doi.org/10.1007/s00107-015-0878-0>
- Straže, A., Novak, K., 2022. Quality and price gain of European oak logs determined by visual and stress grading, in: *Proceedings of the 10th Hardwoods Conference*, Sopron, Hungary. <https://doi.org/10.35511/978-963-334-446-0>
- Tippner, J., Vojáčková, B., Zlámál, J., Kolařík, J., Paulic, V., Funai, J., 2023. The role of geometry precision in frequency-resonance method for non-destructive wood assessment: Numerical case study on sugar maple. *Wood Material Science and Engineering* 18, 792–800. <https://doi.org/10.1080/17480272.2022.2071166>
- Torresan, C., Hilmers, T., Avdagić, A., Di Giuseppe, E., Klopčič, M., Lévesque, M., Motte, F., Uhl, E., Zlatanov, T., Pretzsch, H., 2024. Changes in tree-ring wood density of European beech (*Fagus sylvatica* L.), silver fir (*Abies alba* Mill.), and Norway spruce (*Picea abies* (L.) H. Karst.) in European mountain forests between 1901 and 2016. *Annals of Forest Science* 81, 64. <https://doi.org/10.1186/s13595-024-01264-5>
- Traoré, B., Brancheriau, L., Perré, P., Stevanovic, T., Diouf, P., 2010. Acoustic quality of vène wood (*Pterocarpus erinaceus* Poir.) for xylophone instrument manufacture in Mali. *Annals of Forest Science* 67, 504. <https://doi.org/10.1051/forest/2010054>
- van Rensburg, N.F.J., van der Merwe, A.J., 2006. Natural frequencies and modes of a Timoshenko beam. *Wave Motion* 44, 58–69. <https://doi.org/10.1016/j.wavemoti.2006.06.008>
- Vasques, C.M., Dias Rodrigues, J., 2011. Vibration and Structural Acoustics Analysis. *Springer*, Dordrecht. <https://doi.org/10.1007/978-94-007-1703-9>, ISBN 978-94-007-1702-2

- Viala, R., Placet, V., Cogan, S., 2016. Virtual prototyping: A potential tool for wooden cultural heritage studies. *HAL open archive*. <https://hal.science/hal-03107836>
- Waltham, C., Yoshikawa, S., 2018. Construction of wooden musical instruments, in: Springer Handbook of Acoustics. *Springer*, pp. 63–79. https://doi.org/10.1007/978-3-662-55004-5_4
- Wang, S.Y., Wang, H.L., 1999. Effects of moisture content and specific gravity on static bending properties and hardness of six wood species. *Journal of Wood Science* 45, 127–133. <https://doi.org/10.1007/BF01192329>
- Wegst, U.G.K., 2008. Bamboo and wood in musical instruments. *Annual Review of Materials Research* 38, 323–349. <https://doi.org/10.1146/annurev.matsci.38.060407.132459>
- Wegst, U.G.K., 2006. Wood for sound. *American Journal of Botany* 93, 1439–1448. <https://doi.org/10.3732/ajb.93.10.1439>
- Yang, X., Amano, T., Ishimaru, Y., Iida, I., 2003. Application of modal analysis by transfer function to nondestructive testing of wood II: Modulus of elasticity evaluation of sections of differing quality in a wooden beam by the curvature of the flexural vibration wave. *Journal of Wood Science* 49, 140–144. <https://doi.org/10.1007/s100860300022>
- Yano, H., Kajita, H., Minato, K., 1994. Chemical treatment of wood for musical instruments. *Journal of the Acoustical Society of America* 96, 3380–3381. <https://doi.org/10.1121/1.410600>
- Yoshikawa, S., Waltham, C., 2014. Woods for wooden musical instruments, in: *Proceedings of the International Symposium on Musical Acoustics (ISMA 2014)*, Le Mans, France. <https://doi.org/10.13140/2.1.5067.1369>
- Zhou, X., Miyauchi, R., Inoue, Y., 2022. Effect of wood attributes on the price persistence of acoustic guitars. *Journal of Wood Science* 68, 54. <https://doi.org/10.1186/s10086-022-02054-1>

4 Hypotheses and Objectives

The use of the *ACE* parameter as an overall descriptor of vibro-acoustic properties generally provides a suitable indicator of material quality. However, due to the high natural variability of wood and the numerous factors influencing its properties, the results of existing methods and materials may not always be fully consistent. Moreover, the complex problem of vibrations is often simplified to a single vibration mode, thereby limiting the ability to exploit the full potential of the information contained in the entire vibration spectrum.

Beyond this state-of-the-art, the main objective of this work is to extend the current description of *ACE* by studying the factors that affect its reliable determination and to expand its meaning toward a more comprehensive and robust qualitative assessment of materials intended for use in the construction of musical instruments. The work focuses on the following hypotheses and objectives:

I. Effect of specimen dimensions

H1a: Specimen size affects the ACE determination.

H1b: Specimen geometry affects the ACE determination.

Objective: To carry out FRT on specimens with different dimensions, followed by analysis of the measured data and evaluation of any differences (Paper I).

II. Effect of material characteristics

H2a: The ACE parameter is significantly influenced by MC.

H2b: The ACE parameter is sensitive to changes caused by fungal degradation.

H2c: The ACE parameter detects changes caused by thermal modification.

H2d: The ACE parameter distinguishes solid wood species.

H2e: The ACE parameter distinguishes between solid wood and plywood.

Objective: To perform and evaluate FRT on specimens with varying *MC* (Paper III), on specimens after material modification (Papers II, IV), and on materials with different internal structures (Papers I, III, IV, V).

III. Effect of vibrational mode

H3a: *ACE parameters determined from bending and longitudinal vibration modes are not interchangeable.*

H3b: *ACE parameters determined from higher modes of the same mode shape are not interchangeable.*

Objective: To conduct multimodal FRT on specimens and assess the comparability of the obtained parameters across different mode shapes and mode orders (Papers I, V).

IV. Effect of the measurement technique

H4a: *The choice of sensor affects the evaluation of ACE in FRT.*

H4b: *Vibro-acoustic parameters (MOE_{dyn} , $\tan \delta$) obtained using different testing methods are not interchangeable.*

H4c: *Bending mechanical parameters obtained under different loading speed conditions correlate with each other.*

Objective: To determine vibro-acoustic parameters using different sensors within FRT (Papers I, V) and to evaluate bending mechanical properties (including vibro-acoustic parameters) using established methods under various loading rates (Papers II–IV), followed by an assessment of the consistency and correlations of the obtained results.

V. Characteristics of the vibrational spectrum

H5a: *The vibrational spectrum of a specimen exhibits a specific pattern.*

H5b: *Intervals between vibration modes, defined as frequency ratios, describe differences between specimens in terms of geometry and material properties.*

H5c: *ACE combining vibro-acoustic parameters (dynamic moduli, $\tan \delta$) determined from different mode shapes (bending, longitudinal, torsional) provides a better description of differences between materials than ACE determined from a single mode.*

Objective: To identify individual vibration modes in the spectrum, to analyse the relationship between intervals and specimen properties, and to integrate vibro-acoustic parameters derived from multiple modes into a single descriptor.

5 Materials and Methods

The presented work comprises several partial steps aimed at evaluating the applicability of FRT across different specimen dimensions, materials, and material states, as well as comparing its results with those of other current methods based either on predictions from numerical simulations or on different principles of static or dynamic testing. Each paper in this thesis provides a detailed description of the procedures applied.

The overall methodology of the thesis consists of the following tasks:

1. Specimen preparation (Paper I, II, III, IV, V)
2. Material modification (Paper II, IV)
3. Numerical modal analysis (Paper V)
4. Frequency resonance technique (Paper I, II, III, IV, V)
5. Three-point bending test under static loading (Paper II)
6. Three-point bending test under cyclic loading (Paper III)
7. Three-point bending test under impact loading (Paper IV)

5.1 Specimen Preparation

Specimens were prepared from solid wood and wood-based composites, representing commonly used European species, including both diffuse-porous and ring-porous hardwoods as well as coniferous species. The selection included some traditional materials used in musical instruments and alternative engineered materials, such as plywood. Across the individual studies (Papers I–V), the following species were investigated: Norway spruce (*Picea abies* L.) (Papers I, III–V), European beech (*Fagus sylvatica* L.) (Papers I–IV), ash (*Fraxinus angustifolia* Vahl.; *Fraxinus excelsior* L.) (Papers I, IV), oak (*Quercus robur* L.) (Papers III, IV), small-leaved linden (*Tilia cordata* Mill.) (Papers I, III), alder (*Alnus glutinosa* L.) (Paper I), larch (*Larix decidua* L.) (Paper IV), Norway maple (*Acer platanoides* L.) (Paper V), and wood-based composites including pine, birch, and poplar plywood (Paper V). Special orthotropic specimens were prepared with defined anatomical orientation corresponding to the principal material directions: L, R, T. The specimen dimensions are summarized from the largest to the smallest as follows: $2400 \times 160 \times 50$ mm, $2400 \times 50 \times 50$ mm, $960 \times 64 \times 20$ mm, and $750 \times 50 \times 50$ mm (Paper I); $300 \times 20 \times 20$ mm (Paper IV); $300 \times 20 \times 3.5$ mm (Paper V); $100 \times 7 \times 7$ mm (Paper II); and $61 \times 8.4 \times 1.7$ mm (Paper III). All specimens were visually assessed and selected to minimize the presence of defects, knots, and grain deviation. Using long-term water soaking followed by controlled air-drying to achieve *MC* above the FSP, and a climatic chamber (20 °C, 65% relative humidity) for *MC* below the FSP, the specimens were conditioned to the following levels: $\approx 80\%$ (Paper III), $\approx$

60% (Paper II), $\approx 50\%$ (Paper III), $\approx 12\%$ (Papers I, III, IV, V), $\approx 6\%$ (Paper IV), and $\approx 4\%$ (Paper IV). The *MC* was determined using the oven-dry method.

5.2 Material Modification

Beech specimens intended for the evaluation of degradation by the soft-rot fungus *Kretzschmaria deusta* (Paper II) were acclimatized to a moisture content of 60% ($\pm 10\%$) and subsequently sterilized. Kolle culture flasks (400 mL) were filled with 70 mL of sterile 5% malt extract agar and inoculated with *K. deusta*. After 2 weeks of incubation at 22 °C and 70% relative humidity, when the mycelium had completely covered the surface, the samples were inserted and placed on the radial face. Specimens were tested after 12 weeks of fungal exposure.

To evaluate the effect of thermal modification (Paper IV), boards of ash, oak, beech, larch, and spruce were selected. Boards intended for the preparation of thermally modified samples were divided into two groups: one group was exposed to a final temperature of 180 °C, and the other to 220 °C. The thermal treatment was carried out in a superheated steam environment at atmospheric pressure using a small-scale laboratory closed-system heat modification chamber (Katres s.r.o., Czech Republic). The boards were initially heated at 10 °C/h to 103 °C for approximately 3.5 h to achieve oven-dry conditions, followed by heating at 30 °C/h to the target temperatures. The target temperatures were maintained for 2 h, after which the boards were gradually cooled. Subsequently, specimens were prepared and tested.

To assess both modification processes, the mass loss was calculated as the difference between the mass before and after modification.

5.3 Numerical Modal Analysis

To predict the order of individual modal shapes and their corresponding frequencies of tested solid wood specimens, numerical simulations were performed using ANSYS Mechanical APDL (ANSYS Inc., Canonsburg, PA, USA). Numerical models of maple, spruce, and resonant spruce specimens were developed using material constants obtained from the literature. The 20-node solid element SOLID186, characterized by quadratic displacement behavior, was employed. Each model consisted of 240 elements and 3,267 nodes. The Block Lanczos solver was used for the modal analysis.

5.4 Frequency Resonance Technique

FRT was conducted under free–free boundary conditions, with supports positioned according to ASTM standards E1875 and E1876. To excite bending modes of vibration, elastic supports were placed at the nodal points of the first bending mode (22.4% and 77.6% of the specimen length), and the specimens were impacted at the corresponding antinodes

(either at the centre or at the ends) (Papers I–V). For torsional modes of vibration, the supports were positioned at the nodal points of the second torsional mode (25% and 75% of the specimen length), and excitation was applied at a corner of the specimen (Paper V). Longitudinal modes were excited by impacting the specimens' cross-sections, while the specimens were either suspended, as in the bending configuration, or supported along one face on soft foam pads (Papers I, V). Vibrations were recorded using different sensing techniques depending on the study: a uniaxial accelerometer (KS94B.10, Metra Mess- und Frequenztechnik in Radebeul e.K., Radebeul, Germany) (Paper I), a measurement microphone (TM-1, Audix Microphones Inc., Wilsonville, OR, USA) (Papers I, V), and a laser Doppler vibrometer (PDV-100, Polytec GmbH, Baden-Württemberg, Germany) (Papers I–V). The signals were recorded using a dynamic signal acquisition module (DEWE-41-TDSA, DEWETRON Inc., Grambach, Austria) (Papers I–IV), or an external sound card (UR22MKII, Steinberg Media Technologies GmbH, Hamburg, Germany) (Papers I, V). The sampling frequency was set to 20 kHz (Papers I–IV) and 44.1 kHz (Paper V). All signal analysis was performed in the MATLAB environment (The MathWorks, Inc., Natick, MA, USA).

After determining the resonant frequencies of the respective vibration modes using FFT, the dynamic modulus of elasticity in bending (MOE_{dyn}) was calculated according to Equation 10 (Papers I–V).

$$MOE_{dyn} = \left(\frac{2f_1}{2.25\pi} \right)^2 \frac{ml^3}{I} = \frac{4\pi^2 f_n^2 l^4 \rho}{k^4 i^2} \quad (10)$$

Where m is the specimen mass, I is the moment of inertia, k is a constant corresponding to the mode order ($k = 4.73$ for $n = 1$; 7.853 for $n = 2$; 10.996 for $n = 3$), and i is the radius of gyration.

To determine the dynamic shear modulus (G_{dyn}), from the torsional frequency, Equation 11 was used.

$$G_{dyn} = \frac{4lmf_n^2}{wt} P \quad (11)$$

w and t are the width and thickness of the specimen's cross-section, and P is a correction factor dependent on the width-to-thickness ratio (Paper V).

The dynamic Young's modulus (E_{dyn}) was calculated from the frequencies of longitudinal mode shapes using Equation 12 (Papers I, V).

$$E_{dyn} = 4\rho \left(\frac{f_n}{n} \right)^2 l^2 \quad (12)$$

Z was calculated in accordance with Equation 5 and K according to Equation 6 (Papers I, V). Damping was determined as LDD , i.e., the logarithm of the ratio of successive amplitudes (Papers I–IV), or as η , derived from the bandwidth at half power of the resonance peak and from the exponential decay of the resonance peak amplitude in time (Paper V). The conversion to $\tan \delta$ according to Equation 7 was used to determine ACE (Equation 8) (Papers I, IV, V), and $RACE$ (Equation 9) (Paper V).

To combine dynamic moduli and $\tan \delta$ obtained from different modes, and to express material anisotropy, the aggregated parameters $ACE_{L/T,n}$, $ACE_{B/T,n}$, were introduced (Equation 13).

$$ACE_{L/T,n} = \frac{\sqrt{\frac{E_{dyn,n}}{G_{dyn,n} \cdot \rho^3}}}{\overline{\tan \delta}} ; ACE_{B/T,n} = \frac{\sqrt{\frac{MOE_{dyn,n}}{G_{dyn,n} \cdot \rho^3}}}{\overline{\tan \delta}} \quad (13)$$

Due to the high variability of $\tan \delta$, its mean value over the set of modes considered in the calculation was used (Paper V). An overall descriptor integrating vibro-acoustic parameters derived from all investigated mode shapes and orders was defined as ACE_{combi} (Equation 14) (Paper V).

$$ACE_{combi} = \frac{1}{N} \sum_{n=1}^N (ACE_{L/T,n} + ACE_{B/T,n}) \quad (14)$$

Similarly, the parameters $RACE_{L/T,n}$, $RACE_{B/T,n}$, and $RACE_{combi}$ were defined in Equations 15 and 16 (Paper V).

$$RACE_{L/T,n} = \frac{\sqrt{\frac{E_{dyn,n}}{G_{dyn,n} \cdot \rho}}}{\overline{\tan \delta}} ; RACE_{B/T,n} = \frac{\sqrt{\frac{MOE_{dyn,n}}{G_{dyn,n} \cdot \rho}}}{\overline{\tan \delta}} \quad (15)$$

$$RACE_{combi} = \frac{1}{N} \sum_{n=1}^N (RACE_{L/T,n} + RACE_{B/T,n}) \quad (16)$$

5.5 Three-point Bending Test Under Static Loading

Static three-point bending tests were performed using a universal testing machine (Zwick Z050, 50 kN, Zwick GmbH, Ulm, Germany). The experimental design followed standards for small clear wood specimens (BS 373; ČSN 49 0116). The specimens were loaded at their centre in the T direction with a span-to-thickness ratio of 12. The bending stress (σ_B) was calculated using the Equation 17.

$$\sigma_B = \frac{3Fl_0}{2wt^2} \quad (17)$$

Where F is applied force, l_0 is distance between supports. The bending modulus of rupture (MOR) was determined as the σ_B corresponding to the maximum applied F .

5.6 Three-point Bending Test Under Oscillatory Loading

Three-point bending tests under cyclic loading were conducted using a DMA device (RSA-G2 with TRIOS software, TA Instruments, New Castle, DE, USA). The tests were performed with an initial and minimum axial force of 0.1 N (set 30% higher than the dynamic force) and a displacement of 30 mm. The oscillation frequency was applied as a logarithmic sweep from 0.1 to 100 Hz, with 10 points per decade. Each test lasted approximately 10 min per specimen. MOE_{dyn} was calculated as a complex combination of the measured E' and E'' , in accordance with Equation 18. $Tan \delta$ was determined as the ratio of E'' to E' .

$$MOE_{dyn} = E' + iE'' = \sqrt{E'^2 + E''^2} \quad (18)$$

5.7 Three-point Bending Test Under Impact Loading

Three-point impact bending tests were performed using an impact testing machine (DPFest 400, Labortech s.r.o., Czechia) in accordance with ČSN 49 0115 and ČSN 49 0117. The specimens with a grey-scaled random speckled pattern (enhancing contrast of surface points) were impacted at their centre by a hammer ($m = 4.55$ kg) released from a height of 815.7 mm, corresponding to an impact velocity of 4 m/s and energy of 36.4 J. The testing machine recorded the reaction F and impact bending strength (IBS). The deformation process was monitored using a high-speed camera system (FASTCAM SA-X2 1000K-M2, Photron Cameras, Japan; Micro-Nikkor G, Nikon Corp., Japan). The recorded images were processed using digital image correlation (DIC) software (Vic-2D v. 2010, Correlated Solutions Inc., USA), and the ε tensor was computed using Lagrangian notation. The IBS was calculated as the ratio of the energy required to break the specimen to its cross-sectional area.

6 List of publications

- Paper I** Nop, P., Tippner, J., 2022. Influence of dimensions of wooden samples for determination of acoustic parameters and sound timbre. *Applied Acoustics* 196, 108895. <https://doi.org/10.1016/j.apacoust.2022.108895>
- Paper II** Nop, P., Cristini, V., 2022. Changes in vibro-acoustic properties of green beech wood due to *Kretzschmaria deusta*. In: *Proceedings of the 6th International Conference on Structural Health Assessment of Timber Structures (SHATIS 2022)*, Prague, Czech Republic, 7–9 September 2022. https://utam-api.ssc.avcr.cz/uploads/PROCEEDINGS_SHATIS_22_db5ee1c7e6.pdf
- Paper III** Nop, P., Cristini, V., Zlámal, J., Vand, M.H., Šeda, V., Tippner, J., 2023. Dynamic properties of wood obtained by frequency resonance technique and dynamic mechanical analysis. *Wood and Fiber Science* 55(2), 131–142. <https://doi.org/10.22382/wfs-2023-12>
- Paper IV** Vand, M.H., Nop, P., Tippner, J., 2025. Assessment of the Impact Strength Properties of Thermally Modified Wood by Non-Destructive Testing. *Journal of Nondestructive Evaluation* 45, 17. <https://doi.org/10.1007/s10921-025-01313-7>
- Paper V** Nop, P., Loula, K., Tippner, J., 2026. A multimodal vibro-acoustic evaluation of wood and wood-based materials for musical instrument applications. *Journal of Sound and Vibration*. **Submitted**.

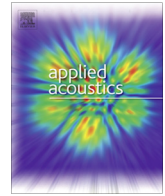
Paper I
**Influence of dimensions of wooden samples for determination of
acoustic parameters and sound timbre**

Patrik Nop, Jan Tippner

Applied Acoustics

<https://doi.org/10.1016/j.apacoust.2022.108895>

Nop, P. (corresponding author): Conceptualization, Methodology, Resources, Investigation, Formal analysis, Writing – original draft



Influence of dimensions of wooden samples for determination of acoustic parameters and sound timbre

Patrik Nop*, Jan Tippner

Mendel University in Brno/Faculty of Forestry and Wood Technology, Czech Republic

ARTICLE INFO

Article history:

Received 21 December 2021

Received in revised form 4 May 2022

Accepted 17 June 2022

Available online 25 June 2022

Keywords:

Frequency resonant technique

Vibro-acoustic parameters

Natural frequency

Mode shape

One-axis vibration measurement

Damping coefficient of overtones

ABSTRACT

Wood is a popular material used in almost all areas of acoustics. The main reason for its vibro-acoustic properties is associated with orthotropic visco-elasticity, describing the ability to conduct, amplify or damp the sound vibrations in different applications. With modern technologies, it is possible to upgrade non-destructive material analysis to more detailed level. This study is dedicated to assess the vibro-acoustic properties of wood using frequency response to an impact excitation. Using one-axis sensors, the parameters such as dynamic modulus of elasticity, radiation coefficient, acoustic impedance, damping coefficient, and acoustic conversion efficiency were determined depending on the anatomical direction of wood. The dependence of sample dimensions and scale of measured parameters was evaluated. Broader description of the damping includes mode shape, its rank, direction and frequency. Comparison of vibro-acoustic behavior across important European species was performed. Results show samples with higher slenderness have smaller differences between MOED calculated from the first bending or first longitudinal mode shape. Analysis of the damping coefficient shows, mode shape, its rank and sample size have greater effect on material damping than frequency of vibrations.

© 2022 Elsevier Ltd. All rights reserved.

1. Introduction

The description of a three-dimensional oscillating body is expressed by general equation of motion in the form (1).

$$[M]\{\ddot{x}(t)\} + [D]\{\dot{x}(t)\} + [K]\{x(t)\} = \{F\} \quad (1)$$

Where $[M]$ represents the mass matrix – expressed by density ρ , $[D]$ is the damping matrix, $[K]$ is the stiffness matrix, and $x(t)$, $\dot{x}(t)$, $\ddot{x}(t)$ are time-dependent vectors of acceleration respectively, velocity, and displacement, and F is the time-dependent force vector [25] of the stress wave propagation. For the full description of the elasticity of orthotropic material a set of samples with the appropriate anatomical orientation must be made for determinate of main 9 parameters ($3 \times$ moduli of elasticity, $3 \times$ shear moduli, $3 \times$ Poisson's ratios) in three directions (T – tangential, R – radial, L – longitudinal) [12], and damping matrix as well. For the basic evaluation of vibro-acoustic behavior of wood, dynamic modulus of elasticity (MOED) is most often used [15]. The speed of sound transmission c through the material is calculated, using MOED and density [26]. Advanced acoustic description includes the radiation coefficient K [20], impedance Z [1], and the damping coefficient

$\tan \delta$ [8]. Summarizing description provides a ratio of radiation and damping called Acoustic Conversion Efficiency (ACE) [22]. In addition to measuring only the velocity of stress-wave propagation, also known as the time of flight (TOF) [14], the frequency resonance test (FRT) is also very often used for its high precision, and bigger amount of acoustic parameters obtained [15]. FRT is a nondestructive method to determine the vibro-acoustic behavior, on measuring resonance frequencies of bending, longitudinal, and torsional modes of vibrations [9]. Most acoustic parameters defining the material are in many cases calculated only from the resonant frequency of the first horizontal transverse (bending) mode shape [1,6]. Calculating the speed of sound transmission through the material from the dynamic modulus of elasticity, determined from the bending mode, may not be the correct way for an anisotropic material. Only the first longitudinal mode shape is used [17], to obtain the speed of sound transmission using FRT and its comparability with the TOF method [3]. To use one material constant in all directions of anisotropy it is necessary to recalculate it using the orthotropic descriptors [18]. This recalculation must be related to the change in orientation and geometry of sample as well. The process of producing differently oriented, same-size samples may not always be advantageous or possible. For clear definition, it is necessary to separate the modulus of elasticity obtained by the bending mode, and obtained by the

* Corresponding author.

E-mail address: xnop@mendelu.cz (P. Nop).

Table 1
Mean density values for species with the coefficient of variation (c.v.).

Species	ρ [kg · m ⁻³] (c.v. [%])
Alder (<i>Alnus glutinosa</i> L.)	546 (2.88)
Ash (<i>Fraxinus angustifolia</i> Vahl.)	714 (4.47)
Beech (<i>Fagus sylvatica</i> L.)	710 (1.51)
Linden (<i>Tilia cordata</i> Mill.)	505 (9.68)
Spruce (<i>Picea abies</i> L.)	525 (9.87)

longitudinal mode of vibrations [15]. The damping coefficient derived from these vibration modes must be distinguished as well because it's also dependent on the direction of orthotropy [9]. In addition damping's dependence on density and MOED (this relationship can be generalized to specific species) [8], it is again necessary to take into account the anatomical direction [7], and mode shape of vibrations [16]. It is known that vibration damping increases exponentially with increasing resonant frequency [23],

but the change also occurs with more complex shapes of higher modes [10]. The complexity of resonant mode shapes and their order is determined by the geometry of tested sample [5,11]. Therefore, the geometry of the sample must be also taken into account to determine the vibro-acoustic properties. Some studies show strong coefficient of determination ($r^2 = 0.94$) between MOED of long logs and short logs [19] because of simple and robust calculation and clearly defined direction of vibration. In comparison, bending is a complex vibration. The ASTM E1876-15 test standard (for MOED calculation from the bending mode, using FRT) states the ratio of a cross-sectional dimension (slenderness) greater than 5 [2]. However, this standard is for isotropic and homogeneous material.

Verification of the comparability of parameters determined from the longitudinal and bending modes of vibration on samples with different sizes is required for wood. To be able to predict the vibro-acoustic behavior of wood in broad-spectrum, it is also necessary to analyze damping of higher modes of vibration.

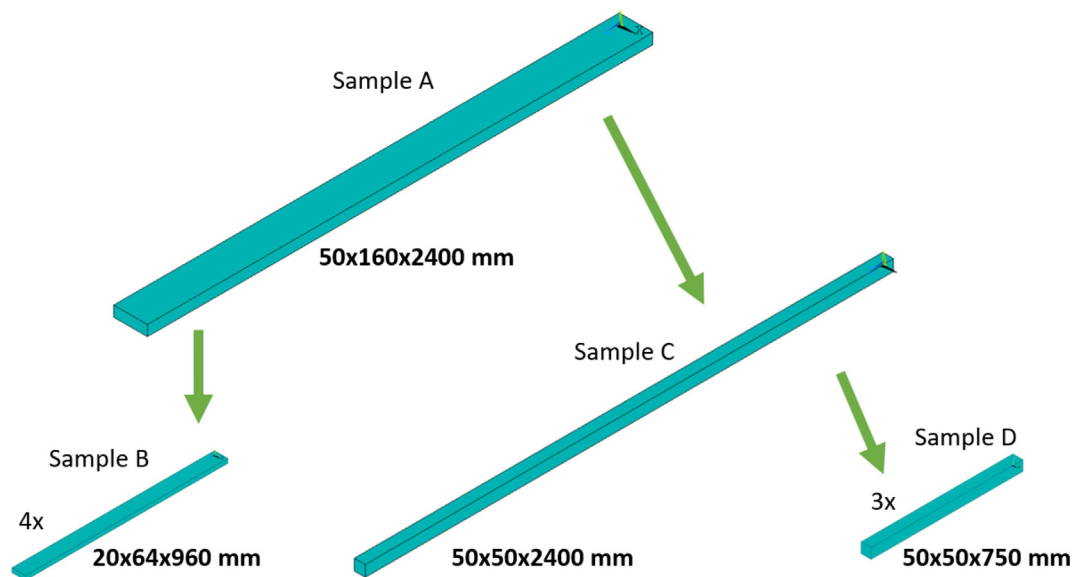


Fig. 1. Group of samples for the multi-scale measurement.

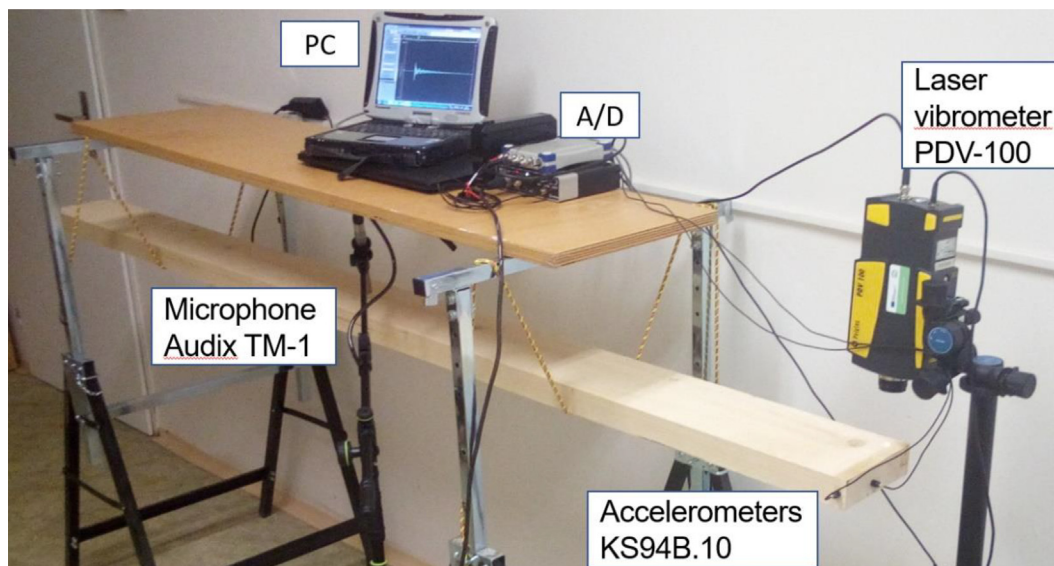


Fig. 2. Experimental setup for one-axis FRT measurement of sample A.

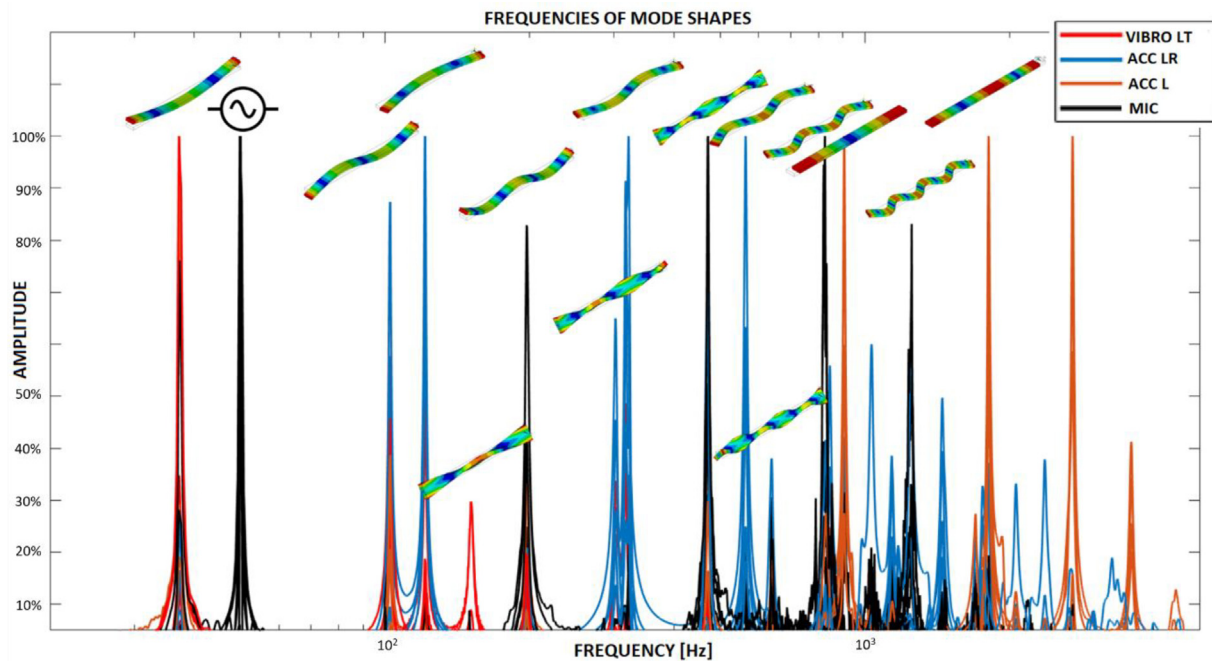


Fig. 3. Resonant frequencies of mode shapes and electrical hum (50 Hz) affecting the microphone measurement – the amplitudes are relative to the ratio comparison of the peaks detected by all sensors after excitation at seven antinodes.

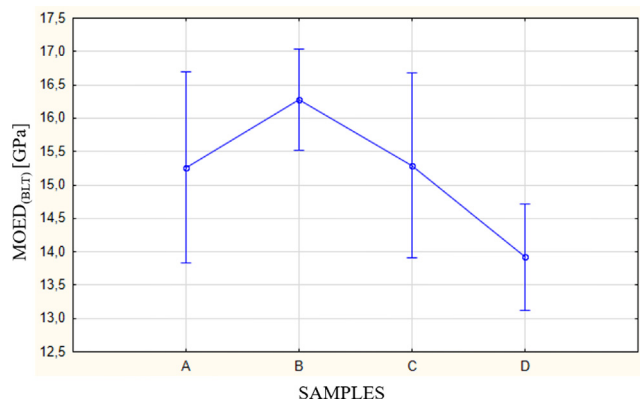


Fig. 4. Comparison of mean ranks of example of $MOED_{BLT}$ and sample sizes across species. Vertical bars indicate 0.95 confidence intervals.

2. Material and methods

This study aims to compare vibro-acoustic parameters obtained from first main modes of vibration of wood. Also the influence of the shape and the size of measured samples on frequency response and dynamic parameters, obtained by FRT. In case of material damping, the influence was extended not only to the frequency, direction, and shape of the mode but also to its rank (in sequential order). A difference was sought for the spectra of natural frequencies (sound timbre) European wood species used for the construction of musical instruments (or their parts).

2.1. Material

Wood beams ($50 \times 160 \times 2400$ mm) were cut from Norway spruce (*Picea abies* L.), European beech (*Fagus sylvatica* L.), Narrow-leaved ash (*Fraxinus angustifolia* Vahl.), European alder (*Alnus glutinosa* L.), and Small-leaved linden (*Tilia cordata* Mill.). Beams were crafted from dried wooden boards which were all of

A quality (the highest quality class) However, due to the size, several knots were present in linden wood. Three samples there were prepared from each species (all with moisture content 12 %). These beams (samples A) were used for big scale measurement. Then three sets of smaller or thinner samples were cut from samples A. Samples B ($20 \times 64 \times 960$ mm), samples C ($50 \times 50 \times 2400$ mm), and samples D ($50 \times 50 \times 750$ mm). Samples B had the same ratio of dimensions as samples A; samples C and D had the same moment of inertia of cross-section. Every sample group of each species had the same dimensional orientation (T: R: L). The slenderness (length/thickness) of samples A and B was 48 in tangential load, and 15 in radial load. The slenderness of samples C was 48, and for samples D was 15. Fig. 1.

2.2. Measurement

Samples of all dimensions were hinged in flexible supports at the location of nodes of the first transversal (bending) mode (Fig. 2). Vibrations were measured in three axes using Doppler's laser vibrometer (PDV-100, Polytec, Inc.) for the bending longitudinal-tangential (BLT) mode, and two accelerometers (KS94B.10, MMF, e.K.), one for the bending longitudinal-radial (BLR) mode, and second for the longitudinal (L) mode in the longitudinal direction. Samples B and D were affected by accelerometers weight and therefore the methodology was modified to contactless, using only the vibrometer for all three axes. Measuring points were chosen at predicted antinodes of transversal and longitudinal modes (ASTM E1876-15). Also for the complex measurement of all resonant frequencies the Audix TM-1 microphone was used. Vibrations were excited with a hammer in the middle, third, and at the end of the sample in T and R load directions. In L direction, the oscillation was excited in the centre of the cross-section. During the sample oscillation, torsion mode shapes were also detected by sensors with a measuring spot in the corner at the end of the sample. They manifested as peaks with similar amplitude for both corner sensors. By change of excitation points to antinodes of higher bending mode shapes (ASTM E1876-15), torsion mode

Table 2Identification of statistically significant differences between groups of samples (evaluated by Kruskal-Wallis test with $\alpha = 0.05$, the significant difference is represented by $\times$).

Species	$Z_{(BLT)}$	$K_{(BLT)}$	$K_{(L)}$	$ACE_{(BLT)}$	$ACE_{(BLR)}$	$ACE_{(L)}$
Alder	$A \times D$	$B \times D$	$A \times B$	$(A, B) \times (C, D)$		$B \times D$
Ash						$B \times D$
Beech	$B \times D$			$(A, B) \times D$	$A \times B$	
Linden	$B \times D$				$A \times D$	
Spruce				$(A, B) \times (C, D)$	$B \times (C, D)$	$(A, C) \times D$
Species	$MOED_{(BLT)}$	$MOED_{(BLR)}$	$\tan \delta_{(BLT)}$	$\tan \delta_{(BLR)}$	$\tan \delta_{(L)}$	
Alder	$(A, B) \times D$		$(A, B) \times C$		$B \times D$	
Ash	$B \times D$		$B \times D$			
Beech	$B \times D$		$(A, B) \times D$	$A \times B$		
Linden	$B \times (A, D)$	$A \times B$			$A \times D$	
Spruce			$(A, B) \times C \times D$	$B \times (C, D)$		

Table 3Medians of MOED, and $\tan \delta$ for first bending and longitudinal mode shapes, across samples and species.

Sample	$MOED_{(BLT)}$ [GPa] (c.v. [%])	$MOED_{(BLR)}$ [GPa] (c.v. [%])	$MOED_{(L)}$ [GPa] (c.v. [%])	$\tan \delta_{(BLT)}$ [-] (c.v. [%])	$\tan \delta_{(BLR)}$ [-] (c.v. [%])	$\tan \delta_{(L)}$ [-] (c.v. [%])
Alder A	17.148 (0.7)	13.980 (2.9)	16.041 (2.5)	0.00509 (6.7)	0.00621 (59.7)	0.00668 (16.7)
Alder B	15.143 (6.7)	14.288 (6.2)	15.400 (6.4)	0.00541 (9.7)	0.00637 (13.1)	0.00668 (10.9)
Alder C	14.882 (3.7)	14.493 (5.4)	14.990 (6.0)	0.00939 (27.2)	0.00955 (15.7)	0.00796 (17.1)
Alder D	13.719 (6.5)	13.968 (7.2)	15.225 (7.1)	0.00672 (39.2)	0.00607 (56.8)	0.00859 (9.6)
Ash A	16.698 (2.5)	15.027 (9.1)	17.710 (1.6)	0.00732 (4.2)	0.00796 (4.2)	0.00764 (20.6)
Ash B	18.830 (7.0)	16.768 (5.4)	19.357 (7.2)	0.00637 (6.0)	0.00993 (18.3)	0.00764 (7.1)
Ash C	17.626 (12.8)	17.814 (6.3)	18.280 (7.5)	0.00764 (11.7)	0.00732 (35.9)	0.00828 (50.0)
Ash D	16.382 (12.7)	16.854 (6.7)	18.379 (7.4)	0.00794 (20.9)	0.00771 (20.9)	0.00828 (11.2)
Beech A	12.126 (4.6)	12.381 (2.0)	13.216 (2.8)	0.00923 (4.3)	0.00987 (1.5)	0.00955 (19.9)
Beech B	13.284 (3.3)	12.661 (3.7)	12.046 (13.6)	0.00987 (7.4)	0.01241 (10.5)	0.01480 (21.5)
Beech C	11.883 (7.0)	12.453 (6.2)	12.702 (6.0)	0.01146 (4.0)	0.00987 (21.6)	0.01019 (7.6)
Beech D	11.404 (8.0)	12.574 (6.2)	13.681 (7.4)	0.01167 (7.4)	0.01107 (10.9)	0.01369 (11.1)
Linden A	13.014 (11.0)	10.154 (13.8)	14.428 (11.9)	0.00653 (8.9)	0.00844 (23.4)	0.01337 (29.7)
Linden B	15.492 (12.6)	14.147 (15.3)	16.501 (12.2)	0.00589 (12.1)	0.00803 (39.7)	0.00955 (53.6)
Linden C	14.346 (15.0)	14.832 (13.7)	15.498 (11.6)	0.00668 (17.3)	0.00732 (9.6)	0.01146 (18.1)
Linden D	12.507 (14.9)	12.531 (13.8)	15.093 (13.9)	0.00650 (17.1)	0.00648 (11.8)	0.00891 (15.9)
Spruce A	17.985 (16.9)	13.187 (14.5)	18.111 (16.3)	0.00605 (4.3)	0.00668 (12.3)	0.00637 (10.4)
Spruce B	17.828 (28.1)	14.401 (16.7)	16.330 (11.4)	0.00573 (13.8)	0.00605 (6.7)	0.01019 (53.0)
Spruce C	20.641 (29.0)	19.647 (27.9)	20.410 (24.3)	0.02196 (5.8)	0.01496 (53.9)	0.00668 (14.1)
Spruce D	18.208 (22.6)	17.428 (24.3)	20.975 (25.3)	0.01066 (17.1)	0.01178 (19.9)	0.01162 (24.0)

Table 4

Intervals between fundamental and frequencies of resonant modes for sample A.

Mode	1.	2.	3.	4.	5.	6.
BLT	1	2.72	5.20	8.42	12.22	16.36
BLR	2.97	7.39	13.00	19.16	25.15	–
L	23.46	46.94	70.96	92.85	–	–

shapes were identified by the decreasing amplitude of the peak for both sensors. Identified torsion mode shapes were excluded from the next analysis. Seven measurements with different excitation

points were performed on each sample using four sensors simultaneously, with sampling frequency of 20 kHz. Total of 150 samples were measured.

2.3. Data analysis

Measured signal was analysed in MATLAB (R2020, Humusoft, ltd.) using Fast Fourier Transformation (FFT) to find resonant frequencies (represented by peaks). Due to the one-axis sensors with separate signals, all acquired frequencies were assigned to individual mode shapes (Fig. 3). Measuring samples with square

Table 5

Intervals between fundamental and frequencies of resonant modes for sample B.

Mode	1.	2.	3.	4.	5.
BLT	1	2.68	5.20	8.40	12.17
BLR	3.03	7.56	12.91	–	–
L	23.52	46.95	–	–	–

cross-section (C, D) causes a problem to analyse individual bending mode shapes. In several cases, the peaks of resonant frequencies approached until they merged into one peak or changed the order. This can cause incorrect frequency assignment to appropriate mode shape. Based on this knowledge, the use of one-axis measurement is required. One-axis sensor detection is influenced by oscillations in other axes, but due to relative amplitude heights, the direction can be assessed correctly Fig. 4.

From the fundamental frequency of each mode shape (longitudinal and both transversals), a parameter for the appropriate anatomical direction was determined. With the known density ρ , the $MOED_{(L)}$ is derived from the fundamental frequency of the longitudinal vibrations using equation (2).

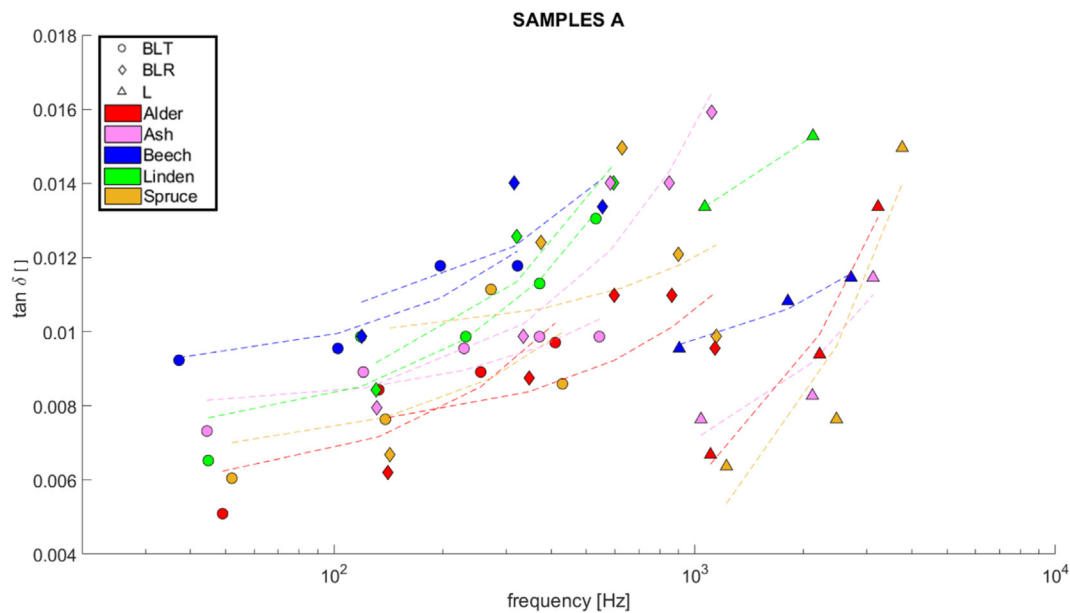
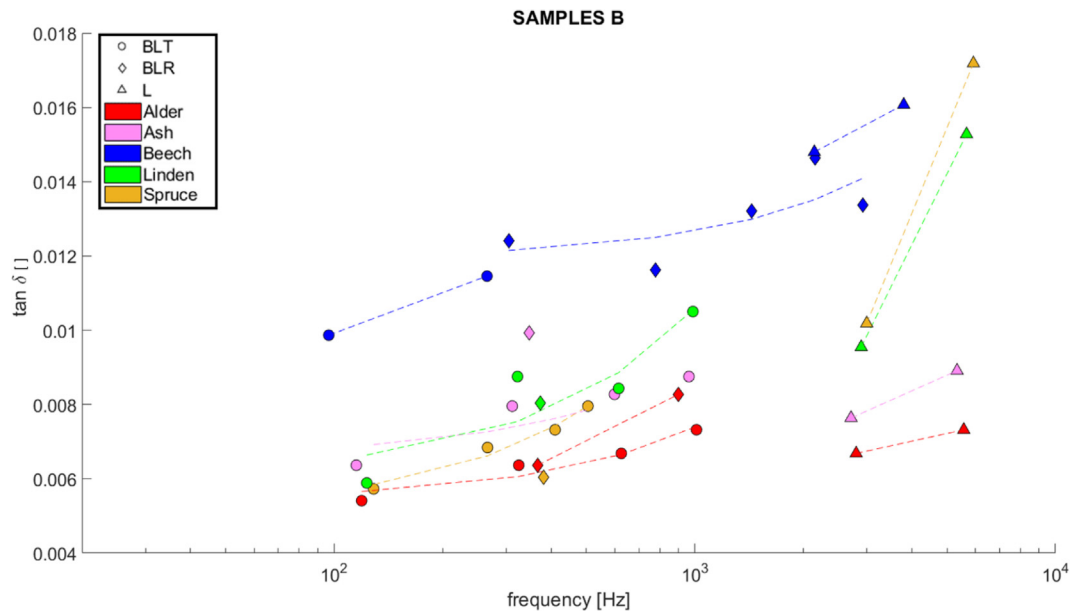
$$MOED = 4\rho \cdot f^2 \cdot l^2 \quad (2)$$

where f = frequency of the first longitudinal mode shape, and l = sample length [9].

$MOED_{(BLR, BLT)}$ is determined from the fundamental frequencies of of transverse mode shapes by equation (3).

$$MOED = \left(\frac{2f}{2.25\pi} \right)^2 \cdot \frac{m \cdot l^2}{I} \quad (3)$$

where m = sample mass, and I = moment of inertia [10].

**Fig. 5.** Influence of the mode shape, rank, frequency, and species to the damping factor.**Fig. 6.** Influence of the mode shape, rank, frequency, and species to the damping factor.

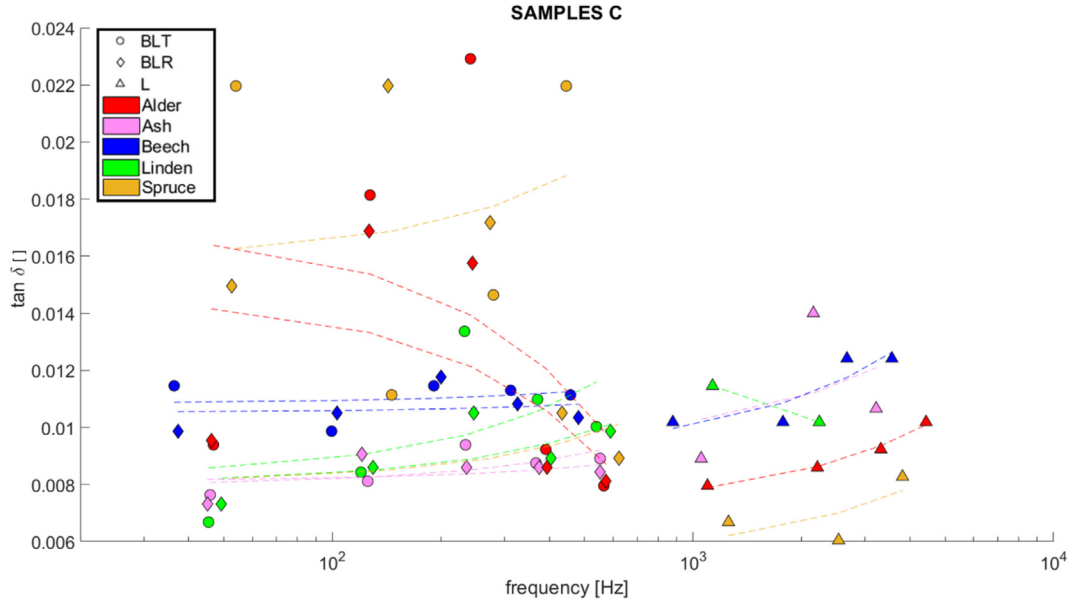


Fig. 7. Influence of the mode shape, rank, frequency, and species to the damping factor.

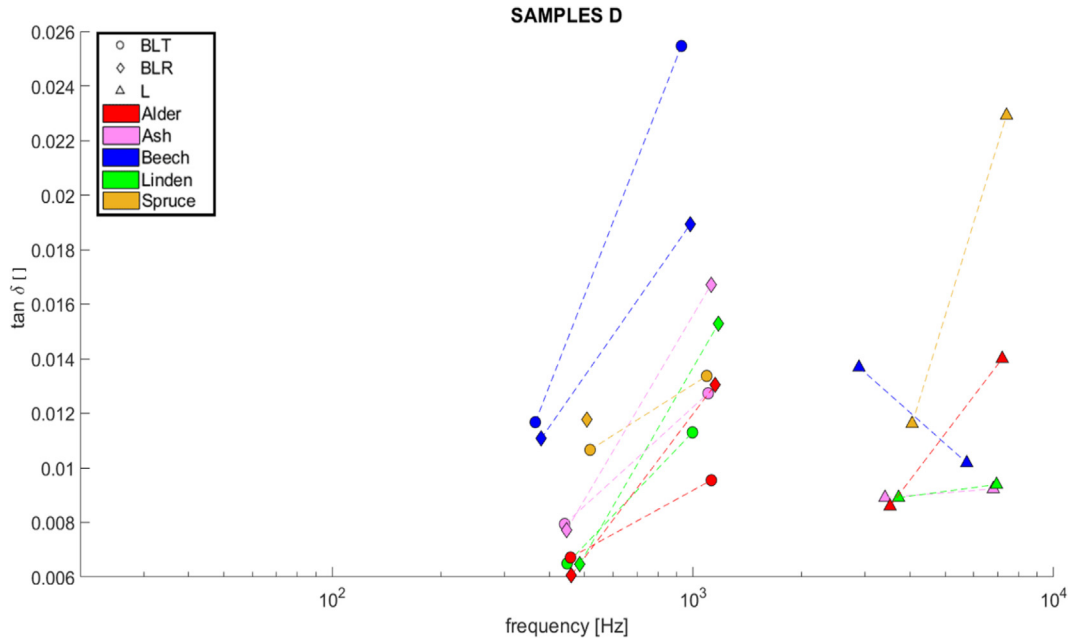


Fig. 8. Influence of the mode shape, rank, frequency, and species to the damping factor.

Formulas for determining the radiation coefficient K and impedance Z using a density ρ and speed of sound transmission c or MOED [26].

$$K = \frac{c}{\rho} = \sqrt{\frac{MOED}{\rho^3}} \quad (4)$$

$$Z = c \cdot \rho = \sqrt{MOED \cdot \rho} \quad (5)$$

The damping coefficient $\tan \delta$ is calculated using the Logarithmic Decrement of Damping (LDD) [4].

$$\tan \delta = \frac{LDD}{\pi} \quad (6)$$

LDD characterizes the decrease in amplitude of damped oscillations per period as the natural logarithm of ratio of successive amplitudes [21].

$$LDD = \ln \frac{a_n}{a_{n+1}} \quad (7)$$

To increase accuracy, regression from at least 10 consecutive periods was used for calculation.

ACE expresses the ratio of radiation and damping [24].

$$ACE = \frac{K}{\tan \delta} \quad (8)$$

Measured and calculated data were analysed with STATISTICA (Statistica Ltd.) and MATLAB (R2020, Humusoft, Ltd.) by one-way analysis of variance (ANOVA) for normal, and nonparametric Kruskal-Wallis test for non-normal data distribution was applied with using the multiple comparison Dunn-Sidak test of mean ranks, with a significance level $\alpha = 0.05$.

3. Results and discussion

Was defined the density ρ Table 1, dynamic modulus $MOED_{(BLT, BLR, L)}$, radiation coefficient $K_{(BLT, BLR, L)}$, impedance $Z_{(BLT, BLR, L)}$, damping coefficient $\tan \delta_{(BLT, BLR, L)}$, and $ACE_{(BLT, BLR, L)}$, for each sample.

Comparability of calculated parameters for samples with different dimensions was determined for each wood species. Significant differences are illustrated in Table 2. Differences can be found in some cases for individual species. The example of more frequent difference of large sample A from other samples for linden wood

shows the issue, that number of defects associated with the certain species in the big scale measuring, is a factor influencing the measurement, and negatively affects the comparison of some vibroacoustic parameters of different shapes and scales.

According to the calculation formulas of Impedance Z , radiation coefficient K , and ACE , a strong correlation dependence was confirmed to $MOED$ and damping coefficient $\tan \delta$. Table 3 shows values of these main parameters, in connection to the measurements of the different sample shapes and the first resonant modes across species. Values of $MOED$ were determined to be around 15 GPa for alder, 17 GPa for ash, 12.5 GPa for beech, 13.5 GPa for linden, and 18 GPa for spruce, but with differences between BLT and L, and between BLR and L, which are related to their compliance to the load direction, ie slenderness.

Samples with slenderness of 48 in the direction of loading showed differences in determining $MOED$ from bending and longitudinal modes between 0.7 % and 10 %. The smallest differences occurred for samples C (0.7 % – 7.4%). For samples with slenderness of 15, the differences were between 5 % and 30 %. The most significant influence of sample dimensions across species was found in $MOED_{(BLT)}$.

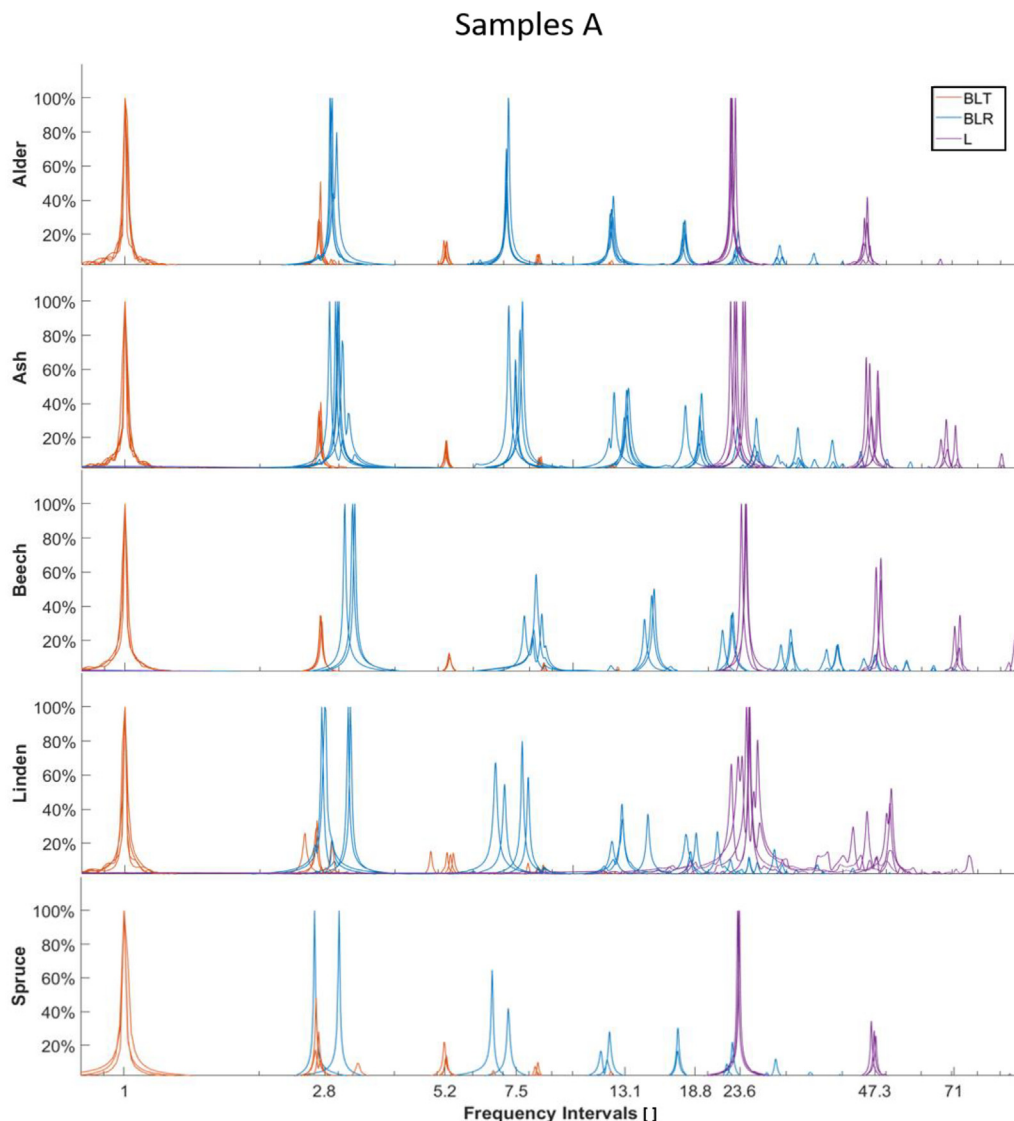


Fig. 9. Intervals of resonant frequencies – comparison of species – samples A Table 4.

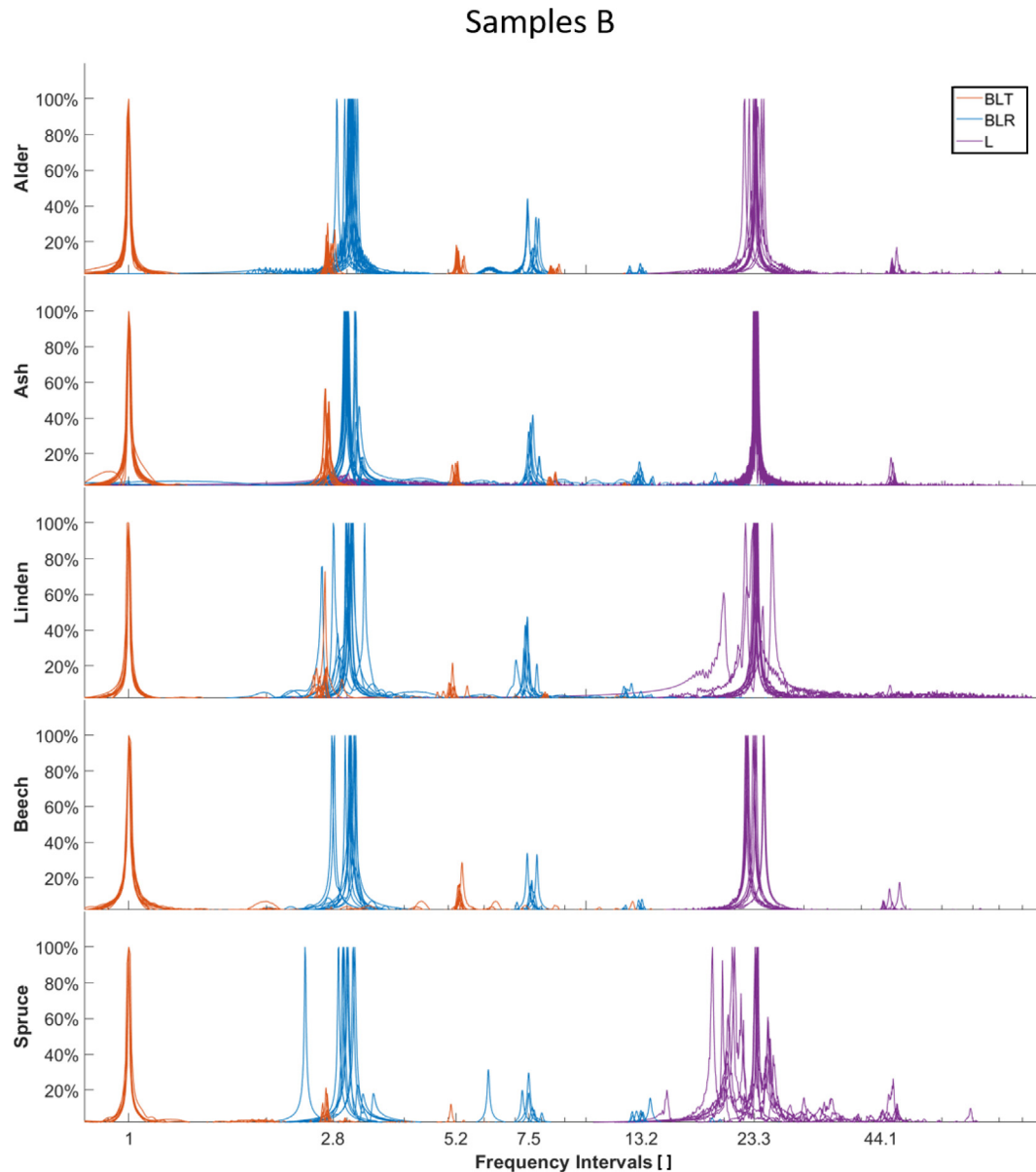


Fig. 10. Intervals of resonant frequencies – comparison of species – samples B Table 5.

Samples B, C and D are statistically comparable to the sample A ($p_{(A,B)} = 0.2629$, $p_{(A,C)} = 0.8854$, $p_{(A,D)} = 0.3320$), demonstrating the vibro-acoustic properties for multiple scales can be tested (for smaller samples made from large sample). Samples B and D are significantly different ($p_{(B,D)} = 0.0040$) across species. This can be caused by fibre angle of wider B samples, which are not so special orthotropic, because of their size. Some differences between samples C and D, which have the same moment of inertia, may be due to the influence of shear mode shapes to oscillations of bending mode shapes of shorter and thicker sample D [24]. Comparison of the influence of parameters on the sample dimensions showed a significant difference between samples B and D.

Correlations between MOED, damping coefficient, and sample slenderness have not been demonstrated for all species. Correlation of LT sample slenderness and $MOED_{(BLT)}$ was found for all hardwoods (from $r = 0.455$ to $r = 0.646$).

Because of wide variability, the damping coefficient was analyzed with even more detail, for individual modes. Figs. 5–8 show

medians of $\tan \delta$ of BLT modes, BLR modes, and L modes, for each species. The first mode (BLT_1 , BLR_1 , L_1) has the lowest frequency in its group.

Samples A (except beech) have damping coefficients of BLT_1 statistically incomparable (p-value from 0.007 to 0.025) to higher ones, although BLT_2 , BLT_3 , BLT_4 are comparable with each other. Beech has $\tan \delta$ of BLT_1 comparable with BLT_2 ($p = 0.376$) but not to others higher modes. Alder, ash, and linden have statistically comparable BLR_1 and BLR_2 (p-value from 0.083 to 0.245). Beech and spruce have not. All the species, the damping coefficients of L_1 and L_2 are statistically comparable (p-value from 0.08 to 0.559). By comparing the damping coefficients of the first modes, it was found that beech and ash have comparable BLT_1 , BLR_1 , and L_1 . Other species have incomparable BLT_1 with other first modes (BLR_1 and L_1 are comparable).

For samples B of all species, except spruce ($p = 0.0595$), $\tan \delta$ of BLT_1 is incomparable with BLT_2 and BLT_3 . Higher BLR modes were only measured for beech and alder. For beech, $\tan \delta$ of BLR modes

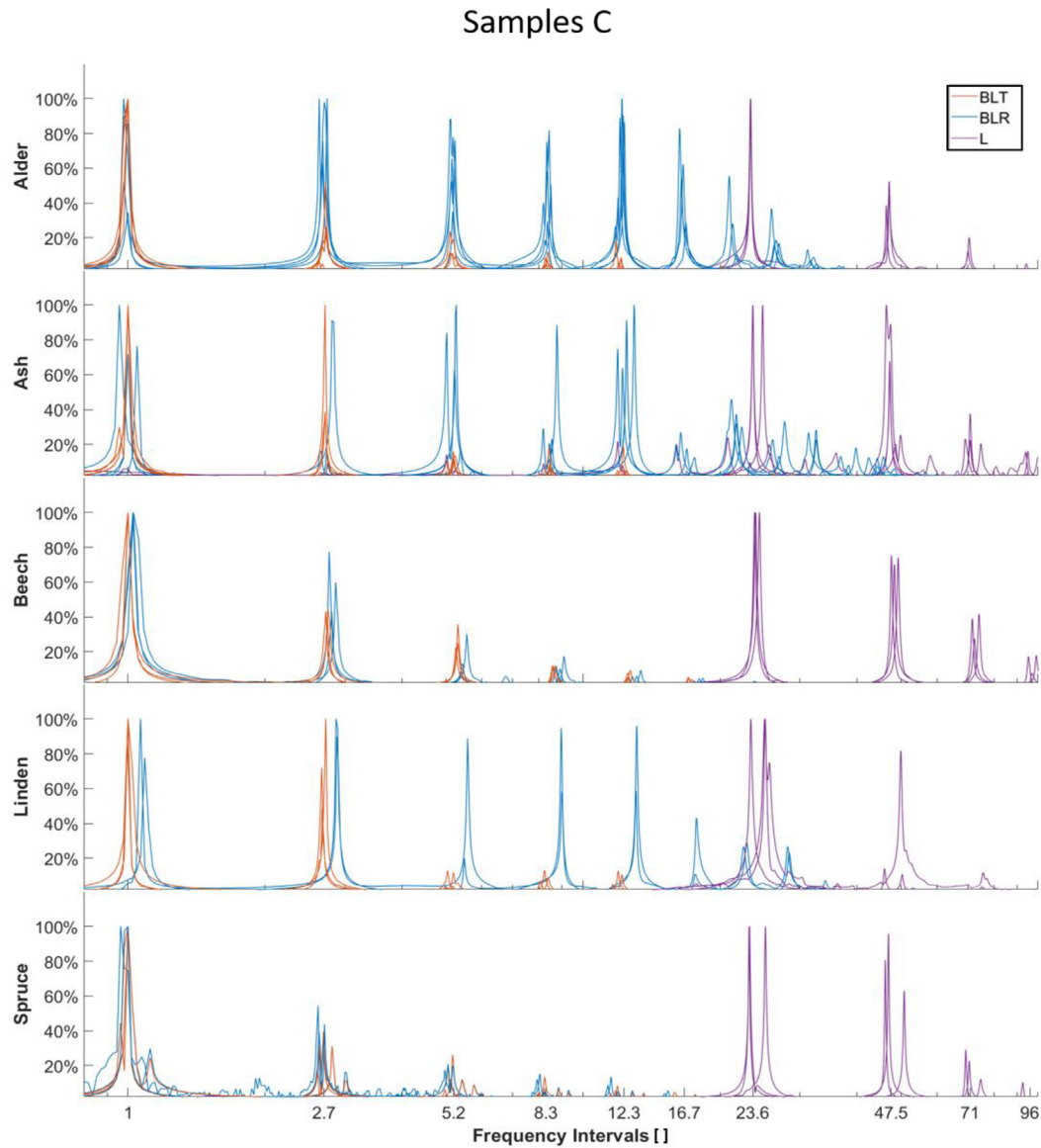


Fig. 11. Intervals of resonant frequencies – comparison of species – samples C Table 6.

are statistically comparable ($p = 0.0217$), not for alder ($p = 0.0008$). All species, except spruce, have comparable $\tan \delta$ of BLR₁ to L₁ (p -value from 0.0927 to 0.266), but both are different to BLT₁. Spruce has comparable $\tan \delta$ of BLR₁ and BLT₁, but both are different to L₁.

Overall, the highest statistical comparability was found in samples C. All species have comparable $\tan \delta$ from BLT₁ to BLT₄, beech up to BLT₅ (p -value from 0.0683 for linden to 0.6928 for beech). All species have comparable $\tan \delta$ from BLR₁ to BLR₄, ash and beech up to BLT₅ (p values from 0.2017 for linden to 0.6945 for beech). Except spruce all species have comparable damping coefficients of all L (p -value from 0.0767 to 0.8932). Spruce has different L₁ and L₂. Spruce has also an incomparable $\tan \delta$ of BLT₁ with L₁. Other species have statistically comparable damping coefficients of all first modes (p values from 0.1384 to 0.6659).

D samples of beech and alder have statistically incomparable $\tan \delta$ of all first and second modes (BLT, BLR and L). Ash has incomparable $\tan \delta$ of both BLT and BLR, but comparable $\tan \delta$ of L₁, L₂ ($p = 0.8376$) Linden has comparable $\tan \delta$ of BLT₁, BLT₂ ($p = 0.2386$), and L₁, L₂ ($p = 0.6343$). Spruce has comparable only

$\tan \delta$ of BLT₁ and BLT₂ ($p = 0.303$), BLR₂ was not measured, and L₁, L₂ are statistically different.

Statistics of the first modes showed that damping coefficients of BLT₁, BLR₁, and L₁, are comparable only for ash samples ($p = 0.1785$) and alder samples ($p = 0.214$). Comparable damping was found of BLT₁ and BLR₁, with different L₁, for beech and linden. Spruce has comparable BLT₁ with BLR₁, and BLT₁ with L₁.

Results showed the most comparable damping coefficients of modes are on samples C. The differences were only for spruce samples, which may be caused by the different anatomy of conifers and different propagation of elastic wave in them [9]. Analysis showed the material damping of vibrations is not as much dependent on the anatomical direction, mode shape, or frequency, but on whether the mode is of a higher order and on how rigid the sample is. For very slender samples of deciduous trees, the results so far show that it is possible to measure only one damping coefficient and thus define damping coefficients for other anatomical directions, ranks and mode shapes (bending and longitudinal).

For some samples, it was not possible to measure frequencies and damping coefficient of higher modes because the amplitudes

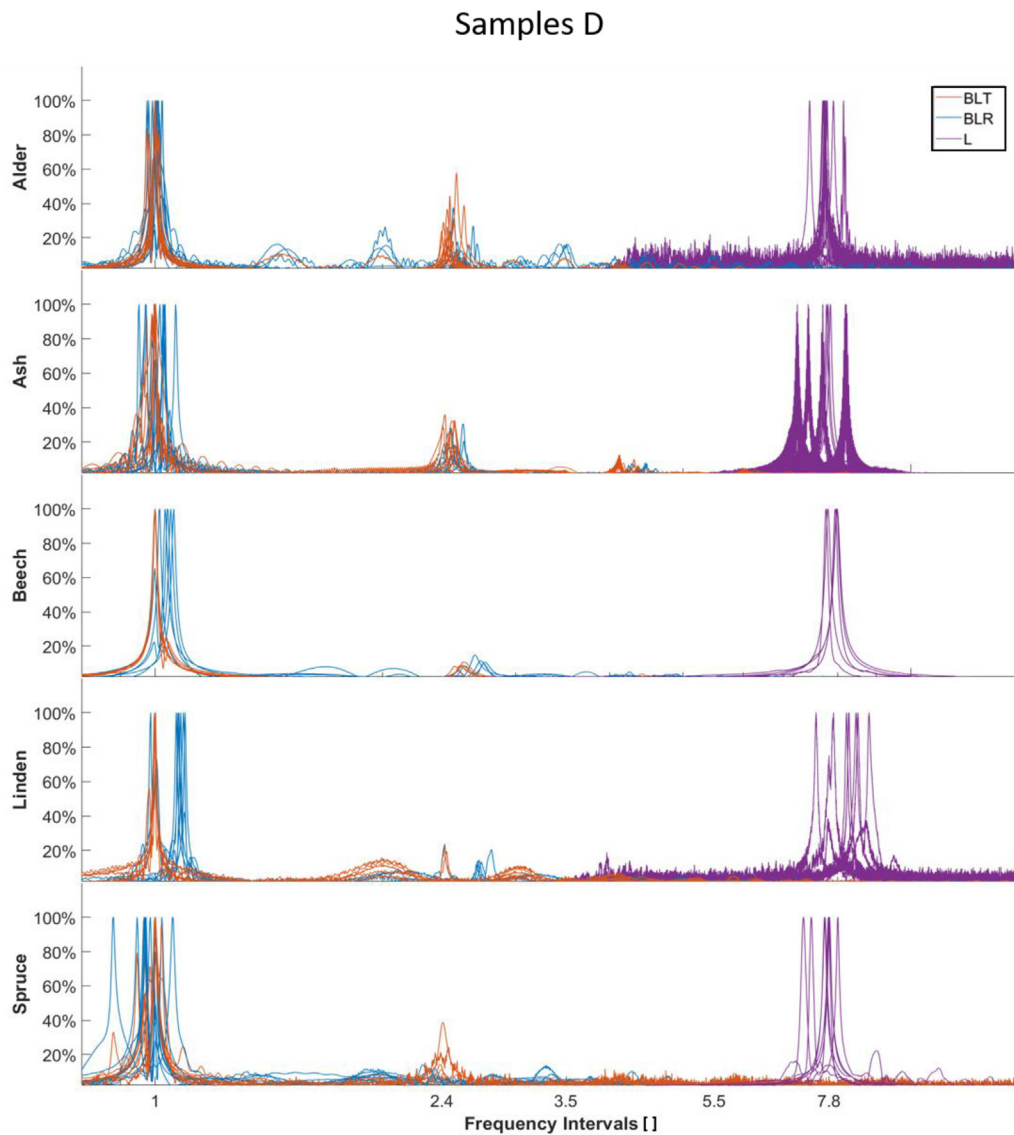


Fig. 12. Intervals of resonant frequencies – comparison of species – samples D [Table 7](#).

Table 6

Intervals between fundamental and frequencies of resonant modes for sample C.

Mode	1.	2.	3.	4.	5.	6.	7.
BLT	1	2.72	5.16	8.36	12.17	16.37	21.19
BLR	0.99	2.72	5.21	8.47	12.31	16.91	21.64
L	23.66	47.47	71.67	95.73	119.75	–	–

Table 7

Intervals between fundamental and frequencies of resonant modes for sample D.

Mode	1.	2.	3.
BLT	1	2.47	3.63
BLR	1.01	2.53	3.50
L	7.79	15.59	–

were too low due to high stiffness (associated with low slenderness).

[Figs. 9–12](#) show the intervals between resonant peaks of frequencies of vibration modes. For better comparison of species, the fundamental (lowest frequency) was assigned the number 1.

Other resonant peaks have interval value corresponding to a multiple of their frequency to the fundamental. This representation shows the vibrations of selected samples have a very similar character, regardless of the species. Differences in higher modes are due to higher damping coefficients.

As expected, due to the shape and anatomical orientation, the fundam (lowest – main frequency) was BLT₁. Except for spruce, the second frequency belongs to mode BLT₂. For spruce, BLT₂ was overlapped by the resonant peak of BLR₁. Without using one-axis sensors, it would be difficult to distinguish which mode it is [Table 6](#), shows medians of intervals between fundamental and other resonant peaks of samples across species.

Compared to samples A, which have the same ratio of dimensions, fewer resonant modes were measured on samples B, because higher modes were damped faster. Defined intervals of resonant peaks are very similar to samples A.

Overlapping of resonant peaks of BLT and BLR mode is a phenomenon associated with samples with square cross-section (C, D). The lowest frequency is usually indicated as the fundament. In this case, BLT₁ will remain marked as the fundament, even if the BLR₁ frequency is slightly lower in resulting median evaluation, but the difference is negligible. Samples C had the highest number of modes measured.

The least number of measured resonant modes were found on samples D. As with samples C, very similar frequencies of BLT and BLR modes can overlap or change order.

The assessment of wood species in terms of natural frequencies showed the effect of the frequency (pitch) on properties of each species according to generally known laws [9], but the intervals of frequencies between individual modes are comparable and very similar across species. Species similarity applies to different sample shapes and sizes. Samples with the slenderness of 48 (A_{LT}, B_{LT}, C) have intervals of bending modes very similar to expected intervals 1, 2.7572, 5.4040, 8.9332, 13.3446 [5]. It can be assumed that with further slimming, the differences between materials will decrease until only geometry prevails.

As expected [13], higher differences occurred at higher frequencies. But the results of this study also show these differences are greater in stiffer samples. It leads to a hypothesis that broader analysis of wood in the field of vibro-acoustics (depending on the properties sought) requires, in addition to one-axis sensors, also a set of samples of different shapes, as for static testing (test standard BS 373). It is necessary to test the greater variability of sample shapes to verify this hypothesis.

4. Conclusions

- As expected, the values of vibro-acoustic parameters are, specific to individual species. Differences in comparability of parameter values of anisotropic material can be significant when using different geometry of measured samples. Differences in the determination of parameters from bending and longitudinal vibrations are smaller when the measured sample is slimmer. Results show that samples with slenderness higher than 48 have negligible differences between MOED calculated from the first bending or first longitudinal mode shape. Orthotropy becomes less significant and geometry predominates. Vibro-acoustic behavior of wood approaches the mathematical assumption of isotropic, homogeneous material.
- More detailed analysis of very variable damping coefficient shows, the most comparable damping coefficients of vibrations were on samples C of deciduous species. In addition to the comparability of the first and higher transverse modes (p values of BLT from 0.0683 to 0.6928, and of BLR from 0.2017 to 0.6945) and the first and higher longitudinal modes (p values from 0.0767 to 0.8932), also first modes (BLT₁, BLR₁, L₁) show statistically significant comparability to each other (p values from 0.1384 to 0.6659). Results also show that the mode shape and its rank have greater effect on material damping than the frequency.
- For assessment of vibro-acoustic properties of wood, smaller samples crafted from a bigger one are comparable. For measuring and comparing samples from different production of material and with different ratios of dimensions, certain maximum range of slenderness in the loading direction is required to be kept.

- For more precise analysis of dynamic properties of wood, especially the properties of samples with the identical moment of inertia, it is necessary to use the one-axis measurement of vibrations in positions of antinodes of measured mode shapes. On these samples, change of order or overlap of resonant peaks of transversal can occur. Without using one-axis sensors, the result may be misinterpreted.
- For further research, it is necessary to focus more on the effect of the slenderness of samples with a square cross-section on parameters such as MOED and damping (both represented by ACE).

Funding

This work was funded by the Internal Grant Agency of Faculty of Forestry and Wood Technology at Mendel University in Brno (Grant No. LDF_VP_2020013).

CRediT authorship contribution statement

Nop Patrik: Conceptualization, Methodology, Software, Formal analysis, Investigation. **Tippner Jan:** Supervision.

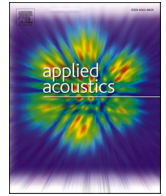
Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationship that could have appeared to influence the work reported in this paper.

References

- [1] Ahmed SA, Adamopoulos S. Acoustic properties of modified wood under different humid conditions and their relevance for musical instruments. *Appl Acoust* 2018;140(May):92–9. <https://doi.org/10.1016/j.apacoust.2018.05.017>.
- [2] ASTM E1876-15. Standard test method for dynamic young's modulus, shear modulus, and poisson's ratio by impulse excitation of vibrations. West Conshohocken, PA: ASTM International; 2015.
- [3] Baar J, Tippner J, Gryc V. The influence of wood density on longitudinal wave velocity determined by the ultrasound method in comparison to the resonance longitudinal method. *Eur J Wood Wood Prod* 2012;70(5):767–9.
- [4] Baar J, Tippner J, Gryc V. Wood anatomy and acoustic properties of selected tropical hardwoods. *IAWA Journal* 2016;37(1):69–83. <https://doi.org/10.1163/22941932-20160121>.
- [5] Bork I. Practical tuning of xylophone bars and resonators. *Appl Acoust* 1995;46(1):103–27. [https://doi.org/10.1016/0003-682X\(95\)93953-F](https://doi.org/10.1016/0003-682X(95)93953-F).
- [6] Brémaud I. Acoustical properties of wood in string instruments soundboards and tuned idiophones: Biological and cultural diversity. *J Acoust Soc Am* 2012;131(1):807–18. <https://doi.org/10.1121/1.3651233>.
- [7] Brémaud I, Gril J, Thibaut B. Anisotropy of wood vibrational properties: Dependence on grain angle and review of literature data. *Wood Sci Technol* 2011;45(4):735–54. <https://doi.org/10.1007/s00226-010-0393-8>.
- [8] Brémaud I, Minato K, & Thibaut B. (2009). Mechanical damping of wood as related to species classification: a preliminary survey. *6th Plant Biomechanics Conference Cayenne*, 1–7. papers3://publication/uuid/80AA77A1-F445-4065-A76F-7AEFF49E3F1C
- [9] Bucur, V. (2006). Acoustics of wood. In *The Acoustics of Wood*. CRC Press. <https://doi.org/10.1201/9780203710128>.
- [10] Corradi R, Miccoli S, Squicciarini G, Fazioli P. Modal analysis of a grand piano soundboard at successive manufacturing stages. *Appl Acoust* 2017;125:113–27. <https://doi.org/10.1016/j.apacoust.2017.04.010>.
- [11] Dumond P, Baddour N. Effects of using scalloped shape braces on the natural frequencies of a brace-soundboard system. *Appl Acoust* 2012;73(11):1168–73. <https://doi.org/10.1016/j.apacoust.2012.05.015>.
- [12] Dünisch O. Relationship between anatomy and vibration behaviour of softwoods and hardwoods. *IAWA Journal* 2017;38(1):81–98. <https://doi.org/10.1163/22941932-20170158>.
- [13] Gerhard S, Loidl D, Stanzl-Tschegg S, Maierhofer J, Puchegger S. Elastic constants of six wood species measured with the resonant beam technique. *Wood Research* 2020;65(2):347–52. <https://doi.org/10.37763/wr.1336-4561/65.2.347352>.

- [14] Grabianowski M, Manley B, Walker JCF. Acoustic measurements on standing trees, logs and green lumber. *Wood Sci Technol* 2006;40(3):205–16. <https://doi.org/10.1007/s00226-005-0038-5>.
- [15] Haines, D. W., Leban, J. M., Herbe, C., & Herb, C. (1996). Determination of Young's modulus for spruce, fir and isotropic materials by the resonance flexure method with comparisons to static flexure and other dynamic methods. 30, 253–263.
- [16] Obataya E, Ono T, Norimoto M. Vibrational properties of wood along the grain. *J Mater Sci* 2000;35(12):2993–3001. <https://doi.org/10.1023/A:1004782827844>.
- [17] Ono T, Norimoto M. Anisotropy of dynamic Young's modulus and internal friction in wood. *Japan J Appl Phys Part 1: Regular Papers Short Notes* 1985;24(8):960–4. <https://doi.org/10.1143/jjap.24.960>.
- [18] Quintavalla M, Gabrielli F, Canevari C. Assessing the quality of materials for stringed instruments soundboards: an approach considering the orthotropic elastic and damping properties. *Appl Acoust* 2022. <https://doi.org/10.1016/j.apacoust.2021.108521>. submitted 108521.
- [19] Rais A, Pretzsch H, van de Kuilen J-W. Roundwood pre-grading with longitudinal acoustic waves for production of structural boards. *Eur J Wood Wood Prod* 2014;72(1):87–98.
- [20] Roohnia M, Kohantorabi M, Tajdini A. Maple wood extraction for a better acoustical performance. *Eur J Wood Wood Prod* 2014;73(1):139–42. <https://doi.org/10.1007/s00107-014-0871-z>.
- [21] Salamak M. Vibration damping identification maximizing adjustment to viscous model in civil structures. *Archiv Civil Eng* 2007;53(3). <https://www.researchgate.net/publication/256472270>.
- [22] Sedik Y, Hamdan S, Jusoh I, Hasan M. Acoustic properties of selected tropical wood species. *J Nondestruct Eval* 2010;29(1):38–42. <https://doi.org/10.1007/s10921-010-0063-7>.
- [23] Spycher M, Schwarze FWMR, Steiger R. Assessment of resonance wood quality by comparing its physical and histological properties. *Wood Sci Technol* 2008;42(4):325–42. <https://doi.org/10.1007/s00226-007-0170-5>.
- [24] Straže A, Mitkovski B, Tippner J, Čufar K, Gorišek. Structural and acoustic properties of African padouk (*Pterocarpus soyauxii*) wood for xylophones. *Eur J Wood Wood Prod* 2015;73(2):235–43. <https://doi.org/10.1007/s00107-015-0878-0>.
- [25] Viala R, Placet V, Cogan S. Identification of the anisotropic elastic and damping properties of complex shape composite parts using an inverse method based on finite element model updating and 3D velocity fields measurements (FEMU-3DVF): Application to bio-based composite violin sou. *Compos A Appl Sci Manuf* 2018;106(December):91–103. <https://doi.org/10.1016/j.compositesa.2017.12.018>.
- [26] Wegst UGK. Wood for sound. *Am J Bot* 2006;93(10):1439–48. <https://doi.org/10.3732/ajb.93.10.1439>.



Corrigendum



Corrigendum to “Influence of dimensions of wooden samples for determination of acoustic parameters and sound timbre” [Appl. Acoust. 196 (2022) 108895]

Patrik Nop^{*}, Jan Tippner

Mendel University in Brno/Faculty of Forestry and Wood Technology, Czech Republic

The authors regret that the equation 3 contains a typographical error (l^2 instead of l^3). Its correct form used for the MOED calculation is:

The authors would like to apologise for any inconvenience caused.

$$MOED = \left(\frac{2f}{2.25\pi} \right)^2 \cdot \frac{m \cdot l^3}{I}$$

DOI of original article: <https://doi.org/10.1016/j.apacoust.2022.108895>.

^{*} Corresponding author.

E-mail address: xnop@mendelu.cz (P. Nop).

<https://doi.org/10.1016/j.apacoust.2024.110206>

Available online 1 August 2024

0003-682X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Paper II
Changes in vibro-acoustic properties of green beech wood due to
Kretzschmaria deusta

Patrik Nop, Valentino Cristini

**Proceedings of 6th International Conference on Structural Health Assessment of
Timber Structures**

https://utam-api.ssc.avcr.cz/uploads/PROCEEDINGS_SHATIS_22_db5ee1c7e6.pdf

Nop, P. (corresponding author): Methodology, Resources, Investigation, Formal analysis, Writing – original draft

CHANGES IN VIBRO-ACOUSTIC PROPERTIES OF GREEN BEECH WOOD DUE TO KRETZSCHMARIA DEUSTA

Patrik NOP¹, Valentino CRISTINI¹

¹ Department of Wood Science and Technology, Mendel University in Brno, 613 00 Czech Republic

ABSTRACT

Dynamic modulus of elasticity (MOED) and vibrational damping are the main parameters defining the vibro-acoustic properties of the material and can be used as important indicators in selecting a suitable wood for structures and also in the health assessment of timber constructions. Parameters are determined using a non-destructive frequency resonance technique. Degradation caused by wood-decaying fungi significantly affects observed parameters. The study presents dynamic bending properties of artificially inoculated samples with soft-rot fungus *Kretzschmaria deusta*, and their correlation to the static bending modulus of rupture (MOR) which was evaluated by standard mechanical test on small samples of beech wood (*Fagus sylvatica* L.). Data comparison with intact specimens and a relationship between mechanical properties and mass loss are also included. Presented results can be used to improve vibro-acoustic non-destructive timber assessment.

KEYWORDS: Non-destructive testing, Wood-decaying fungi, Modulus of elasticity, Vibrational damping, Vibro-acoustic properties

INTRODUCTION

Wood-decaying fungi cause the breakdown of components of the wood cell wall (lignin, cellulose and hemicellulose) which may lead to a decrease in mechanical properties [1]. The degree of wood degradation is often assessed according to mass reduction. Nevertheless, strength reduction can occur considerably sooner than weight loss [2]. However, changes in physical-mechanical properties cannot be generalized. It depends on the wood and fungus species, rot type, but on the incubation time as well [3]. A reliable method for assessing the change in physico-mechanical properties of wood caused by fungal degradation is provided by non-destructive techniques (NDT) [4]. It was demonstrated that enzymatic activity of the white-rot fungus *Physisporinus (P) vitreus* on spruce wood can reduce the density relatively more than stiffness [5]. It was presented as a modification improving the acoustic properties of wood for stringed musical instruments, but overall these parameters (and with them the velocity of sound propagation) were slightly reduced. Other studies with *P. vitreus* [6, 7] reported changes in vibro-acoustic parameters, which were evident mostly on the vibrational damping. Using the frequency resonance method, the vibrational mode shapes (longitudinal, transverse, torsional) are exciting, sensed and analyzed. From the frequency of the vibration of the first mode of each shape, dynamic moduli of elasticity (longitudinal, bending) and dynamic shear modulus (torsion) were calculated [8]. The damping coefficient ($\tan \delta$) manifests an amplitude attenuation of these vibrations [9]. According to the current state of the arts, changes of MOED and $\tan \delta$ of green wood caused by fungal degradation are insufficiently explored. A detailed description of the abovementioned relationships can lead to an improvement of non-destructive methods used for timber health assessment.

METHODOLOGY

Special orthotropic samples (7×7×100 mm) of European beech (*Fagus sylvatica* L.) intact sapwood were crafted from fresh (two weeks after felling) timber. After obtaining oven-dry weight (moisture content = 0%), samples were acclimatized to moisture content of 60% (± 10%), and sterilised. To consider the influence of drying, climatization, and sterilisation process on dynamic properties (MOED and tan δ), specimens were tested in every stage of the process. The culture of soft-rot fungus *Kretzschmaria deusta* used for specimens' degradation was obtained from the wood of a mature standing beech tree and cultivated for one month on beech wooden blocks before the experiment to stimulate its enzymatic activity. Kolle culture flasks (400 ml) were filled with 70 ml of sterile malt extract agar medium (5%) and inoculated with a *K. deusta*. After 2 weeks of incubation at 22 °C and 70% relative humidity, when the mycelium completely covered the flask surface, samples were inserted and laid on the radial face. After 12 weeks of fungal exposure, dynamic and also static tests were carried out on samples (cleaned of superficial mycelium). To determine dynamic parameters, the frequency resonance technique of free-free-flexural vibration was used. Specimens were supported by foam pads at the nodes of the first bending mode (22.4% and 77.6% of length). Oscillations of longitudinal-tangential and longitudinal-radial bending mode shapes (B_{LT} and B_{LR}) were excited by a small hammer, sensed using vibrometer PDV-100 (Polytec, Inc., Germany), and recorded using dynamic signal acquisition module DEWE-41-T-DSA with DEWESoft (DEWETRON, Inc., USA) at sampling frequency 20 kHz. The measured signal of vibrations was decomposed into an amplitude-frequency domain using the Fast Fourier Transform (FFT) processed using MATLAB® (The MathWorks, Inc., USA). Frequencies of first bending modes were used for the MOED calculation, using an equation (1).

$$MOED = \left(\frac{2f}{2.25\pi} \right)^2 \frac{ml^2}{I} \quad [\text{Pa}] \quad (1)$$

Where f = frequency, m = sample mass, l = sample length, and I = moment of inertia [8].

The vibrational damping was determined from the amplitude (a) decrease in consecutive oscillations per period as the logarithmic decrement of damping (LDD) [10].

$$LDD = \ln \frac{a_n}{a_{n+1}} \quad (2)$$

Using the LDD, the damping coefficient tan δ is calculated [11].

$$\tan \delta = \frac{LDD}{\pi} \quad (3)$$

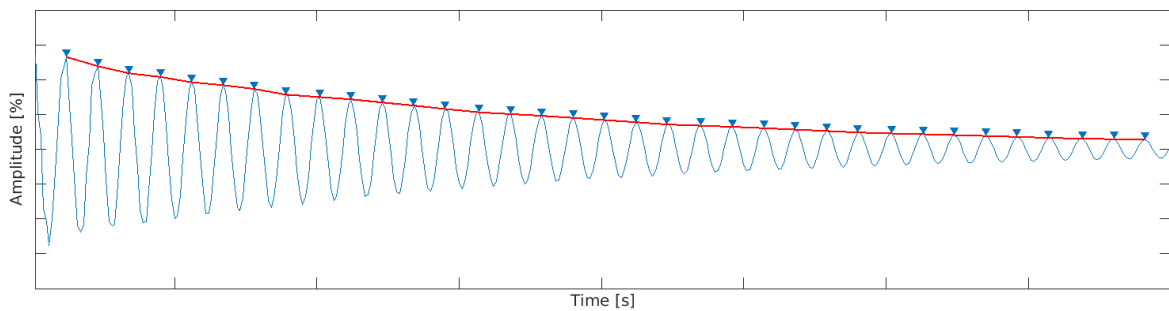


Figure 1: Damped oscillation, which decreasing amplitudes are used to calculate the LDD.

To determine MOR, static three-point bending tests were carried out on a universal testing machine (ZWICK® Z050, 50 kN). Bending stress in the tangential direction (σ_b) was calculated according to the equation (4).

$$\sigma_b = \frac{3Fl_o}{2bh^2} \quad [\text{Pa}] \quad (4)$$

Where F = applied force, l_o = distance between supports, and b , h = the base and the height of the specimen's cross section. Modulus of rupture (MOR) was calculated as the bending stress at the highest applied force (F_{max}). For the analysis, only specimens with a statistically comparable moisture content before and after inoculation were chosen. Of the 108 samples measured, 60 samples were analysed.

Measured data was statistically processed in MATLAB, using the Kruskal-Wallis test with a multiple comparison.

RESULTS

Dynamic measurements during the inoculation process (drying, climatization, sterilisation) did not show any statistically relevant difference ($p < 0.01$). All changes in material properties were caused only by *K. deusta*. Mass loss of specimens was an average of 30%. For MOED and $\tan \delta$, a statistically relevant difference was proven ($p < 0.05$) between intact and degraded samples. Figure 2A shows an increased data variation after specimens' degradation. Due to degradation, the median of MOED_{BLT} decreased by 65% and MOED_{BLR} decreased by 61%. A relevant correlation coefficient between MOED_{BLT} and MOR was proven ($r = 0.84$). After degradation, the median of damping coefficients raised over 288 % ($\tan \delta_{\text{BLT}}$) and 279 % ($\tan \delta_{\text{BLR}}$). Figure 2B compares the decrease of amplitudes of bending vibrations over the same time period. A conclusive relationship between $\tan \delta$ and any other observed parameter wasn't found.

Table 1: Median values of measured data

	Intact	Degraded
MOED_{BLT} [MPa]	11 462.6	4 062.3
MOED_{BLR} [MPa]	11 861.3	4 640.7
MOR_{BLT} [MPa]	63.4	28.3
$\tan \delta_{\text{BLT}}$	0.0231	0.0899
$\tan \delta_{\text{BLR}}$	0.0227	0.0861

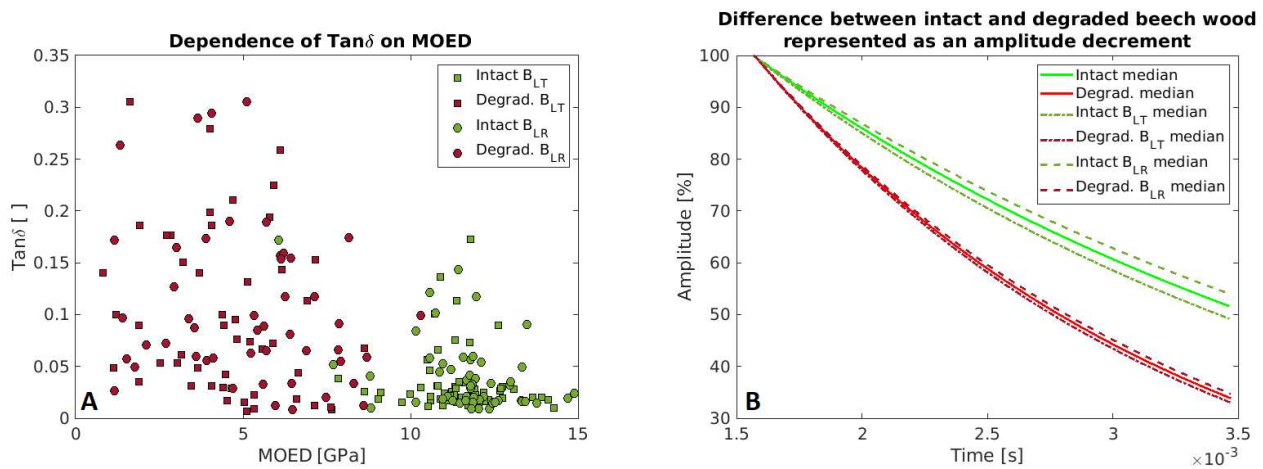


Figure 2: Comparison between MOED and $\tan \delta$ for longitudinal-tangential and longitudinal-radial bendings (A), comparison of vibrational damping between intact and degraded specimens (B)

CONCLUSION

Degradation of European beech wood cell components caused by the wood-decaying fungus *Kretzschmaria deusta* led to a decrease of dynamic bending moduli of elasticity and an increase of the vibrational damping. These parameters obtained through non-destructive frequency resonance method can serve as indicators for timber health assesment.

ACKNOWLEDGEMENTS

This outcome was supported by the Internal Grant Schemes of Mendel University in Brno, registration no.: CZ.02.2.69/0.0/0.0/19_073/0016670, funded by the ESF.

REFERENCES

- [1] Malakani M, Khademieslam H, Hosseinihashemi SK, Zeinaly F (2014) Influence of fungal decay on chemi-mechanical properties of beech wood (*Fagus orientalis*). *Cellul Chem Technol* 48:97–103
- [2] Witomski P, Olek W, Bonarski JT (2016) Changes in strength of Scots pine wood (*Pinus silvestris* L.) decayed by brown rot (*Coniophora puteana*) and white rot (*Trametes versicolor*). *Constr Build Mater* 102:162–166. <https://doi.org/10.1016/j.conbuildmat.2015.10.109>
- [3] Spycher, M. (2008). The Application of Wood Decay Fungi to Improve the Acoustic Properties of Resonance Wood for Violins, Ph.D. Thesis, Albert Ludwig University of Freiburg, Freiburg, Germany.
- [4] Niemz P, Mannes D (2012) Non-destructive testing of wood and wood-based materials. *J Cult Herit* 13:S26–S34. <https://doi.org/10.1016/j.culher.2012.04.001>
- [5] Schwarze, F. W. M. R., Spycher, M., & Fink, S. (2008). Superior wood for violins - Wood decay fungi as a substitute for cold climate. *New Phytologist*, 179(4), 1095–1104. <https://doi.org/10.1111/j.1469-8137.2008.02524.x>
- [6] Danihelová, A., Spišiak, D., Reinprecht, L., Gergel, T., Vidholdov, Z., & Ondrejka, V. (2019). Acoustic properties of Norway spruce wood modified with staining fungus (*Sydowia polyspora*). *BioResources*, 14(2), 3432–3444. <https://doi.org/10.15376/biores.14.2.3432-3444>
- [7] Gilani, M. S., Neuenschwander, J., Heeb, M., Furrer, R., Sanabria, S. J., Stoel, B. C., & Schwarze, F. W. M. R. (2016). Influence of incubation time on the vibration and mechanic properties of mycowood. *Holzforschung*, 70(6), 557–565. <https://doi.org/10.1515/hf-2015-0128>
- [8] Bucur, V. (1995). *Acoustics of Wood* (1st ed.). CRC Press. <https://doi.org/10.1201/9780203710128>
- [9] Brémaud I, Minato K, Langbour P, Thibaut B (2010) Physico-chemical indicators of inter-specific variability in vibration damping of wood. *Ann For Sci* 67:707–707. <https://doi.org/10.1051/forest/2010032>
- [10] Salamak, M. (2007). Vibration damping identification maximizing adjustment to viscous model in civil structures. In *Archives of Civil Engineering* (Vol. 53, Issue 3). <https://www.researchgate.net/publication/256472270>
- [11] Baar, J., Tippner, J., & Gryc, V. (2016). Wood anatomy and acoustic properties of selected tropical hardwoods. *IAWA Journal*, 37(1), 69–83. <https://doi.org/10.1163/22941932-20160121>

Erratum

Patrik Nop*, Valentino Cristini

Erratum to: CHANGES IN VIBRO-ACOUSTIC PROPERTIES OF GREEN BEECH WOOD DUE TO KRETZSCHMARIA DEUSTA

Erratum to: Nop, P.; Cristini, V.; Changes in Vibro-acoustic Properties of Green Beech Wood Due to Kretzschmaria Deusta. Proceedings of SHATIS 2022, 2022, pp. 87-90. (DOI: 10.21495/54-3):

Equation 1 contains a typographical error (l^2 instead of l^3). Its correct form used for the *MOED* calculation is:

$$MOED = \left(\frac{2f}{2.25\pi} \right)^2 \frac{ml^3}{I}$$

***Corresponding author: Patrik Nop**, Department of Wood Science and Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Zemědělská 1, 61300 Brno, Czech Republic, E-mail: patrik.nop@mendelu.cz

Valentino Cristini, Department of Wood Science and Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Zemědělská 1, 61300 Brno, Czech Republic

Paper III

Dynamic properties of wood obtained by frequency resonance technique and dynamic mechanical analysis

Patrik Nop, Valentino Cristini, Jan Zlámál, Mojtaba Hassan Vand, Vít Šeda, Jan Tippner

Wood and Fiber Science

<https://doi.org/10.22382/wfs-2023-12>

Nop, P. (corresponding author): Conceptualization, Methodology, Resources, Investigation, Formal analysis, Writing – original draft

DYNAMIC PROPERTIES OF WOOD OBTAINED BY FREQUENCY RESONANCE TECHNIQUE AND DYNAMIC MECHANICAL ANALYSIS

*P. Nop**

Researcher

E-mail: patrik.nop@mendelu.cz

*V. Cristini***

Researcher

E-mail: valentino.cristini@mendelu.cz

*J. Zlámal***

Researcher

E-mail: jan.zlamal@mendelu.cz

M. Hassan Vand

Researcher

E-mail: mojtaba.vand@mendelu.cz

V. Šeda

Researcher

E-mail: vit.seda@mendelu.cz

*J. Tippner***

Assistant Professor

Faculty of Forestry and Wood Technology
Department of Wood Science and Technology

Mendel University in Brno
Brno, Czech Republic

E-mail: jan.tippner@mendelu.cz

(Received May 2023)

Abstract. The study of the viscoelastic properties of wood involves the determination of dynamic parameters. Different methods can be used to determine these parameters, which bring the question of whether these parameters can be considered identical depending on the method used. This study compares the frequency resonance technique and dynamic mechanical analysis for determining the bending dynamic modulus of elasticity (MOED) and damping coefficient $\tan \delta$ of dry and green wood. Groups of specimens of European beech (*Fagus sylvatica* L.), small-leaved linden (*Tilia cordata* Mill.), European oak (*Quercus robur* L.), and Norway spruce (*Picea abies* L.) wood at three levels of MC were tested with both methods. The bending dynamic modulus of elasticity decreased with increasing MC until the FSP. There were no significant changes in dynamic modulus with increasing MC above FSP. A strong linear correlation between MOED obtained through both methods was found ($r = 0.92$, $r^2 = 0.84$). For the damping coefficient $\tan \delta$, the relationship was weaker ($r = 0.57$, $r^2 = 0.32$), and each method showed a different influence of MC on the damping coefficient above the FSP, which leads to the conclusion that the damping coefficient is sensitive to the measurement method.

Keywords: Frequency resonance technique, dynamic mechanical analysis, dynamic modulus of elasticity, damping coefficient, wood.

* Corresponding author

** SWST member

INTRODUCTION

Among the main parameters describing the response of a material under dynamic loading are dynamic moduli of elasticity and damping coefficients (Bucur 2006). Due to their high correlation with static moduli ($r = 0.82$) (Karlinasary et al 2008), and with the modulus of rupture ($r = 0.90$) (Hassan et al 2013), dynamic moduli of elasticity are used to predict the strength and stiffness of wood (Hein and Brancheriau 2018; Fernández-Serrano and Villasante 2021), wood-based composites (Hamdan et al 2010; Wang et al 2012), and standing trees (Wang et al 2001; Mora et al 2009). Density (ρ) and dynamic moduli of elasticity are used to calculate the acoustic impedance and the sound radiation coefficient (Wegst 2006; Ahmed and Adamopoulos 2018), and (in combination with the damping coefficient $\tan \delta$) to determine the acoustic conversion efficiency (Hossen et al 2018; Danihelová et al 2019). All these vibro-acoustic parameters are used for the selection of quality material for musical instruments (Wegst 2006; Bucur 2016; Ahmed and Adamopoulos 2018; Hossen et al 2018; Tronchin et al 2020). The dynamic parameters for the above-mentioned applications are determined using nondestructive tests (NDT) (Wang et al 2012). Among these NDT methods, the frequency resonance technique (FRT) is one of the most used. This method excites, detects, and analyzes the natural frequencies of the mode shapes arising on the oscillating specimen (Bucur 2006; Brémaud 2012; Baar et al 2016; Hamdan et al 2018; Teles et al 2018). The frequency of these oscillations depends on the material properties, the geometry of the tested specimen, and the way it is fixed. The bending dynamic modulus of elasticity (MOED) of longitudinal-tangential and longitudinal-radial bending mode shape, shear modulus, and Young's modulus is calculated from the first modes of certain oscillation shapes (bending, torsion, and longitudinal) (Bucur 2006; ASTM E1875–20 2020). From the amplitude decrease of free oscillations over time, the logarithmic decrement of damping (LDD) is determined, and the damping coefficient $\tan \delta$ is further derived from it (Baar et al 2016). The FRT can be

used on specimens of various dimension ratios, but standard ASTM E1876 states that the ratio of length to minimum cross-sectional dimension with a value of at least five (5) must be maintained, and a ratio of 20:25 is preferred (ASTM E1876–15 2016). Depending on the sensitivity of the measuring device, and the method of oscillation excitation, the FRT can be used on small samples (Cristini et al 2022), through sizes corresponding to standards for static laboratory tests (Yang et al 2003), and up to beams and logs (Papandrea et al 2022). Besides the described FRT, which is used for small to large samples, the dynamic mechanical analysis (DMA) method can serve for the assessment of small to micro specimens (Salmén et al 2016; Pizzo et al 2018). During the DMA, the oscillating force is applied to the clamped sample, and its response in time is analyzed (Menard 2006). From the force and displacement data, the storage modulus (E_S), which represents the elasticity, and the loss modulus (E_L), which represents energy loss by the internal friction, are determined (Pizzo et al 2018). These two components are used to determine the dynamic modulus of elasticity (complex modulus) as the ratio of stress and deformation (Menard 2006). The phase shift between the stress input and the strain response is the damping coefficient $\tan \delta$, and it is calculated as the ratio of E_L and E_S (Brémaud et al 2011). The main use of DMA is to observe the effect of environmental changes (temperature and RH) on the measured parameters of tested materials (Menard 2006). The test parameters (span length, deflection, loading force, and loading frequency) are also factors influencing the results (Ashaduzzaman et al 2020). The size of DMA samples is defined by the type of loading and the associated span of clamps, the internal dimension of a heating or climatic chamber, and the type of device used (Peng et al 2008; Chowdhury et al 2010; Kaboorani and Blanchet 2014). The MC is the factor frequently observed affecting the viscoelastic properties of wood. Below the FSP (approximately 30% MC), an increase of MC of wood causes a rapid decrease in dynamic moduli, and a significant increase in the damping coefficient (Bucur 2006; Matsunaga et al 2000; Lu et al 2012). Further increases above the FSP will not

cause significant changes in the dynamic moduli (Bucur 2006, 2016; Barrett and Hong 2010; Nocetti et al 2014). For the FRT, the influence of increments of MC above FSP on the damping coefficient $\tan \delta$ has not been described yet (Barré et al 2018). For the DMA, an increase of the damping coefficient $\tan \delta$ was found when the MC increased above the FSP (Goken et al 2018). The main difference between the above-mentioned methods is the frequency at which the sample oscillates. The FRT is focused on individual resonant frequencies and the area of interest starts at the frequency of the first mode shape. The response in the case of impulse excitation is a combination of all the rising mode shapes (depending on the sensing and excitation point, and sample support). During continuous excitation, a frequency sweep is used to find the frequencies of resonance peaks, so the individual modes are separated (ASTM E1876–15 2016; ASTM E1875–20 2020). The frequency range of DMA starts basically in the static area (0.001 Hz) and ends, depending on the device used, at a frequency of approximately 100 Hz. Using a frequency sweep, the goal is generally not to find resonance peaks (on the contrary, resonance peak is undesirable for data collection), but to show the change in material response (Menard 2006). As the loading frequency increases, the material loses time to relax, and its behavior changes from viscous to elastic. Higher frequencies result in an increase in the dynamic modulus, while causing a reduction in the damping coefficient (Menard 2006).

In this paper, MOED and corresponding damping coefficient $\tan \delta$, obtained using FRT and DMA, were compared on the same specimens made from different species and with different MC. The chosen sample size was a compromise meeting the maximum sample size for a DMA device and the ratio of the length to minimum cross-sectional dimension stated by the ASTM standard for FRT. In the case of comparable results, it would be possible to combine these methods in the future and predict parameters for a wide scale of wooden products with various dimensions. For an even wider use including green wood, it is also necessary to

focus on deepening the knowledge about the influence of MC on dynamic parameters.

MATERIALS AND METHODS

Materials

Special orthotropic specimens with dimensions of $61 \times 8.4 \times 1.7$ mm (with coefficient of variation [cv] of dimensions = $0.9 \times 2.2 \times 9.6\%$), longitudinal $\times$ radial $\times$ tangential (Fig 1), were cut from European beech (*Fagus sylvatica* L.), Small-leaved linden (*Tilia cordata* Mill.), European oak (*Quercus robur* L.), and Norway spruce (*Picea abies* L.). Specimens were allocated into three groups of twenty pieces, and every group was conditioned to a different MC. One group had a low MC (LMC), and two groups were above FSP—a medium MC group (MMC), and a high MC group (HMC) (Table 1). The MC was determined using the oven-dry method. Basic density (ρ_b) was determined for all 240 specimens from the ratio of oven-dry mass and volume above the FSP (Table 2).

Measurements

Specimens for both measuring methods were loaded at the center span by longitudinal-tangent bending. To preserve the MC, measurements were done right after each other, FRT first, then immediately DMA, and FRT again (to monitor the impact of MC loss after each measurement). Before each stage of measurement, specimens were weighed. The entire measurement time for one sample was 12 min. Measurements were carried out in laboratory conditions at 24°C and 54% of RH.

Frequency resonance technique. Specimens were supported by flexible pads at the nodes of the first bending mode shape (0.224% and 0.776% of length). Specimen oscillations were excited at the specimen's midpoint by the rubber band. This excitation method was used due to the size and weight of specimens. The hit of a rubber band was sufficient to induce the first bending mode shape but also soft enough so that the loose sample did not reflect out or move on the pads. Vibrations were sensed in the middle of the

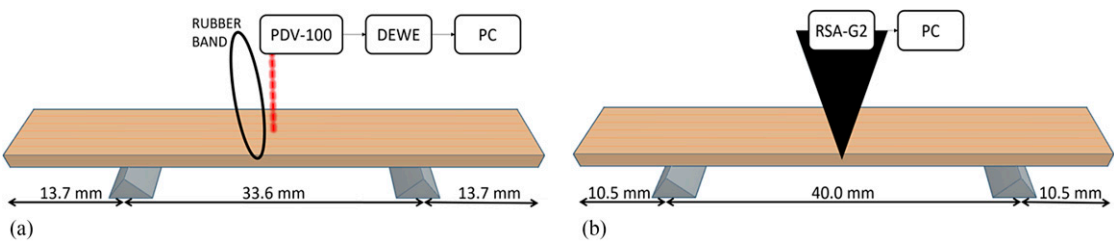


Figure 1. Setups of frequency resonance technique (a) and dynamic mechanical analysis (b).

specimen using a Doppler laser vibrometer PDV-100 (Polytec, Inc., Baden-Württemberg, DE), and recorded using a dynamic signal acquisition module DEWE-41-T-DSA and DEWESoft (DEWETRON Inc., Grambach, AT), with a sampling frequency of 20 kHz. A minimum of ten hits on the specimen were recorded. The test lasted under a minute.

Dynamic mechanical analysis. For the DMA, three-point bending was carried out using an RSA-G2 device (TA Instruments, New Castle, DE). Specimens were loaded by a spike in the middle and supported by a 40-mm-wide span. The samples were tested using the TRIOS software (TA Instruments), with an initial and minimum axial force value of 0.1 N (the axial force was set 30% higher than the dynamic force), and a displacement of 30 μm. The oscillation frequency was set as a logarithmic sweep from 0.1 to 100 Hz, with 10 points per logarithmic decade. This test was performed for 10 min per sample.

Data Analysis

Frequency resonance technique. For every specimen, the entire length of the signal recorded from FRT was transformed from the time domain into a frequency domain using the fast-Fourier transform processed using MATLAB

(The MathWorks, Inc., Natick, MA). The highest peak with the lowest frequency belongs to the first bending mode. The bending dynamic modulus of elasticity obtained by this method ($MOED_{FRT}$) was calculated from the frequency f using the following equation:

$$MOED_{FRT} = \left(\frac{2f}{2.25\pi} \right)^2 \frac{ml^2}{I} \tag{1}$$

where m is the mass and l is the length of the specimen. The moment of inertia I was calculated from width (w) and thickness (t) as a :

$$I = \frac{wt^3}{12} \tag{2}$$

Subsequently, five separate records (each for one of the five selected hits with the highest amplitude) were exported from the signal. For each record with a minimum of 25 periods of oscillation, the LDD was determined. From the LDD, the damping coefficient $\tan \delta$ was calculated using Eq 3. As a representative damping coefficient of the specimen, a median from five $\tan \delta$ was determined.

$$\tan \delta = \frac{LDD}{\pi} \tag{3}$$

Dynamic mechanical analysis. The main results obtained by DMA were storage modulus

Table 1. Medians of MC of each sample group.

Group	LMC		MMC		HMC	
	MC (%)	cv (%)	MC (%)	cv (%)	MC (%)	cv (%)
Beech	11.7	4.6	49.7	6.4	78.8	6.2
Linden	10.2	7.0	47.9	10.0	81.5	7.1
Oak	11.1	3.0	50.5	8.1	82.8	10.6
Spruce	11.1	2.9	47.3	8.3	84.3	15.5

Table 2. Medians of basic density values of species.

Species	ρ_b (kg · m ⁻³)	cv (%)
Beech (<i>Fagus sylvatica</i> L.)	558.8	3.5
Linden (<i>Tilia cordata</i> Mill.)	360.8	4.8
Oak (<i>Quercus robur</i> L.)	526.9	5.8
Spruce (<i>Picea abies</i> L.)	412.3	11.1

(E_L) and loss modulus (E_S), which are components of the stress ($\sigma - E_L$ is 90° out of phase with the strain ε , and E_S is in the phase with the strain).

$$E_L = \frac{\sigma \sin \delta}{\varepsilon} \quad (4)$$

$$E_S = \frac{\sigma \cos \delta}{\varepsilon} \quad (5)$$

The damping coefficient $\tan \delta$ represented a phase lag between the stress and the strain and was calculated as the ratio of E_L to E_S .

$$\tan \delta = \frac{E_L}{E_S} \quad (6)$$

The bending dynamic modulus of elasticity (MOED_{DMA}) is a complex combination of E_L and E_S .

$$\text{MOED}_{\text{DMA}} = \frac{\sigma}{\varepsilon} = E_S + iE_L = \sqrt{E_S^2 + E_L^2} \quad (7)$$

The MC of specimens decreased during the 10-min-long DMA depending on the moisture group, and (in the case of MC above the FSP) also depending on species (Table 3). Based on the weight method, MC loss was determined from known MC before and MC after the DMA test. For every specimen from groups MMC and HMC, the average of MC before and MC after the DMA test is given for the next data evaluation. DMA results were reported for each testing frequency. From all sample groups, the median for each testing frequency was calculated (Figs 2 and 3). For further statistical evaluation, the median value across frequencies was determined for each sample. The statistical comparison of

data obtained by both methods was done using one-way ANOVA, with a significance level of 0.05 and 0.01, in the MATLAB environment. The steepness of the function (stp) was determined according to the first term (a) of the linear regression $y = ax + b$.

RESULTS AND DISCUSSION

The MC loss during the DMA test was comparable for species in LMC groups, up to 1%. In groups with MC above the FSP, the differences between the species were observed. Wood species with lower ρ_b had a bigger MC loss during the experiment.

The DMA showed that the change from viscous to elastic behavior when changing frequency (Menard 2006) occurred earlier in dry wood than in wood at a higher MC (Fig 2). Resonance peaks at the same frequencies (39.8 and 63 Hz) were found in the data of E_L for all specimens, which affected the determination of MOED_{DMA} and the damping coefficient at these frequencies. Since it concerns the same frequencies for all specimens (regardless of MC and species), which are far from their resonance frequencies found by FRT (from 1240 to 2708 Hz), a measurement error caused by a device or clamps resonance was considered. According to Bucur (2006), and Nocetti et al (2014), the modulus of elasticity decreases with increasing MC. The results from DMA show that compared with LMC, moduli of MMC and HMC decreased by: 40% and 52% for beech, 28% and 46% for linden, 27% and 43% for oak, and 37% and 52% for spruce respective. So, for species with a higher modulus of elasticity, there is a more significant decrease with increasing MC.

All samples showed a drop or lack of the damping coefficient values at the lowest frequencies, caused by very low or negative values of E_L (Fig 3). From the erroneous resonance peaks, described earlier, all damping coefficients followed an increase and convergence of all samples at the same value. The middle area of the measurement (0.5-10 Hz) showed a trend of a slight decrease with increasing frequency, which agrees with

Table 3. Medians of MC loss during the DMA of each sample group.

Group	LMC		MMC		HMC	
	MC loss (%)	cv (%)	MC loss (%)	cv (%)	MC loss (%)	cv (%)
Beech	1.08	25.9	10.38	15.4	11.88	20.6
Linden	1.00	179.6	16.50	12.1	18.07	10.3
Oak	0.85	23.1	9.33	19.8	12.19	13.6
Spruce	0.87	24.1	15.81	14.9	20.61	15.3

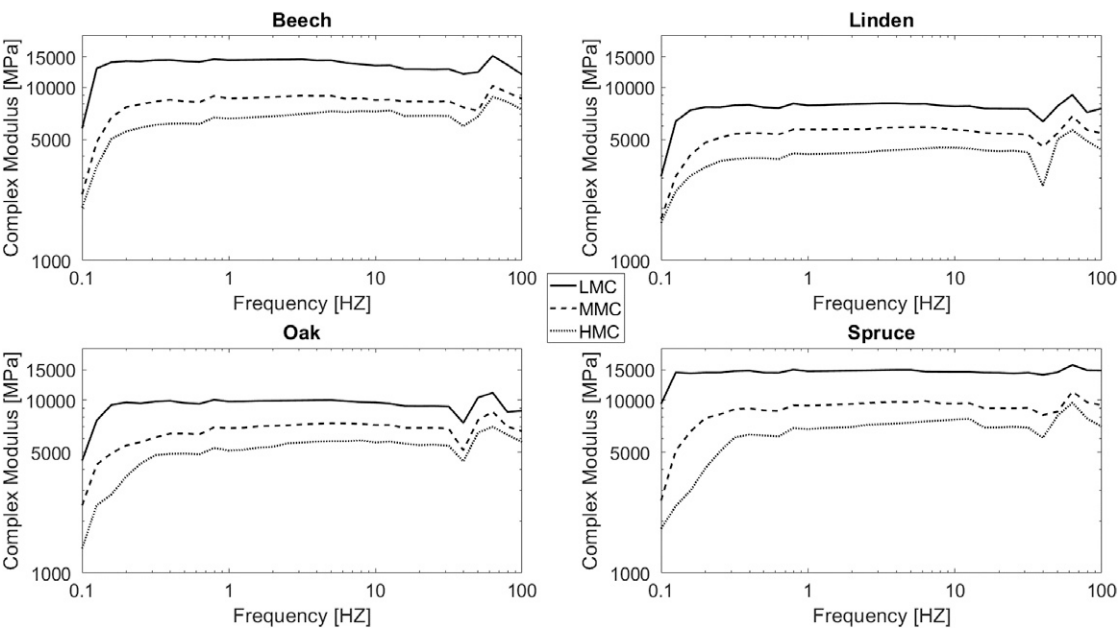


Figure 2. Medians of complex dynamic moduli of elasticity of species groups with different MC at varying load frequency—determined by dynamic mechanical analysis.

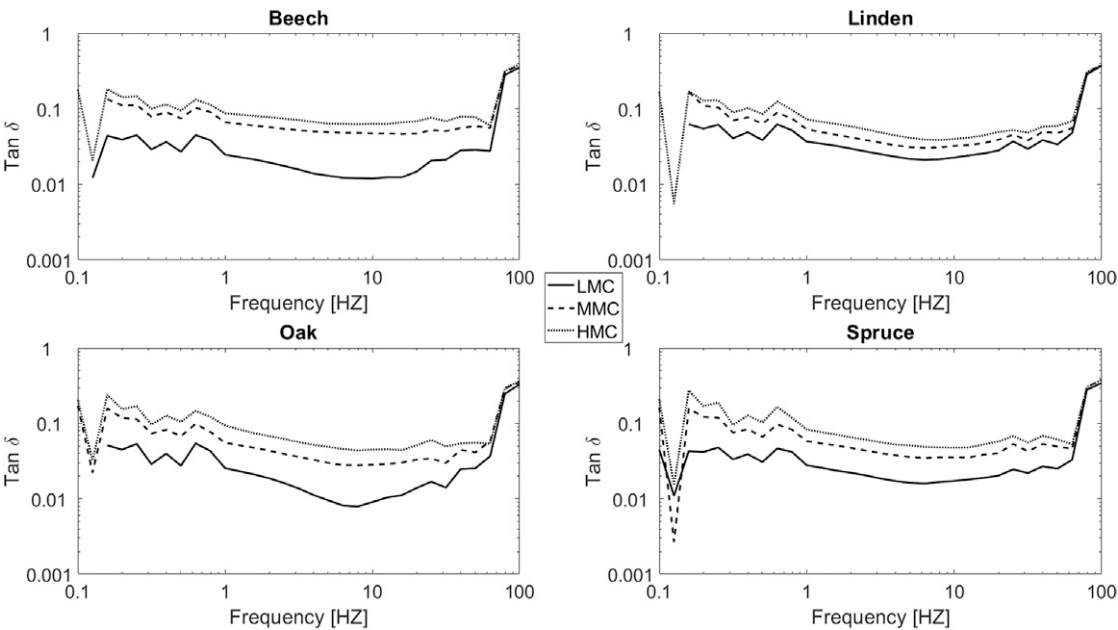


Figure 3. Medians of damping coefficients determined by dynamic mechanical analysis at varying load frequency.

Menard's (2006) description of the influence of the loading frequency on material behavior determined by DMA.

As expected for DMA above FSP, the damping coefficient increased with rising MC (Goken et al 2018). For individual species, the DMA revealed a rising damping coefficient for MMC and HMC compared with LMC by: 161% and 253% for beech ($\text{stp} = 11 \text{e}^{-4}$), 39% and 74% for linden ($\text{stp} = 4 \text{e}^{-4}$), 107% and 204% for oak ($\text{stp} = 9 \text{e}^{-4}$), and 95% and 167% for spruce ($\text{stp} = 6 \text{e}^{-4}$) respective. This could lead to the conclusion that with higher ρ_b , the damping coefficient $\tan \delta$ increases rapidly with increasing MC. Nevertheless, by focusing on individual species and their MC groups, no significant change was observed when the ρ_b changed. In case, there was any slight change observed, it was a decrease in the damping coefficient with increasing ρ_b (Fig 4). For the DMA, species with a higher ρ_b showed a faster increment of the damping coefficient with increasing MC. Brémaud et al (2009, 2010) stated that extractives content, chemical composition, and properties related to MC have a significant influence on the damping coefficient

$\tan \delta$. This confirms that the steepness of the damping coefficient increases with increasing MC and is influenced not only by ρ_b , but also by the properties of the species itself.

Since the results were comparable for FRT in individual MC groups before and after DMA tests, values are stated (Table 4) as the unified medians from the average of LMC, MMC, and HMC before and after performing the DMA.

The bending dynamic modulus of elasticity obtained by DMA was statistically comparable within the individual species for both groups above FSP (MMC and HMC) with ($p = 0.01$), beech even with ($p = 0.05$). No species had statistically comparable MOED_{DMA} of the LMC group with MMC and HMC groups. Similar results were obtained for MOED_{FRT} . Individual species had comparable MMC and HMC groups with ($p = 0.01$), beech and oak even with ($p = 0.05$). No species had statistically comparable MOED_{FRT} of the LMC group with groups above FSP. Both methods agreed with the assumption that there is no statistically significant change in MOED above FSP, with increasing

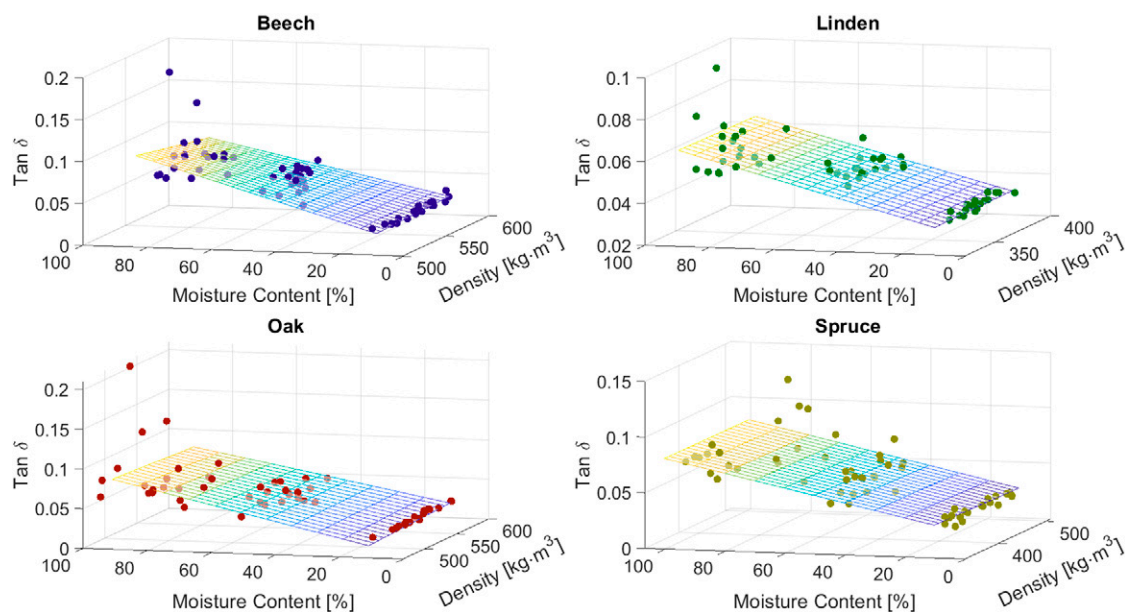


Figure 4. Increasing damping coefficient with increasing MC for individual species.

Table 4. Medians of MOED and tan δ for species with different MC.

Species	MC	FRT				DMA			
		MOED (MPa)	cv (%)	tan δ (–)	cv (%)	MOED (MPa)	cv (%)	tan δ (–)	cv (%)
Beech	LMC	17 297	8.9	0.0093	15.7	14 117	14.7	0.0227	12.9
	MMC	11 563	8.9	0.0206	30.6	8 448	14.2	0.0592	15.3
	HMC	10 721	12.6	0.0195	27.3	6 768	33.5	0.0800	35.5
Linden	LMC	10 222	9.9	0.0124	7.1	7 758	12.4	0.0339	5.3
	MMC	7 876	12.2	0.0162	16.2	5 616	12.1	0.0473	9.3
	HMC	6 467	17.0	0.0179	16.8	4 192	19.8	0.0589	17.7
Oak	LMC	12 088	19.9	0.0087	14.6	9 697	19.8	0.0222	10.1
	MMC	9 160	16.6	0.0137	24.8	7 048	15.4	0.0459	13.6
	HMC	7 976	17.5	0.0139	23.5	5 524	57.2	0.0676	49.5
Spruce	LMC	17 704	21.0	0.0073	12.5	14 595	24.0	0.0258	15.8
	MMC	11 541	10.9	0.0134	31.4	9 137	11.9	0.0502	16.2
	HMC	10 753	23.6	0.0148	31.2	6 935	42.4	0.0688	32.5

MC (Bucur 2006; Nocetti et al 2014). A comparison of MOED between both methods showed no statistical difference only for the spruce LMC group ($p = 0.01$), otherwise, the methods do not have comparable MOED within the species and their individual MC groups. Medians of $MOED_{DMA}$ were smaller than $MOED_{FRT}$, and were similar for all species, depending on the MC group: LMC of 18.4% for beech, 24.1% for linden, 19.8% for oak, and 17.6% for spruce; MMC of 26.9% for beech, 28.7% for linden, 23.1% for oak, and 20.8% for spruce; HMC of 36.9% for beech, 35.2% for linden, 30.7% for oak, and 35.5% for spruce. The data from both methods show that, as a function of MC, $MOED_{FRT}$, and $MOED_{DMA}$ copy a similar trend (Fig 5).

The damping coefficients obtained by DMA are higher than those obtained by FRT. Similar results were found in one of the few papers comparing DMA and FRT results of the damping coefficient tan δ on the mulberry wood with MC of 1.4% (Se Golpayegani et al 2012). Thus, these results for LMC groups were expected. Above FSP, a further increase was expected for DMA (Goken et al 2018), which again would be similarly steep for both methods (for FRT, it has not been described yet [Barré et al 2018]). This has been confirmed for DMA data only (Fig 6). The FRT results showed that above the FSP, the damping coefficients had similar trends as

dynamic moduli, ie that there were no statistically significant changes with increasing MC. For all species, the damping coefficients tan δ were comparable between their MMC and LMC groups ($p = 0.05$). For the damping coefficient obtained by DMA, MMC and HMC groups were comparable only for spruce ($p = 0.01$).

As the results indicate, the methods are very similar in determining MOED, just with shifted values (Fig 7). Through multiple linear regression from all data (using function “regress”—Matlab) the coefficient $c = 0.7567$ to convert $MOED_{FRT}$ to $MOED_{DMA}$, and $c = 1.2813$ to convert $MOED_{DMA}$ to $MOED_{FRT}$ were calculated. $MOED_{DMA}$ and $MOED_{FRT}$ had strong linear correlations ($r = 0.92$, $r^2 = 0.84$). This correlation is very similar to the correlation between static bending MOE and dynamic bending $MOED_{FRT}$ ($r = 0.98$, $r^2 = 0.96$) (Chauhan and Sethy 2016). The DMA operates at the boundary between statics and dynamics, with loading frequencies ranging from 0.1 to 100 Hz. As the static moduli have lower values than dynamic moduli, DMA corresponds to lower MOED values. A comparison of both methods for determining the damping coefficient tan δ showed a weaker correlation ($r = 0.57$, $r^2 = 0.32$). When comparing only the LMC groups (disregarding the influence of MC) the correlation coefficient remained almost unchanged. The correlation between MOED and the

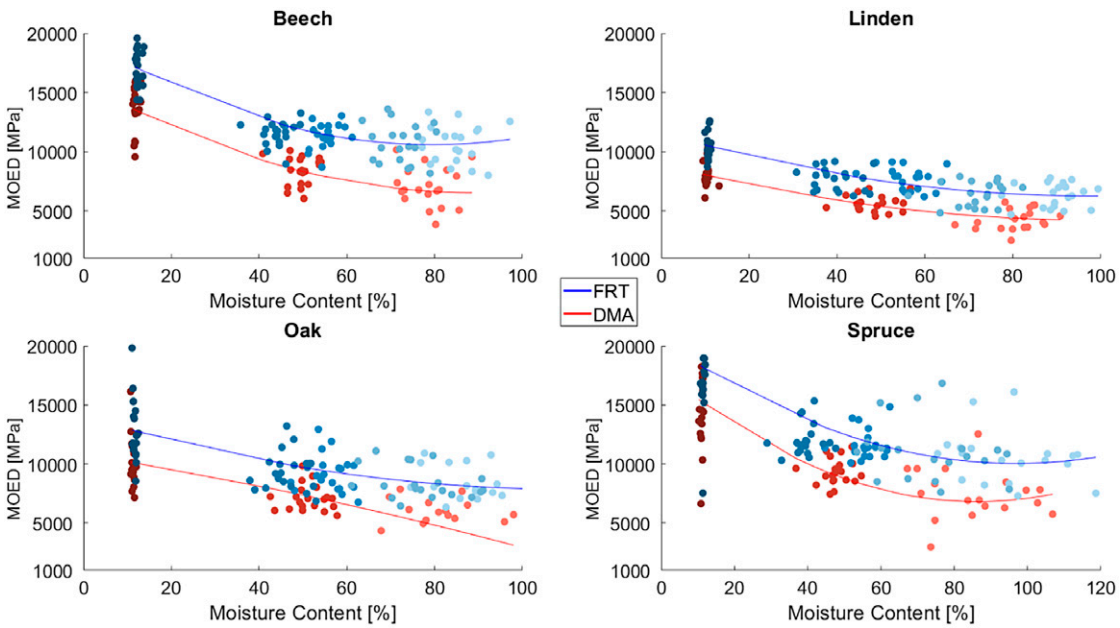


Figure 5. Comparison of bending dynamic modulus of elasticity (MOED) obtained by frequency resonance technique (FRT) and dynamic mechanical analysis (DMA).

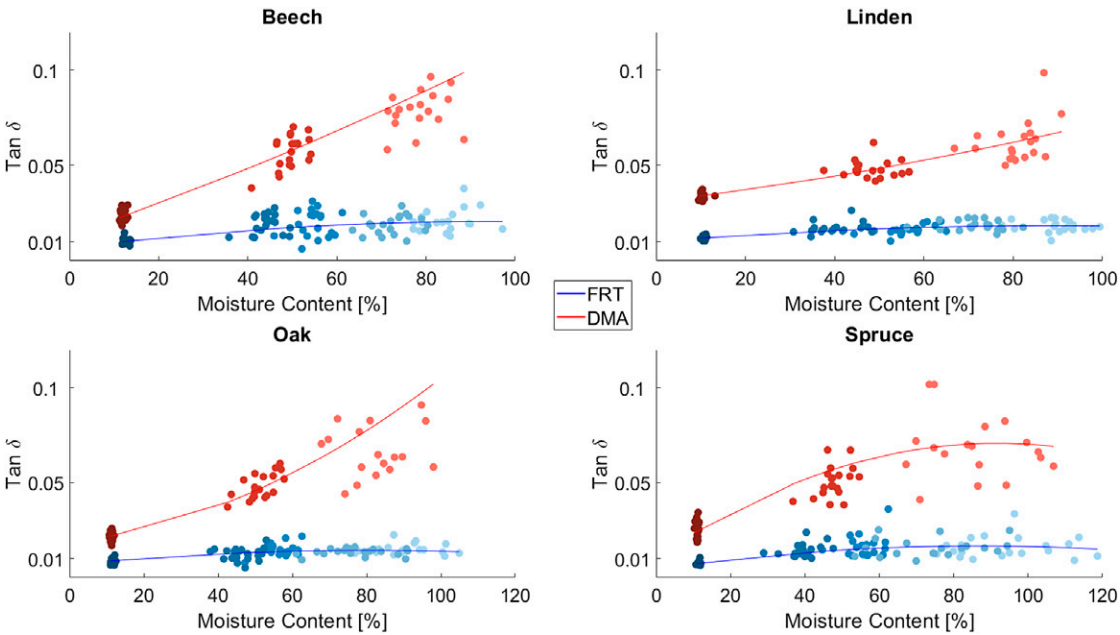


Figure 6. Comparison of damping coefficient obtained by frequency resonance technique (FRT) and dynamic mechanical analysis (DMA).

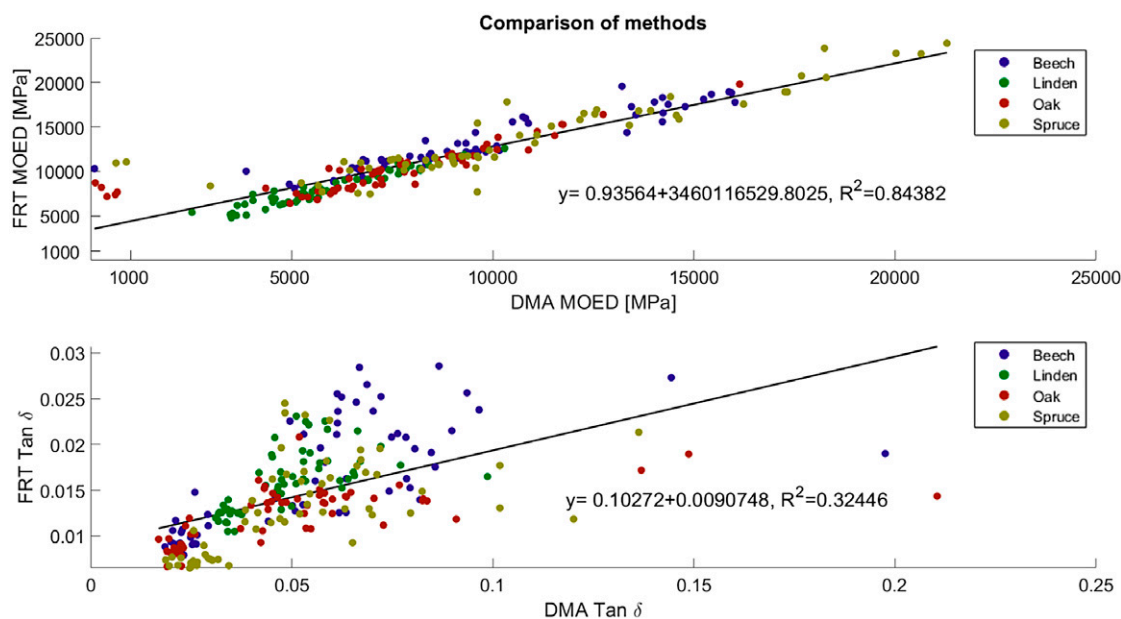


Figure 7. Comparison of dynamic parameters obtained by frequency resonance technique (FRT) and dynamic mechanical analysis (DMA).

damping coefficient $\tan \delta$ was moderate for FRT ($r = 0.42$) and stronger for DMA ($r = 0.65$).

CONCLUSIONS

- The DMA showed that the damping coefficient increased with increasing MC, and increased faster with increasing MC for species with a high ρ_b (oak, beech with $\text{stp} = 9\text{e}^{-4}$; 11e^{-4}) than for species with lower ρ_b (linden, spruce with $\text{stp} = 4\text{e}^{-4}$; 6e^{-4}).
- The FRT showed that the damping coefficient values do not change significantly above the FSP.
- Same-size specimens showed that MOED determined using FRT was 24–28% more than MOED determined using DMA, with a strong linear correlation ($r = 0.92$, $r^2 = 0.84$). The influence of the scale while maintaining the ratio of dimensions and furthermore, the influence of the change of geometry should be verified.
- For further research, it is necessary to focus in more detail on the damping coefficient $\tan \delta$ and the factors influencing it. Based on the

results of this paper, the damping coefficient $\tan \delta$ obtained by DMA and the damping coefficient $\tan \delta$ obtained by FRT should not be confused.

ACKNOWLEDGMENTS

This work was funded by the Internal Grant Schemes of Mendel University in Brno, registration no.: CZ.02.2.69/0.0/0.0/19_073/0016670, funded by the ESF.

The authors thank the University of Primorska and InnoRenew CoE for the loan of the measurement equipment.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- Ahmed SA, Adamopoulos S (2018) Acoustic properties of modified wood under different humid conditions and their

- relevance for musical instruments. *Appl Acoust* 140:92-99. <https://doi.org/10.1016/j.apacoust.2018.05.017>
- American Society for Testing and Materials, ASTM E1875 – 20 (2020). Standard test method for dynamic Young's modulus, shear modulus, and Poisson's ratio by sonic resonance. Philadelphia. <https://doi.org/10.1520/E1875-20>
- American Society for Testing and Materials, ASTM E1876 - 15 (2016). Standard test method for dynamic Young's modulus, shear modulus, and Poisson's ratio by impulse excitation of vibration. Philadelphia. <https://doi.org/10.1520/E1876-08.2>
- Ashduzzaman M, Hale MD, Ormondroyd GA, Spear MJ (2020) Dynamic mechanical analysis of Scots pine and three tropical hardwoods. *Int Wood Prod J* 11(4):189-203. <https://doi.org/10.1080/20426445.2020.1799910>
- Baar J, Tippner J, Gryc V (2016) Wood anatomy and acoustic properties of selected tropical hardwoods. *IAWA J* 37(1):69-83. <https://doi.org/10.1163/22941932-20160121>
- Barré JB, Bourrier F, Brancheriau L, Bertrand D, Rey F (2018) Effects of fungal decay on elasticity and damping of small-diameter silver fir logs assessed by the transverse vibration resonant method. *Wood Sci Technol* 52(2):403-420. <https://doi.org/10.1007/s00226-017-0961-2>
- Barrett JD, Hong JP (2010) Moisture content adjustments for dynamic modulus of elasticity of wood members. *Wood Sci Technol* 44(3):485-495. <https://doi.org/10.1007/s00226-009-0292-z>
- Brémaud I (2012) Acoustical properties of wood in string instruments soundboards and tuned idiophones: Biological and cultural diversity. *J Acoust Soc Am* 131(1):807-818. <https://doi.org/10.1121/1.3651233>
- Brémaud I, Gril J, Thibaut B (2011) Anisotropy of wood vibrational properties: Dependence on grain angle and review of literature data. *Wood Sci Technol* 45(4):735-754. <https://doi.org/10.1007/s00226-010-0393-8>
- Brémaud I, Minato K, Langbour P, Thibaut B (2010) Physico-chemical indicators of inter-specific variability in vibration damping of wood. *Ann For Sci* 67(7):707. <https://doi.org/10.1051/forest/2010032>
- Brémaud I, Minato K, Thibaut B (2009) Mechanical damping of wood as related to species classification: A preliminary survey. Pages 1–7 in *Proceedings, 6th Plant Biomechanics Conference Cayenne, November 16–21, Cayenne, French Guiana*. <https://doi.org/10.1007/978-3-319-32080-9>
- Bucur V (2006) Acoustics of wood. Theory of and Experimental Methods for the Acoustic Characterization of Wood. Pages 39–104 in *TE Timell, R Wimmer, eds CRC Press, Boca Raton, FL*. <https://doi.org/10.1201/9780203710128>
- Bucur V (2016) Handbook of materials for string musical instruments. *Mechanical Characterisation of Materials for String Instrument*. Pages 93–126. Springer International Publishing, Switzerland. <https://doi.org/10.1007/978-3-319-32080-9>
- Chauhan S, Sethy A (2016) Differences in dynamic modulus of elasticity determined by three vibration methods and their relationship with static modulus of elasticity. *Maderas Cienc Tecnol* 18(2):373-382. <https://doi.org/10.4067/S0718-221X2016005000034>
- Chowdhury S, Fabiyi J, Frazier CE (2010) Advancing the dynamic mechanical analysis of biomass: Comparison of tensile-torsion and compressive-torsion wood DMA. *Holzforschung* 64(6):747-756. <https://doi.org/10.1515/HF.2010.123>
- Cristini V, Tippner J, Nop P, Zlámál J, Hassan Vand M, Šeda V (2022) Degradation of beech wood by *Kretzschmaria deusta*: Its heterogeneity and influence on dynamic and static bending properties. *Holzforschung* 76(9):813-824. <https://doi.org/10.1515/hf-2022-0039>
- Danihelová A, Spišiak D, Reinprecht L, Gergel T, Vidholdov Z, Ondrejka V (2019) Acoustic properties of Norway spruce wood modified with staining fungus (*Sydowia polyspora*). *BioResources* 14(2):3432-3444. <https://doi.org/10.15376/biores.14.2.3432-3444>
- Fernández-Serrano Á, Villasante A (2021) Longitudinal, transverse and ultrasound vibration for the prediction of stiffness using models incorporating features in *Pinus sylvestris* timber. *Eur J Wood Wood Prod* 79(6):1541-1550. <https://doi.org/10.1007/s00107-021-01707-0>
- Goken J, Fayed S, Schafer H, Enzenauer J (2018) A study on the correlation between wood moisture and the damping behaviour of the tonewood spruce. *Acta Phys Pol A* 133(5):1241-1260. <https://doi.org/10.12693/APhysPolA.133.1241>
- Hamdan S, Rahman MR, Jusoh I, Wahid HA (2018) Dynamic Young's modulus and moisture content of tropical wood species across Sap, median, and internal wood regions. *BioResources* 13(2):2907-2915. <https://doi.org/10.15376/biores.13.2.2907-2915>
- Hamdan S, Talib ZA, Rahman MR, Ahmed AS, Islam MS (2010) Dynamic Young's modulus measurement of treated and post-treated tropical wood polymer composites (WPC). *BioResources* 5(1):324-342.
- Hassan KTS, Horáček P, Tippner J (2013) Evaluation of stiffness and strength of scots pine wood using resonance frequency and ultrasonic techniques. *BioResources* 8(2):1634-1645. <https://doi.org/10.15376/biores.8.2.1634-1645>
- Hein PRG, Brancheriau L (2018) Comparison between three-point and four-point flexural tests to determine wood strength of Eucalyptus specimens. *Maderas Cienc Tecnol* 20(3):333-342. <https://doi.org/10.4067/S0718-221X20180050003401>
- Hossen MF, Hamdan S, Rahman MR (2018) Investigation of the acoustic properties of chemically impregnated Kayu Malam wood used for musical instrument. *Adv Mater Sci Eng* 2018. <https://doi.org/10.1155/2018/7829613>
- Kaboorani A, Blanchet P (2014) Determining the linear viscoelastic region of sugar maple wood by dynamic

- mechanical analysis. *BioResources* 9(3):4392-4409. <https://doi.org/10.15376/biores.9.3.4392-4409>
- Karlinasary L, Wahyuna ME, Nugroho N (2008) Non-destructive ultrasonic testing method for determining bending strength properties of Gmelina wood (*Gmelina arborea*). *J Trop For Sci* 20(2):1-12.
- Lu J, Jiang J, Wu Y, Li X, Cai Z (2012) Effect of moisture sorption state on vibrational properties of wood. *Forest Prod Journal* 62(3):171-176. <https://doi.org/10.13073/0015-7473-62.3.171>
- Matsunaga M, Obataya E, Minato K, Nakatsubo F (2000) Working mechanism of adsorbed water on the vibrational properties of wood impregnated with extractives of pernambuco (*Guilandina echinata* Spreng.). *J Wood Sci* 46(2):122-129. <https://doi.org/10.1007/BF00777358>
- Menard KP (2006) Dynamic mechanical analysis, 2nd edition. CRC Press, Boca Raton, FL. https://doi.org/10.1007/978-3-540-29805-2_1224
- Mora CR, Schimleck LR, Mahon JM, Isik F, Mahon JM, Clark A, Daniels RF (2009) Relationships between acoustic variables and different measures of stiffness in standing *Pinus taeda* trees. *Can J For Res* 39(8):1421-1429. <https://doi.org/10.1139/X09-062>
- Nocetti M, Brunetti M, Bacher M (2014) Effect of moisture content on the flexural properties and dynamic modulus of elasticity of dimension chestnut timber. *Eur J Wood Wood Prod* 73(1):51-60. <https://doi.org/10.1007/s00107-014-0861-1>
- Papandrea SF, Cataldo MF, Bernardi B, Zimbalatti G, Proto AR (2022) The predictive accuracy of modulus of elasticity (MOE) in the wood of standing trees and logs. *Forests* 13(8):1273. <https://doi.org/10.3390/f13081273>
- Peng Y, Shi SQ, Kim MG (2008) Effect of temperature on the dynamic mechanical properties of resin film and wood. *For Prod J* 58(12):68-74.
- Pizzo B, Pecoraro E, Lazzeri S (2018) Dynamic mechanical analysis (DMA) of waterlogged archaeological wood at room temperature. *Holzforschung* 72(5):421-431. <https://doi.org/10.1515/hf-2017-0114>
- Salmén L, Stevanic JS, Olsson AM (2016) Contribution of lignin to the strength properties in wood fibres studied by dynamic FTIR spectroscopy and dynamic mechanical analysis (DMA). *Holzforschung* 70(12):1155-1163. <https://doi.org/10.1515/hf-2016-0050>
- Se Golpayegani A, Brémaud I, Gril J, Thevenon MF, Arnould O, Pourtahmasi K (2012) Effect of extractions on dynamic mechanical properties of white mulberry (*Morus alba*). *J Wood Sci* 58(2):153-162. <https://doi.org/10.1007/s10086-011-1225-7>
- Teles RF, De Souza MR, Wimmer P (2018) Acoustical properties of 10 Amazonian hardwoods. Pages 1675-1681 in *Proceedings 25th International Congress on Sound and Vibration 2018, ICSV 2018: Hiroshima Calling, July 3, 2018, Hiroshima, Japan*.
- Tronchin L, Manfren M, Vodola V (2020) Sound characterization through intensity of acoustic radiation measurement: A study of Persian musical instruments. *Appl Sci* 10(2):633. <https://doi.org/10.3390/app10020633>
- Wang X, Ross RJ, McClellan M, Barbour RJ, Erickson JR, Forsman JW, McGinnis GD (2001) Nondestructive evaluation of standing trees with a stress wave method. *Wood Fiber Sci* 33(4):522-533.
- Wang Z, Li L, Gong M (2012) Measurement of dynamic modulus of elasticity and damping ratio of wood-based composites using the cantilever beam vibration technique. *Constr Build Mater* 28(1):831-834. <https://doi.org/10.1016/j.conbuildmat.2011.09.001>
- Wegst UGK (2006) Wood for sound. *Am J Bot* 93(10):1439-1448. <https://doi.org/10.3732/ajb.93.10.1439>
- Yang JL, Ilic J, Wardlaw T (2003) Relationships between static and dynamic modulus of elasticity for a mixture of clear and decayed eucalypt wood. *Aust For* 66(3):193-196. <https://doi.org/10.1080/00049158.2003.10674911>

Errata

Nop, P., Cristini, V., Zlámál, J., Vand, M. H., Šeda, V., and Tippner, J. Dynamic Properties of Wood Obtained by Frequency Resonance Technique and Dynamic Mechanical Analysis. *Wood and Fiber Science*, 55(2), 2023; pp. 131-142. (<https://doi.org/10.22382/wfs-2023-12>):

Equation 1 contains a typographical error (l^2 instead of l^3). The correct formula used for the $MOED_{FRT}$ calculation is:

$$MOED_{FRT} = \left(\frac{2f}{2.25\pi} \right)^2 \frac{ml^3}{I}$$

***Corresponding author:** Patrik Nop, Department of Wood Science and Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Zemědělská 1, 61300 Brno, Czech Republic, E-mail: patrik.nop@mendelu.cz

Valentino Cristini, Jan Zlámál, Mojtaba Hassan Vand, Vít Šeda, Jan Tippner, Department of Wood Science and Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Zemědělská 1, 61300 Brno, Czech Republic

Paper IV
**Assessment of the Impact Strength Properties of Thermally
Modified Wood by Non-Destructive Testing**

Mojtaba Hassan Vand, Patrik Nop, Jan Tippner

Journal of Nondestructive Evaluation

<https://doi.org/10.1007/s10921-025-01313-7>

Nop, P.: Methodology, Investigation, Formal analysis, Writing – review & editing



Assessment of the Impact Strength Properties of Thermally Modified Wood by Non-Destructive Testing

Mojtaba Hassan Vand¹ · Patrik Nop¹ · Jan Tippner¹

Received: 27 February 2025 / Accepted: 8 December 2025 / Published online: 22 December 2025
© The Author(s) 2025

Abstract

This article examines the effectiveness of non-destructive testing (NDT) in assessing wood under impact loadings. Our research was to evaluate the feasibility of using the frequency resonance technique (FRT), to predict the behaviour under impact of thermally modified timber (TMT) compared with a control sample of untreated wood. Wooden planks from five different species were subjected to a thermal modification process (TMP) under two different regimes. Both the TMT and control samples were evaluated using NDT to measure their dynamic modulus of elasticity (MOED), logarithmic decrement of damping (LDD) and acoustic conversion efficiency (ACE). Subsequently, wood samples from the same species were tested using drop-weight impact tests to measure their inflicted maximum force and impact bending strength (IBS), while high-speed cameras recorded the impacts to measure the maximum deflection of the specimens. The results revealed that the only relatively efficient prediction of FRT was the relationship between MOED and IBS. The ACE and LDD results did not show any acceptable correlations with impact tests, indicating that NDT is not reliable for assessing maximum force and deflection in the wood species under impact. Our study also found that the efficiency of the results and predictions were influenced by the wood species and the TMP conditions, necessitating a large number of samples for each species and heat modification temperature to achieve accurate NDT results. Our study found that the efficiency of NDT predictions was significantly influenced by both wood species and the TMP conditions. Specifically, oak showed a relatively higher coefficient of determination, while ash had the lowest. The thermal treatment also had a varied effect on NDT's ability to determine IBS, increasing its efficiency for larch specimens while decreasing it for ash and beech, with no significant effect on oak and spruce. These findings imply that future NDT methodologies must be developed with a species-specific approach and calibrated for each unique modification condition. Consequently, achieving accurate NDT results will require comprehensive data sets with a large number of samples for each species and heat modification temperature.

Keywords Wood · Thermal modification · Non-destructive test · Impact test · Frequency resonance technique · Digital image correlation method

1 Introduction

Wood modification describes all the various methods, processes and operations employed to improve the performance of wood. One such method is thermal modification [1]. In the thermal modification process (TMP), applicable to a wide range of species, wood is exposed to temperatures above 160 °C in the absence of oxygen, which is replaced with steam, nitrogen or oil [2]; the process alters the composition of the wood's cell walls and its physical properties. Thermal modification improves the dimensional stability, water vapour sorption and biohazard resistance making the wood more durable against environmental effects [3]. TMP also increases resistance to weathering agents such as

✉ Mojtaba Hassan Vand
xhassanv@mendelu.cz

Patrik Nop
patrik.nop@mendelu.cz

Jan Tippner
jan.tippner@mendelu.cz

¹ Department of Wood Science and Technology, Faculty of Forestry and Wood Technology, Mendel University in Brno, Zemědělská 3, 613 00 Brno, Czech Republic

sunlight, moisture gradients, temperature changes, chemical agents, atmospheric pollution and abrasion from windblown particles [4]. Despite its advantages, TMP has several detrimental effects on wood, including damage to its tensile, bending and impact strength; it also intensifies the negative effects of knots and defects in the wood [5]. TMP changes the wood's chemical properties, drastically reducing the hemicellulose content and relatively doesn't decrease the content of cellulose, lignin and extractives. It also causes de-acetylation and forms new carbonyl and carboxyl groups due to oxidation [6]. The results of TMP depend on several variables, such as its duration, temperature, treatment atmosphere, wood species, moisture content before TMP and the wood dimensions; among these parameters, the temperature and treatment time are the most critical [7]. The severity of the negative effect of TMP on mechanical properties at lower temperatures (up to around 160 °C) is minimal [8], but as the temperature increases, the mechanical properties decrease significantly, the effect becoming catastrophic at 200 °C [9].

Non-destructive testing (NDT) using dynamic loading can be employed to determine the mechanical properties of wood such as its Young's modulus of elasticity (E), bending modulus of elasticity (MOE) and shear modulus of elasticity (G) [10–12]. Using this method, it has proved possible to detect defects inside wood [13] and can determine the lifespan of timber poles by identifying internal damage without the need for sampling or service interruptions [14]. This approach is suitable for materials with limited availability, such as modified materials, as it allows the same specimens to be tested at various stages of modification [15] or even during the modification process [16, 17] as well as under different types of loading (axial, bending, shear) [18]. Anisotropic materials can be tested in multiple structural-anatomical directions simultaneously [19]. One commonly used method is the frequency resonance technique (FRT), in which the specimen is vibrated and the natural frequencies of individual vibration modes (longitudinal, flexural, torsional) are determined [18, 20, 21]. Based on these frequencies, along with geometry and density, the dynamic moduli of elasticity (Edyn, MOED, Gdyn) are calculated, which show a strong correlation with the static moduli of elasticity [22, 23].

Another parameter that can be determined non-destructively using FRT is internal friction, represented by a damping coefficient such as the logarithmic decrement of damping (LDD) [11, 12, 24, 25]. Although elastic moduli and damping coefficients are generally related [21, 26], their relationship is not easily determinable since the damping coefficient exhibits high variability and is significantly influenced by micro-structural modifications such as heat treatment, chemical treatment, dynamic loading of the specimen, temperature, moisture content and extractive content [27, 28]. Since the damping coefficient indicates the material's

ability to absorb dynamic energy, determining the relationship between LDD and impact strength is a key aspect of this study. When using FRT, the summary parameter called the acoustic conversion efficiency (ACE) is often also determined, expressing the ratio of stiffness, density and the damping coefficient in one variable [29–31].

Various impact testing devices have been proposed over the years for assessing the impact behaviour of different materials under loadings with different loading rates in shape of tensile, compression, shear, punch and bending tests [32]. Compared with other materials, wood can absorb tremendous amounts of energy inflicted during an impact loading; another advantage is its damage tolerance by crack isolation and the weak interface between fibres [33]. In assessing the efficiency of materials under impact loadings, impact bending strength (IBS) shows the ability to absorb and dissipate energy through impact loading. IBS refers to the amount of work required to break the wood under given conditions divided by the cross-section area [34]. Some studies have showed that the absorption of energy is directly related to the amount of inflicted energy [35], while others have found their values irrelevant [36]. Generally, the IBS of wood increases with its density while it decreases with the width of rings [37]. The angle of the grain can affect the IBS of wood by affecting its crack initiation strength [38]. In general, a wide range of material and environmental properties can affect the impact behavior of wood [39, 40].

TMP significantly reduces two key mechanical properties of wood: its resistance to bending under static (modulus of rupture) and dynamic (IBS) loadings. The degree of reduction varies with the wood species and the specific conditions of the treatment process [41]. Hemicellulose acts as a bonding agent between the wood's fibres, giving an overall integrity to the material, thereby enhancing its resistance to sudden shocks and impacts. Since TMP deteriorates hemicellulose much more than other materials in the wood, it makes the wood weaker against impact loadings [42]. In some cases, the dynamic bending strength can decrease by half, showing the devastating effect of TMP on the IBS [43]. Some studies have stated that IBS is the mechanical property most affected by TMP [44]. However, the severity of IBS deterioration strongly depends on the temperature; at lower temperatures it is much less than at higher temperatures (in some cases it even increases) [45–48]. The effect of temperature on the impact strength of wood is not only at the high temperatures of TMP; much lower temperatures can also affect the impact strength of wood [49]. Some studies have stated that IBS has no clear relation to MOED and whether they are related or not is defined by the species [50], however, studies showed that the NDT effectively can detect the defects around the impacted zone of wooden composites [51]. TMP decreases the deflection, maximum normal strain and IBS of the TMT and this decrease is intensified by

increasing the temperature of the TMP [52]. Table 1 presents a list of the selected sources of the literature review about on the key studies that are most relevant to this research—especially those related to non-destructive testing (NDT), impact loading, and thermally modified timber (TMP).

The effect of TMP on the mechanical properties of wood, particularly related to impact loadings such as IBS, is significant. For this study we aimed to determine if NDT methods could accurately predict the impact-related mechanical properties of wood. Specifically, we investigated which parameters could be predicted using NDT, which could not, and the accuracy of these predictions. We also examined how different wood species and types of heat modification influenced the efficiency of NDT in assessing TMT. Our study provides a clear and comprehensive view of the effectiveness of NDT in evaluating various aspects of the impact-related behaviour of TMT from different species under different TMP.

2 Materials and Methods

2.1 Thermal Modification and Preparation of the Specimens

Air-dried boards with a thickness of 5 cm were obtained from various trees of five species within enterprises of the South Moravia region, Czech Republic. All stages of sample preparation and testing were conducted at the Josef Ressel Research Centre, affiliated with Mendel University in Brno, Czech Republic. The selected species were ash (*Fraxinus excelsior* L.), oak (*Quercus robur* L.), beech (*Fagus sylvatica* L.), larch (*Larix decidua* L.) and spruce (*Picea abies* L.).

The boards from each species were placed into three groups: Group N—the unmodified control group; Group L—subjected to thermal modification at a final temperature of 180 °C for two hours; and Group H—subjected to thermal modification at a final temperature of 220 °C for two hours. The TMP was done in a superheated steam environment at atmospheric pressure in a small-scale laboratory closed system heat modification chamber (Katres s.r.o., Czech Republic). The steps of the thermal modification process are outlined below:

- A. Initially, the boards were heated for around 3.5 h at a rate of 10 °C/h up to 103 °C to reduce the moisture content to zero.
- B. Following drying, the weight of each group was immediately measured, without allowing for cooling, to determine the dry weight of the unmodified material, which is essential for determining the mass loss.
- C. The material underwent monitored controlled re-heating at a rate of 30 °C/h for around 3.5 h to reach the maxi-

mum temperature (180 and 220 °C), with cautions to prevent combustion within the chamber.

- D. The thermal-modification temperatures (180 or 220 °C) were maintained for two hours, after which the materials were gradually cooled to around 100 °C over approximately two hours and the risk of combustion dissipated.
- E. The weight of the planks was once again measured to get the dry weight of the TMT for determination of the mass loss.

The mass loss serves as a valuable criterion for evaluating the effectiveness of TMP, and was measured by calculating the difference in mass of the materials before and after undergoing the thermal modification process.

Rectangular beam test specimens (20 × 20 × 300 mm) were cut from the boards, ensuring they were specially orthotropic. These specimens underwent thorough inspection, and those exhibiting visible defects – mainly knots and cracks (partially attributed to thermal modification) – were discarded. Finally, 318 samples were selected as test specimens and were placed in an environmental room set at 20 °C and 65% relative humidity. Table 2 shows the number of specimens for each group. The equilibrium moisture content of the specimens was monitored until stability was reached, confirmed through periodic weighing of the specimens. Figure 1 depicts the thermal modification chamber before and after the thermal modification.

2.2 Non-destructive Tests

The frequency resonance technique (FRT) was used in compliance with ASTM standards E1875 and E1876 to determine the dynamic modulus of elasticity (MOED). Vibration excitation was applied at the centre of each sample using a rubber hammer while it was positioned on foam supports at the nodes of the first bending mode, which corresponded to 22.4% and 77.6% of the specimen length. Each specimen received ten hits in the tested direction in both longitudinal-tangential (LT) and longitudinal-radial (LR) bending directions. Vibrations of the specimens were detected at the centre of the specimen using a laser vibrometer (PDV-100, Polytec, Inc., Baden-Württemberg, Germany) with signal recording done using the dynamic signal acquisition module DEWE-41-T-DSA and DEWESoft software (DEWETRON Inc., Grambach, Austria) at a sampling frequency of 20 kHz. Resonant frequency was determined by analysing the entire ten-hit signal using fast Fourier transform in MATLAB (The MathWorks, Inc., Natick, USA). Figure 2 shows the setting of the test.

The $MOED_{LT}$ and $MOED_{LR}$ (for LT and LR direction of bending) was then calculated from the frequency of the first bending mode, as per Eq. 1.

Table 1 Summary of key literature on NDT and thermally modified wood

Author(s) and Year	Wood Species	Treatment(s)	Main Method(s)	Key Findings
Boonstra et al. 2007	Softwood species	Thermal modification	Static and dynamic tests	TMP leads to a reduction in the strength properties of softwoods due to changes in wood's structure
Menezzi et al. 2014	Tropical species	Thermal modification	Non-destructive testing (NDT)	All evaluated properties of TMT could be modeled at a reasonable level
Kutnar et al. 2013	Norway spruce and Beech	Thermal modification	Non-destructive testing (NDT)	DMA can be used to TMP and simultaneously provide data about the modification process
Hassan et al. 2013	Scots pine	No modification	Non-destructive testing (NDT)	All three studied dynamic MOE methods correlated well with the static MOES and MOR
Viala et al. 2020	Norway spruce	No modification	Non-destructive testing (NDT)	The study found that material anisotropy decreases as the wood grade decreases
Nop et al. 2024	Beech and Oak	No modification	Non-destructive testing (NDT)	The beech generally has higher values for MOED, MOE, and Aw
Fernández-Serrano and Vil-lasante 2021	Scots Pine	No modification	Non-destructive testing (NDT)	The vibration techniques showed a better MOES prediction than the ultrasound techniques
Chauhan and Sethy 2016	Tropical species	No modification	Non-destructive testing (NDT)	All three dynamic methods showed a near-perfect correlation with the MOE, making them suitable for prediction
Mania and Skrodzka 2020	Norway spruce	Thermal modification	Non-destructive testing (NDT)	The treatment decreased the logarithmic decrement in the highest quality spruce wood
Brémaud et al. 2010	Different species	No modification	Non-destructive testing (NDT)	By combining data on a species' specific modulus of elasticity, moisture-related properties, and color, it's possible to create simple predictive models
Brémaud et al. 2009	Different species	No modification	Non-destructive testing (NDT)	morphological and biochemical taxonomic markers can affect dynamic mechanical properties of wood
Baar et al. 2016	Tropical species	No modification	Non-destructive testing (NDT)	woods with longer fibers and more slender rays tend to have better acoustic properties
Gašparík et al. 2016	Beech and aspen	Thermal densification	Impact test and hardness test	Wood densification did not have a significant effect on IBS
Yamada et al. 2018	Japanese Cedar	No modification	Impact test	Determination of methods for improvement of roof panels against impacts
Bučar and Merhar 2015	Norway Spruce	No modification	Impact test	The accelerometer-based method is a comparable and effective way to determine impact bending strength

Table 1 (continued)

Author(s) and Year	Wood Species	Treatment(s)	Main Method(s)	Key Findings
Hassan Vand and Tippner 2024	Scots Pine	No modification	Impact test	At high moisture content, both sapwood and heartwood of scots pine are equally strong
Leijten 2004	Different species	Thermal modification	Impact test	The TMP affects toughness
Borůvka et al. 2018	Beech and Birch	Thermal modification	Static and dynamic tests	Comparing the effect of TMP on beech in comparison to birch
Rautkari et al. 2014	Scots Pine	Thermal densification	Static and dynamic tests	Measurement of effect of heartwood vs sapwood in pine TMP
Gaff et al. 2019	Oak and spruce	Thermal modification	Chemical analysis and impact tests	Correlation of IBS to chemical change of TMTs
Bragov et al. 2022	Aspen	Temperature elevation	Compressive impact test	the effect of elevated temperature on strength and deformation properties of aspen is estimated
Hassan Vand and Tippner 2023	different species	Thermal modification	Impact test	The duration of modification can affect toughness of wood

Table 2 The number of specimens for each groups

Number	Ash	Beech	Larch	Oak	Spruce	Treatments
Group N	28	20	29	20	22	No Treatment
Group L	21	21	30	8	29	Thermal Modification at 180 °C for 2 h
Group H	20	19	15	17	19	Thermal Modification at 200 °C for 2 h

Fig. 1 The thermal modification chamber before and after the thermal modification

$$MOED = \frac{4\pi^2 f^2 L^4 \rho}{\beta^4 i^2}$$

(1) where f is frequency of the first bending mode, L is the length of the specimen, ρ is the density, β is the constant corresponding to the first bending mode ($\beta = 4.73$) and i is



Fig. 2 Vibration sensing setup of a freely supported specimen, excited by a rubber hammer

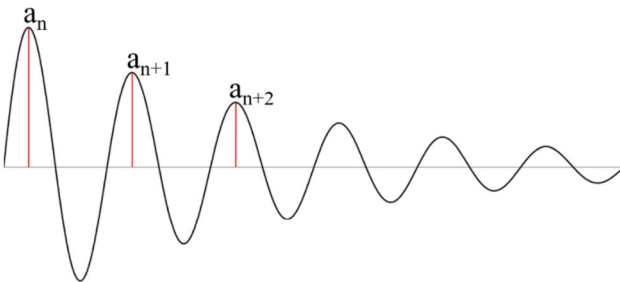


Fig. 3 Amplitudes of successive vibration cycles, used for LDD calculation

the radius of gyration which, for a specimen with rectangular cross-section, can be calculated using Eq. 2.

$$i^2 = \frac{h^2}{12} \quad (2)$$

where h is the side of the square. For each hit, both the maximum amplitude and the amplitude closest to 5% of the maximum were identified. Amplitudes of individual oscillation cycles within this range were identified (Fig. 1), and their LDDs were computed as the logarithm of the ratio of two successive amplitudes, following Eq. 3. The LDD for one hit was determined as the median of these LDDs. The specimen's resulting LDD was calculated as the median of the LDDs from ten hits (Fig. 3).

$$LDD_1 = \ln \frac{a_n}{a_{n+1}} \quad LDD_2 = \ln \frac{a_{n+1}}{a_{n+2}} \quad LDD_m = \ln \frac{a_{n+m-1}}{a_{n+m}} \quad (3)$$

The ACE was determined by using the ratio of stiffness, density and the damping coefficient in one variable by Eq. 4 [29].

$$ACE = \frac{\pi \sqrt{\frac{MOED}{\rho^3}}}{LDD} \quad (4)$$

2.3 Impact Tests

A DPfest 400 impact testing machine (Labortech s.r.o., Czechia) was used for the three-point impact bending test in compliance with standards ČSN 490115 (1979) and ČSN 490117 (1980). The tests were done at room temperature (20 °C). The testing machine's 4.55 kg hammer was released from a height of 815.7 mm, reaching a velocity of 4 m/s at the point of impact with the longitudinal centre of the specimens, thereby delivering 36.4 J of energy. The position and reaction force on the hammer were measured with an accuracy of 0.01 mm and a sample frequency of 1 MHz. The testing machine recorded both the maximum reaction force and the IBS of the specimens. It is important to note that some samples of unmodified ash and beech did not break, thus the definition of IBS may not exactly align with the generally accepted definition. In such cases, the total energy absorbed by the specimens without failure was considered as their IBS.

To record the results of the impact tests, a high-speed camera FASTCAM SA-X2 1000K-M2 (Photron Cameras, Japan) was used to capture images at a resolution of $1,024 \times 512$ pixels in grey scale, with a frame rate of 50,000 frames per second. Additionally, a Micro-Nikkor G lens (Nikon, Japan), two high-speed MultiLED QT equipment and two teleconverters (Nikon, Japan) were integrated into the set-up. The camera was positioned perpendicular to the surface of the specimens. The specimens were painted with a grey-scaled random speckled pattern, with the aim of increasing the contrast between points on the surface of the specimens and consequently enhance the efficiency of the digital image correlation (DIC) method. Figure 4 presents a representative image recorded by the camera following crack initiation. Figure 5 depicts a schematic diagram of the DIC software's process for measuring sample deflection, with a specific example shown at a deflection of 6 mm.

The recorded images were processed using Vic-2D v. 2010 DIC software (Correlated Solutions Inc., USA). A 1D-scale calibration method was employed to determine the conversion factor. A simple scale calibration was conducted to determine the conversion factor. The strain tensor was computed using the Lagrange notation. The displacement field and strain filter size were set at 3×3 points and 5×5 points, respectively. Through DIC analysis, the software successfully determined the deflection of the beams up to crack initiation, allowing the measurement of each specimen's flexibility.

3 Results and Discussion

Sample measurements and the results of the non-destructive and impact tests are presented in Table 3, showing median values alongside their standard deviations for all groups.

Fig. 4 An output recorded image of the impact

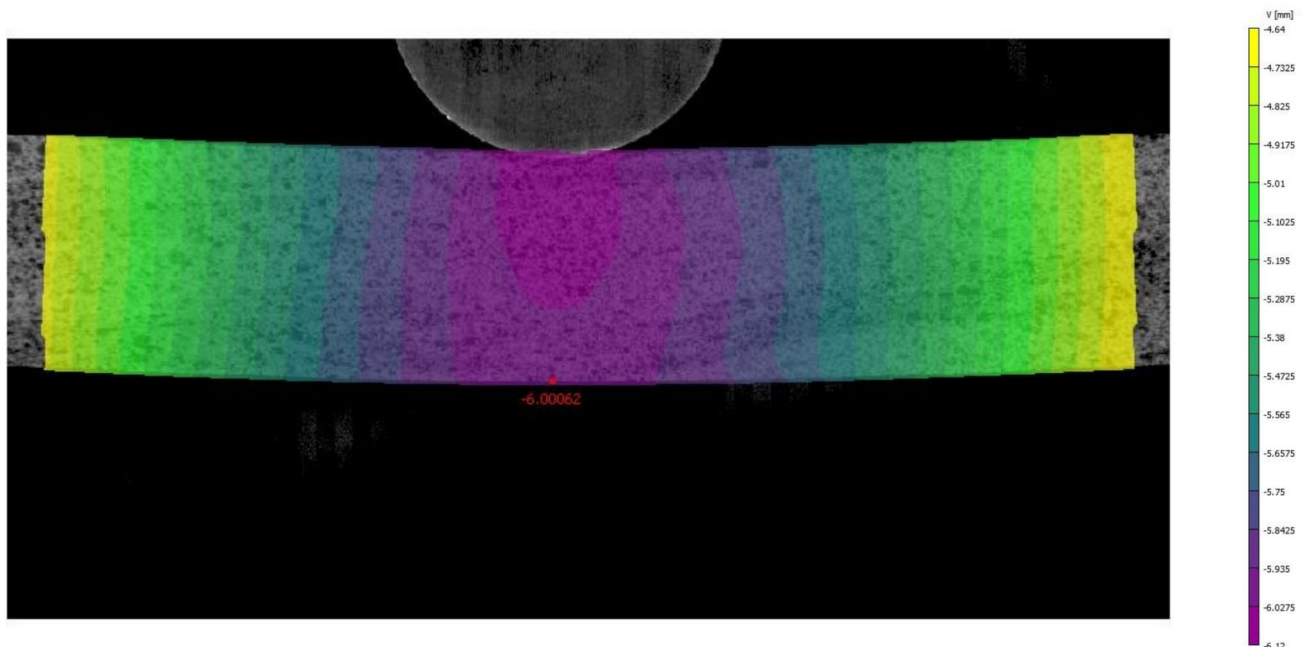
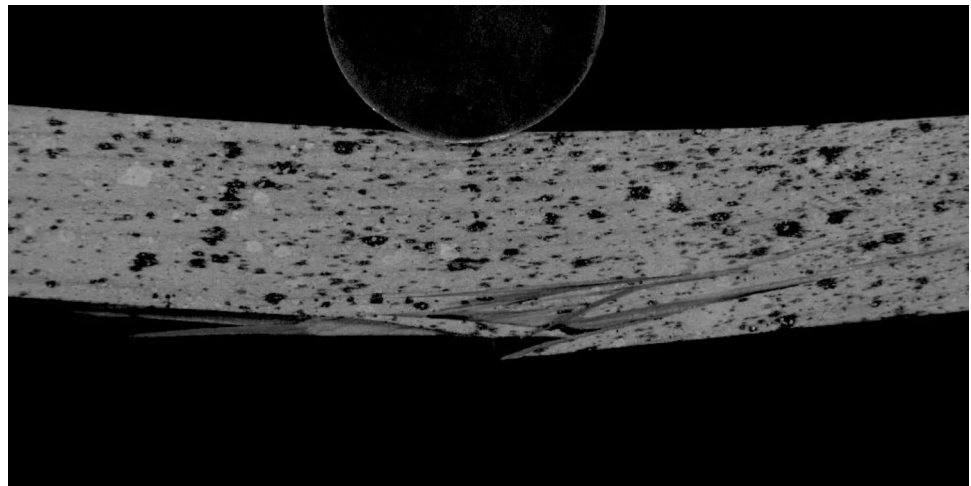


Fig. 5 A schematic of DIC software's process for measuring deflection

The impact bending strength (IBS) results from this study show both similarities and differences when compared to previous research on thermally modified wood. For spruce, the 15% decrease in IBS for the L-group in our study is significantly better than the 57% decrease reported by Boonstra et al. (2007) for the same temperature but a longer treatment duration (6 h). The absolute IBS values of the modified spruce between our study and theirs have 27% difference. The IBS of non-treated spruce in our research differed by only 13% from the value reported by Bučar and Merhar (2015), indicating good agreement for the baseline property. When comparing our results for beech to those of Borůvka et al. (2018), who used a 210 °C treatment for

3 h, we found a more significant decrease in density but a less significant decrease in IBS. Their study reported a 55% decrease in density and an 80% decrease in IBS, while our results showed a less drastic reduction in IBS. Finally, our findings for IBS change for spruce (14.5% decrease for L groups and 40% decrease for H group) are consistent with the study by Gaff et al. (2019), showing a 18.5% decrease in IBS for the L-group and a 39% decrease for the H-group. However, there was a considerable difference in the IBS of oak between the two studies. Both our study and Gaff et al. (2019) agree that the decrease in oak's IBS is relatively similar for both the L and H groups, while spruce's IBS decreases more significantly for the H group.

Table 3 Median values and standard deviations of results

Median (STDV.S)	No of samples	Density (kg/m ³)	Moisture content (%)	MOED _{LT} (GPa)	MOED _{LR} (GPa)	LDD _{LT} (%)	LDD _{LR} (%)	ACE (m ⁴ /kg.s)	Fmax (N)	IBS (kJ/m ²)	Deflection (mm)
ASH N	28	707 (27)	10.66 (0.44)	15.25 (1.17)	15.67 (1.40)	2.77 (0.52)	2.74 (0.57)	5,523 (903)	3,442 (1,593)	45.4 (1.6)	8.17 (0.68)
ASH L	22	691 (15)	6.08 (0.69)	15.95 (0.99)	16.45 (0.96)	2.69 (0.73)	2.68 (0.87)	6,434 (1,012)	3,776 (1,189)	44.4 (2.9)	6.61 (0.47)
ASH H	20	614 (22)	3.22 (0.29)	14.06 (1.23)	14.48 (1.15)	2.14 (0.54)	2.31 (1.01)	10,060 (2,119)	5,850 (1,925)	20.3 (8.2)	4.43 (0.84)
BEECH N	20	735 (29)	12.2 (0.19)	14.76 (0.97)	15.12 (1.25)	3.47 (0.70)	3.86 (1.43)	3,653 (1,070)	3,257 (653)	45.9 (1.5)	7.88 (0.66)
BEECH L	21	728 (16)	6.1 (0.46)	17.49 (0.94)	18.28 (1.09)	3.24 (1.57)	2.95 (0.66)	5,373 (1,053)	5,374 (1,725)	42.0 (4.2)	6.59 (0.6)
BEECH H	19	619 (57)	4.02 (0.37)	14.23 (2.65)	14.61 (2.77)	2.43 (0.37)	2.62 (0.77)	7,436 (1,656)	2,861 (1,164)	30.9 (13.8)	4.54 (0.96)
LARCH N	28	539 (20)	11.28 (0.33)	12.48 (1.57)	12.31 (1.79)	2.95 (0.92)	2.81 (0.89)	10,116 (2,356)	4,319 (1,108)	24.1 (10.2)	5.37 (0.82)
LARCH L	29	504 (21)	5.66 (0.63)	12.9 (1.78)	12.96 (2.13)	3.0 (1.06)	2.72 (0.46)	12,525 (2,428)	2,659 (913)	28.4 (9.5)	4.91 (0.66)
LARCH H	14	483 (23)	4.29 (0.94)	11.05 (3.86)	11.52 (3.71)	4.94 (3.19)	5.47 (5.99)	1,2633 (5,240)	2,274 (1,097)	11.8 (8.1)	3.51 (0.6)
OAK N	19	715 (45)	13.26 (2.36)	11.29 (2.24)	11.65 (2.24)	3.80 (1.82)	3.46 (0.96)	3,531 (1,098)	3,396 (1,280)	28.2 (11.6)	6.08 (1.03)
OAK L	9	639 (27)	4.8 (1.1)	10.71 (2.21)	11.85 (1.72)	3.70 (1.92)	3.27 (0.95)	4,219 (992)	3,374 (963)	14.2 (3.3)	4.13 (0.67)
OAK H	18	560 (39)	3.83 (1.05)	7.95 (2.5)	9.62 (1.83)	5.72 (3.96)	4.00 (2.93)	6,910 (3,538)	1,085 (343)	11.6 (4.3)	3.23 (1.05)
SPRUCE N	23	450 (55)	10.34 (0.67)	13.39 (2.25)	13.57 (2.26)	2.43 (0.46)	2.54 (1.03)	18,223 (6,587)	2,150 (586)	38.5 (11.8)	5.94 (1.23)
SPRUCE L	30	455 (65)	5.6 (0.82)	13.91 (3.66)	13.99 (2.81)	2.87 (1.00)	2.90 (1.35)	19,186 (6,316)	2,720 (1,449)	32.9 (16.3)	4.44 (1.27)
SPRUCE H	31	431 (40)	3.66 (0.67)	12.66 (2.46)	12.92 (2.55)	2.87 (1.00)	2.79 (1.28)	20,345 (6,159)	1,430 (1,807)	22.9 (11.2)	2.64 (1.33)

We determined the density and moisture content of samples by measuring their weight before and after testing and depict these in Figs. 6 and 7. It is evident that TMP decreased both moisture content and density. In ash, beech and spruce, the density of the H group was significantly

different from the control N group, while the L group was not, indicating the limited influence of TMP on density at the lower temperature. However, in the moisture content chart, the N group is notably distant from the other two,

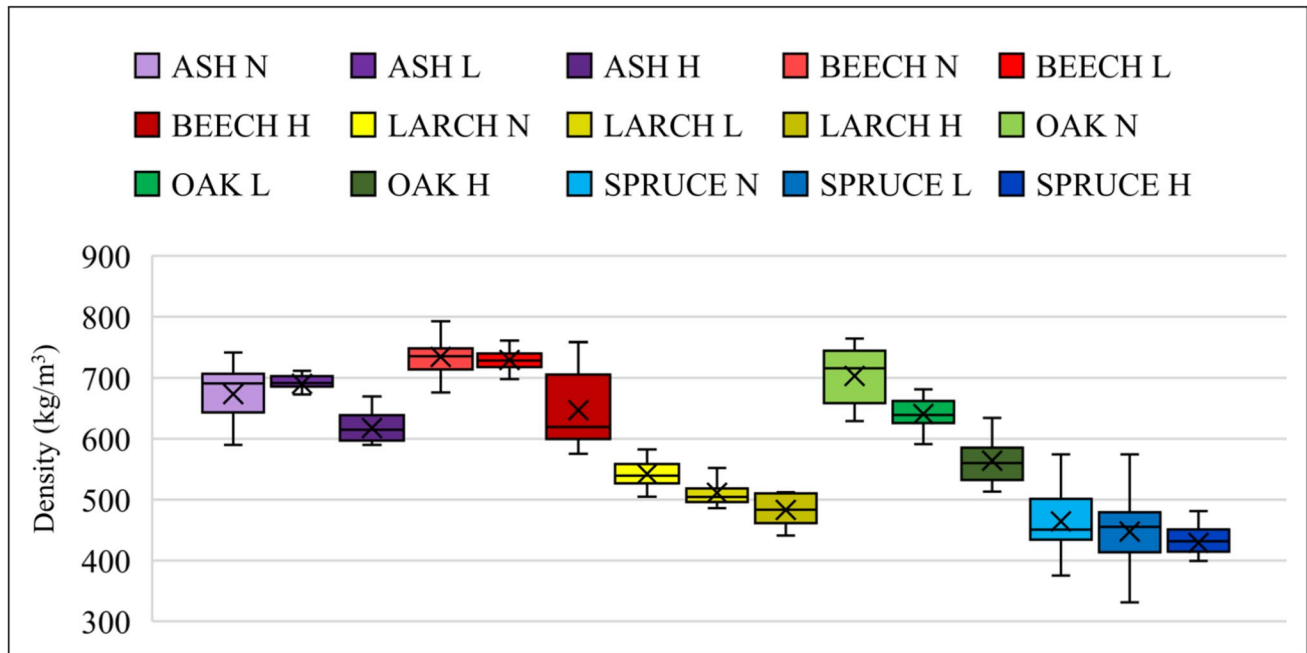


Fig. 6 Densities of wood samples by species with different levels of TMP

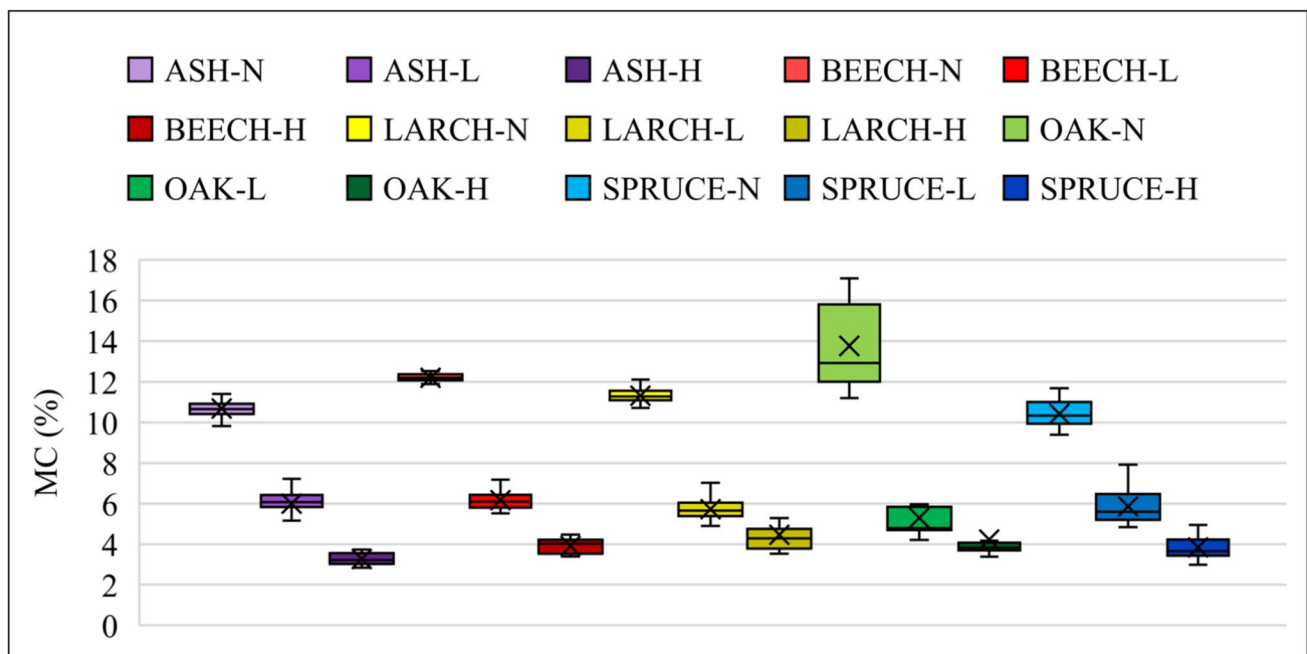


Fig. 7 Moisture content of wood samples by species with different levels of TMP

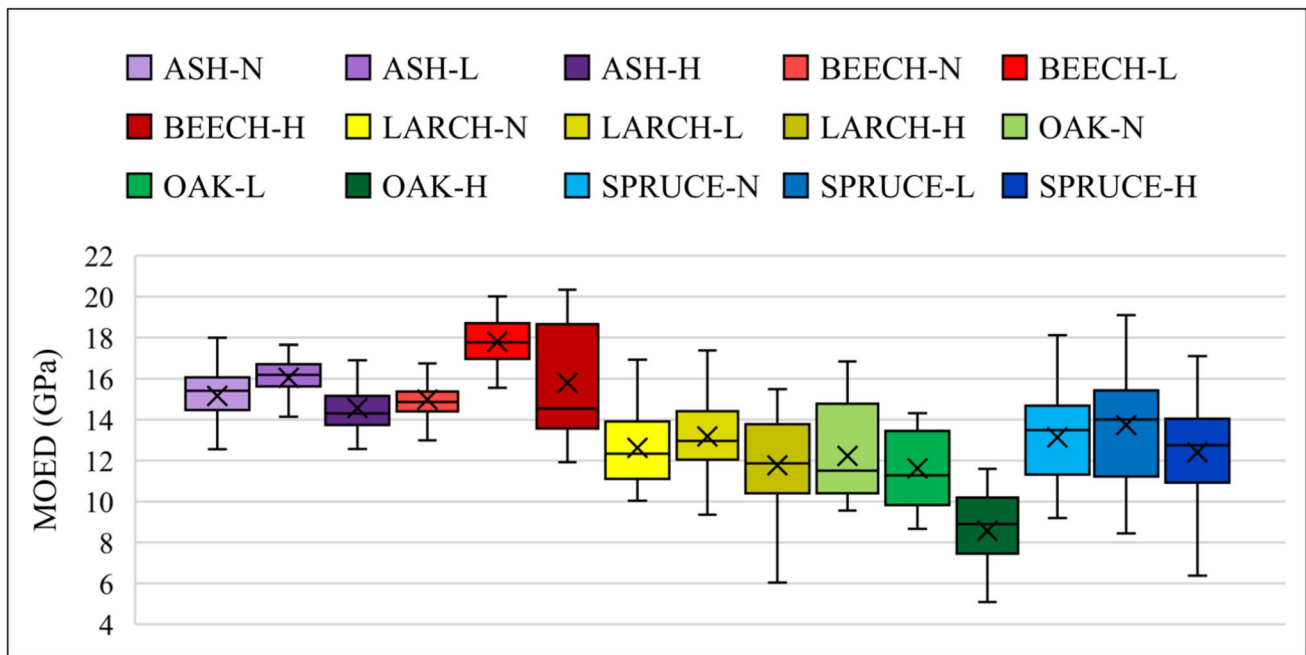


Fig. 8 MOED of wood samples by species with different levels of TMP

indicating the substantial effectiveness of TMP in reducing the wood's moisture content even at the lower temperature.

An ANOVA test confirmed that the size of each group did not significantly affect any of the parameters measured, and

the results were independent of the number of specimens in the groups.

Furthermore, the p-values derived from ANOVA tests on all groups (N, L and H), as well as their combination (H + L + N) about their $MOED_{LT}$ and $MOED_{LR}$, were

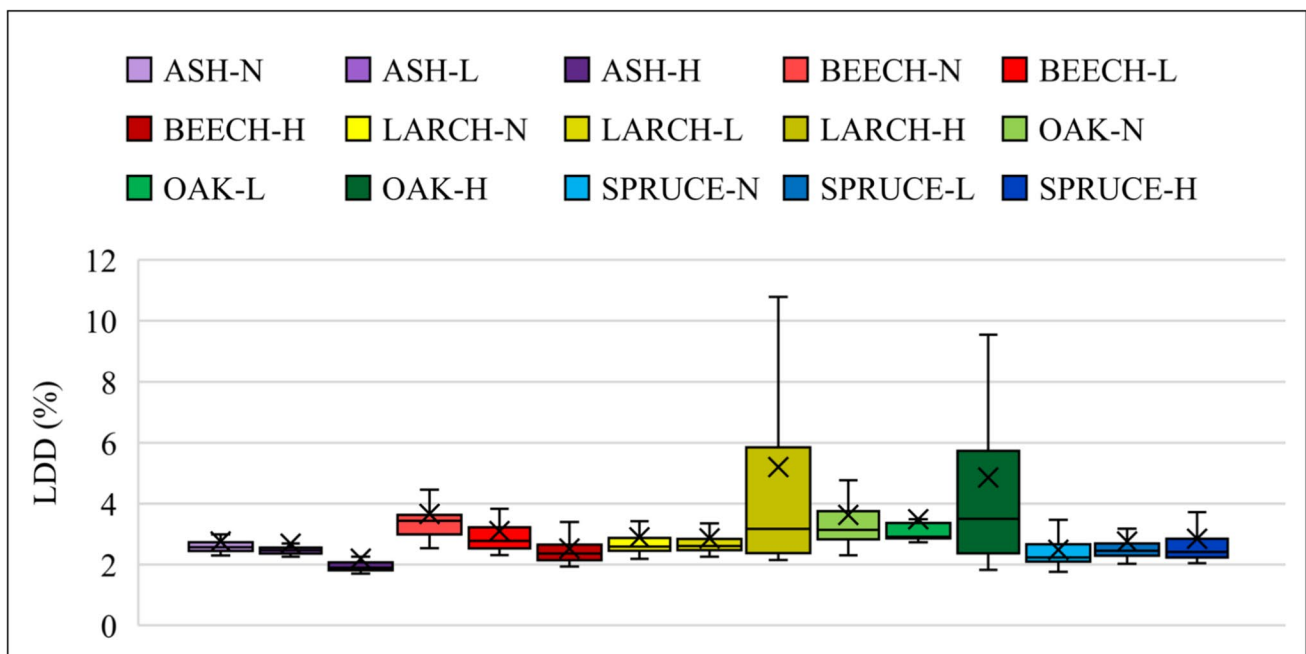


Fig. 9 LDD of wood samples by species with different levels of TMP

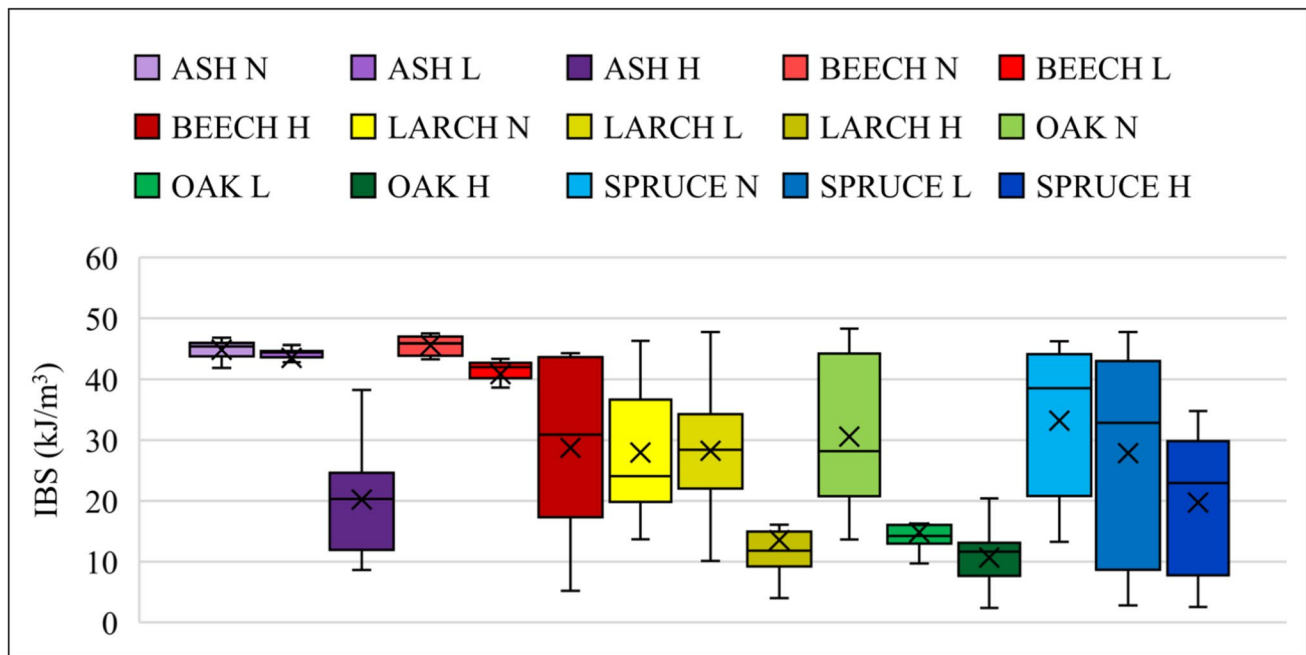


Fig. 10 IBS of wood samples by species with different levels of TMP

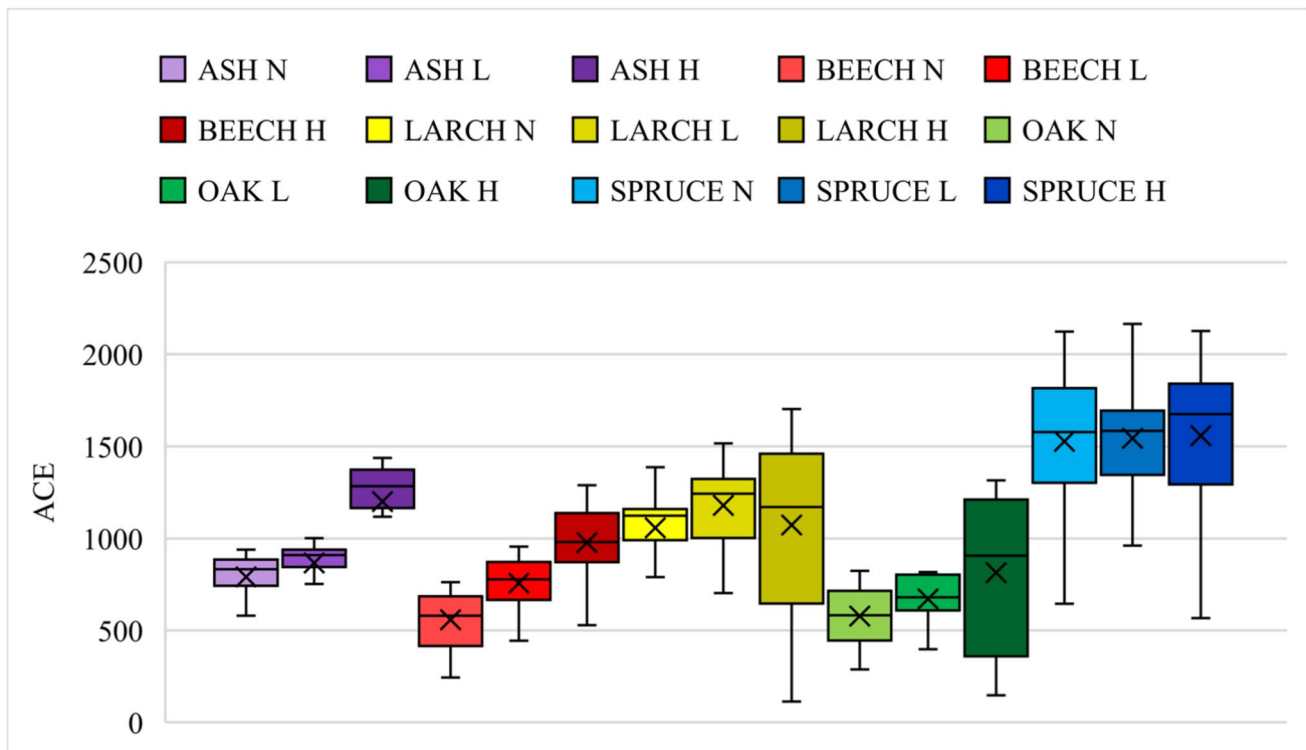
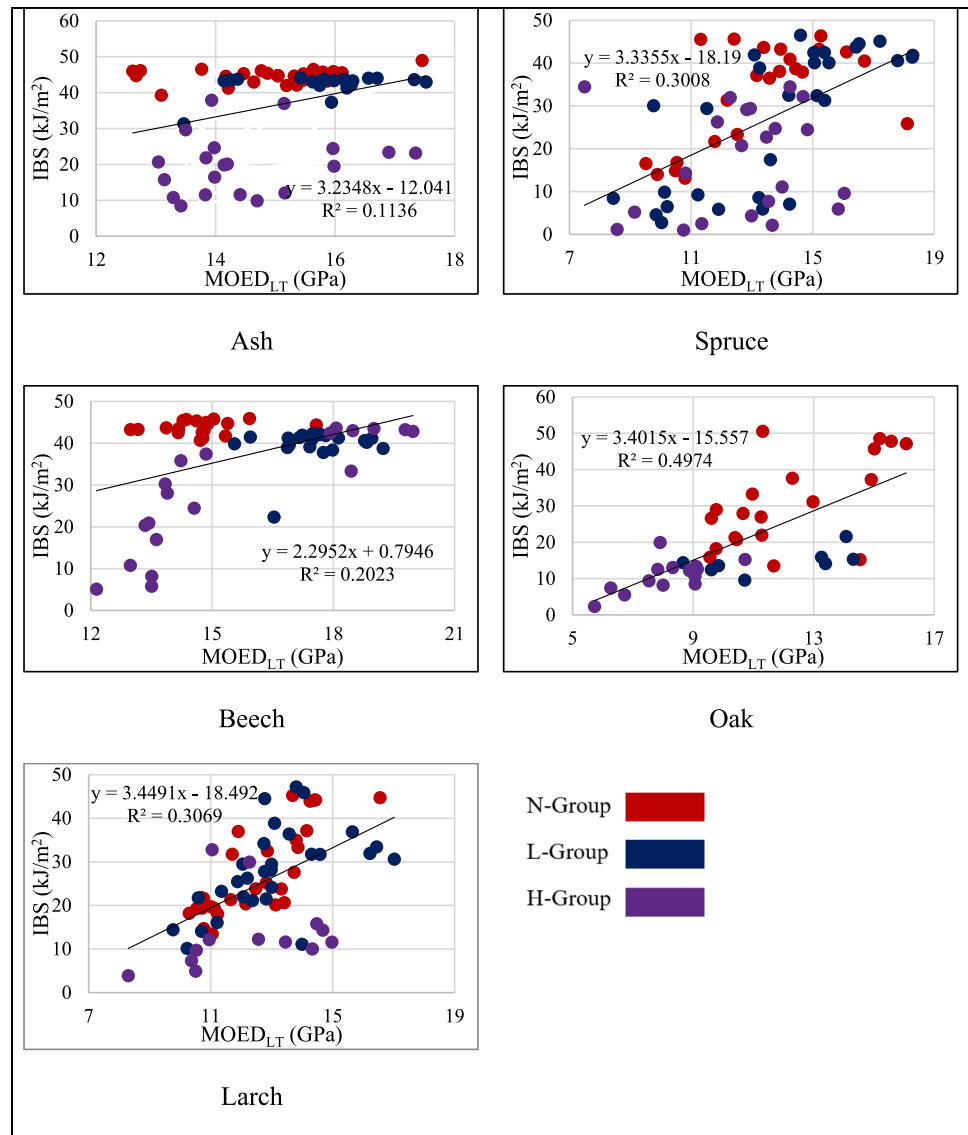


Fig. 11 ACE of wood samples by species with different levels of TMP

Fig. 12 IBS vs MOED for wood samples by species

observed to exceed 0.05, except for the oak H group. This outcome suggests that the MOED_{LT} and MOED_{LR} exhibited statistically equivalent values, and there is no necessity to distinguish between MOED_{LT} and MOED_{LR}.

The p-values derived from Kruskal–Wallis tests on all groups (N, L and H), as well as their combination (H+L+N) about their LDD_{LT} and LDD_{LR}, were observed to exceed 0.05, except for three groups (larch N). Thus, the assumption of the equality of LDD_{LT} and LDD_{LR} is correct. For LDD, the Kruskal–Wallis test was used because of the non-normal distribution within all groups.

Figures 8 and 9 are box-plot charts representing the MOED and LDD of each species, as well as their

combination categorised as hardwood, softwood, and all wood species included in this study. The MOED of the L groups (except for oak) was higher than N groups, indicating a slight improvement of MOED by low-temperature TMP, while for higher temperatures, the MOED decreased significantly. The LDD showed different behaviour in different species. It decreased in ash and beech and increased in larch, oak and spruce.

IBS test results for each specimen are given as box-plots in Fig. 10 to allow species to be compared. It should be noted that the IBS for ash-N and beech-N are not exact as most of the specimens did not break; they simply absorbed the impact energy.

Table 4 Regression analysis of MOED, LDD, and ACE with maximum deflection and maximum force

R^2	For determining Max Deflection			For determining Max Force		
	MOED	LDD	ACE	MOED	LDD	ACE
ASH N	0.024	0.028	0.036	0.047	0.040	0.007
ASH L	0.213	0.013	0.025	0.001	0.018	0.013
ASH H	0.010	0.087	0.065	0.035	0.166	0.134
ASH ALL	0.042	0.087	0.505	0.001	0.110	0.136
BEECH N	0.155	0.328	0.261	0.284	0.033	0.021
BEECH L	0.011	0.001	0.000	0.000	0.080	0.233
BEECH H	0.082	0.022	0.008	0.610	0.242	0.051
BEECH ALL	0.047	0.163	0.569	0.358	0.038	0.048
LARCH N	0.086	0.050	0.112	0.005	0.037	0.064
LARCH L	0.004	0.031	0.001	0.027	0.008	0.050
LARCH H	0.010	0.002	0.002	0.423	0.325	0.262
LARCH ALL	0.001	0.030	0.013	0.086	0.092	0.002
OAK N	0.121	0.141	0.216	0.397	0.031	0.000
OAK L	0.233	0.083	0.009	0.190	0.066	0.301
OAK H	0.040	0.036	0.000	0.000	0.005	0.033
OAK ALL	0.158	0.011	0.067	0.544	0.032	0.008
SPRUCE N	0.042	0.264	0.481	0.414	0.060	0.102
SPRUCE L	0.364	0.079	0.022	0.186	0.002	0.129
SPRUCE H	0.019	0.116	0.168	0.160	0.070	0.036
SPRUCE ALL	0.091	0.155	0.037	0.148	0.011	0.015
WOOD ALL	0.128	0.008	0.104	0.255	0.039	0.038

Acoustic conversion efficiency results are given in Fig. 11 and show the difference between the species tested. TMP increased the ACE for all species, and at the higher temperature, ACE results were more variable.

We plotted charts depicting IBS and maximum deflection versus MOED, LDD and ACE to assess the linear regression between these variables. The regression analysis revealed a correlation between IBS and MOED. However, the other parameters did not demonstrate any significant correlation, particularly LDD, which exhibited not even a

remote relationship with deflection and IBS at any level. The scatter graphs are given in Fig. 12 and the regression equations and their coefficients of determination are presented in Table 5. The coefficients of determination of LDD, ACE and MOED with Maximum deflection and reaction force were also presented in Table 4 as a form of null results. The Table 4 depicts that the NDT surely is not able to measure the max deflection and reaction force robustly. Almost all of the coefficients of determination for the regressions of the charts are less than 0.5, indicating a weak relationship

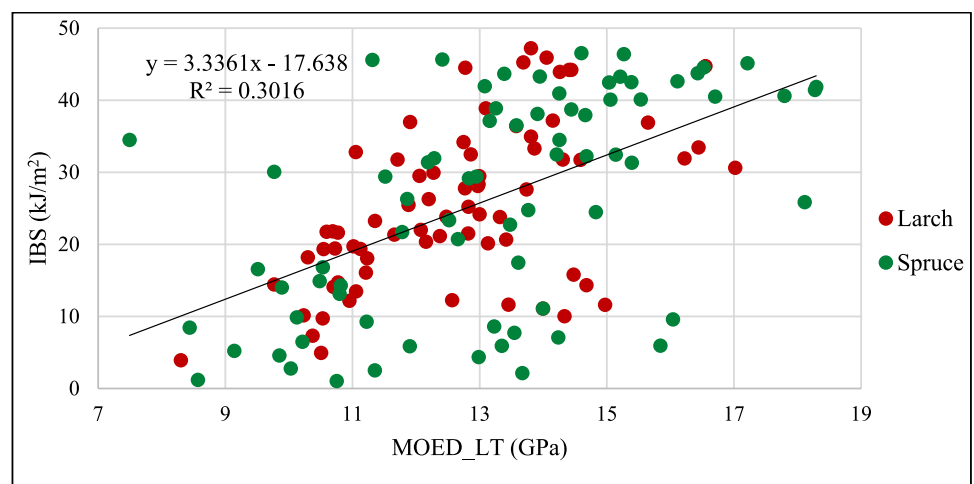
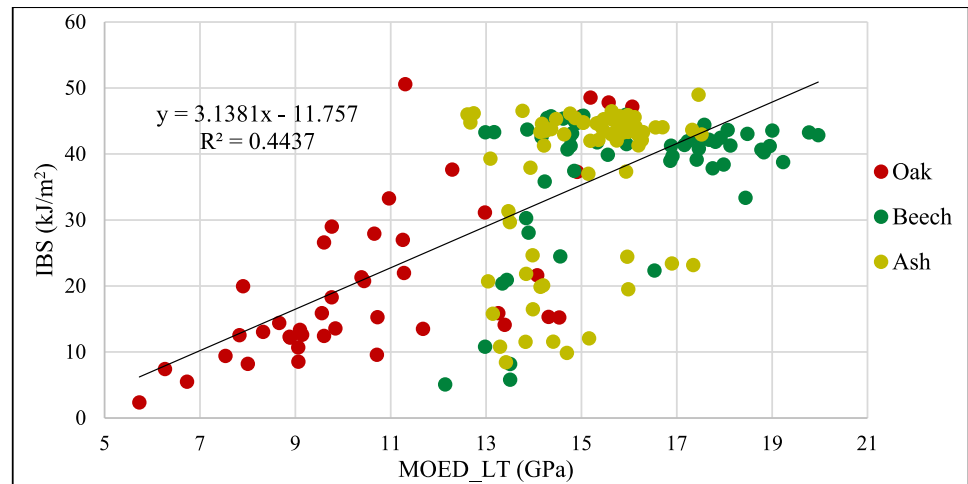
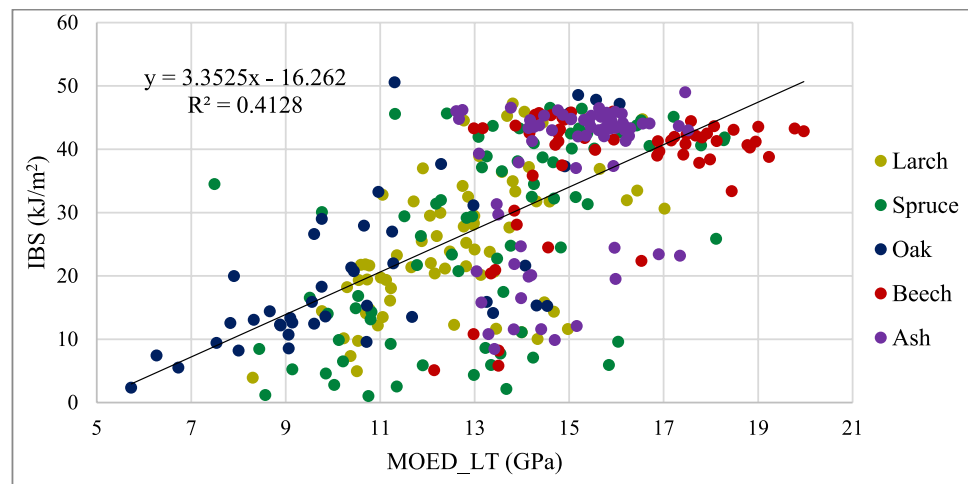
Fig. 13 IBS vs MOED for the softwood species

Fig. 14 IBS vs MOED for the hardwood species**Fig. 15** IBS vs MOED for the all wood species

between IBS and MOED. Nonetheless, regression analysis can still assist in assessing the impact strength of wood to some extent via NDT, albeit with low precision.

Due to some similarities between the softwood species and hardwood species, each in aggregate, we examined whether their combined groups could offer any information about the relationship between MOED and IBS. Figures 13, 14 and 15 show the results for these aggregations. Larch and spruce show a good matching pattern and overall have similarities; however, among the hardwoods, oak seems to behave differently from ash and beech, which have similar patterns in their scatter graphs.

Additionally, NDT proved to be useful for assessing the IBS of non-modified oak, larch and spruce. In Fig. 16, scatter charts depict the relationship between IBS and MOED for these species, as well as for both softwood species in this study (larch and spruce).

By linear regression of the relationship between IBS and MOED, equations determining the IBS through MOED can be achieved. Table 5 presents the equations and coefficients of determination R^2 . The efficiency of the linear regression of the relationship between IBS and MOED showed a considerable variation between the groups and species.

Fig. 16 IBS vs MOED for non-modified specimens of oak and the two softwoods (larch and spruce) separately and together

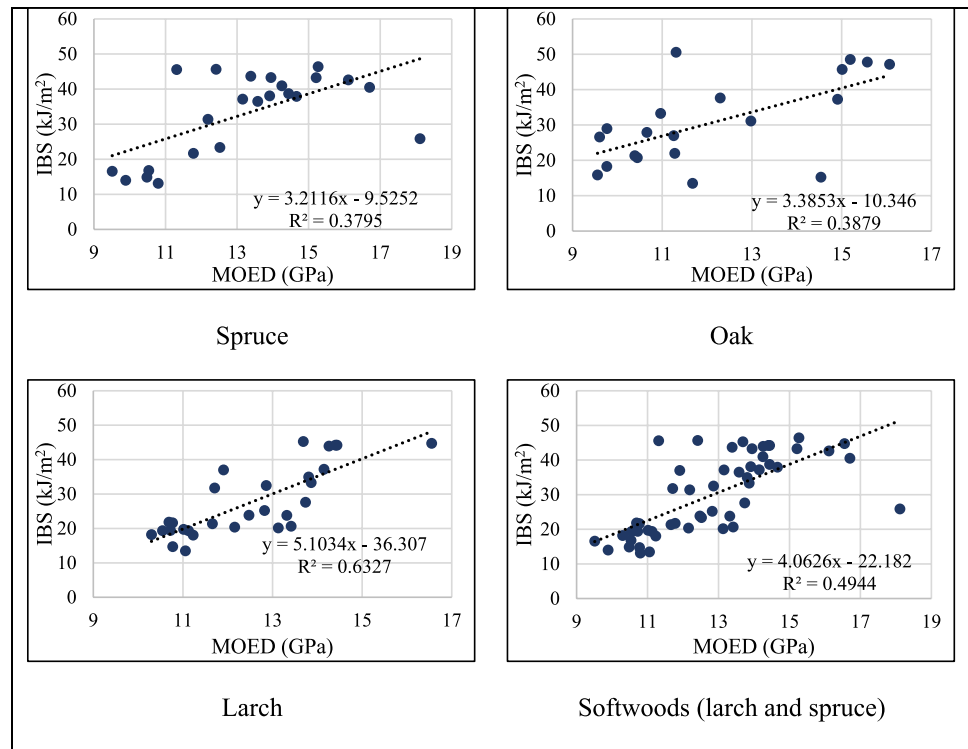


Table 5 Regression equation and coefficients of determination between MOED and IBS of the samples

	Treated	Untreated	All
Ash	$y = 5.4896x - 51.84$ $R^2 = 0.3034$	$y = 0.3784x + 38.918$ $R^2 = 0.0534$	$y = 3.2348x - 12.041$ $R^2 = 0.1136$
Beech	$y = 4.3735x - 38.333$ $R^2 = 0.6793$	$y = 0.3207x + 39.153$ $R^2 = 0.0396$	$y = 2.2952x + 0.7946$ $R^2 = 0.2023$
Larch	$y = 2.8335x - 12.57$ $R^2 = 0.2218$	$y = 5.1034x - 36.307$ $R^2 = 0.6327$	$y = 3.4491x - 18.492$ $R^2 = 0.3069$
Oak	$y = 1.2039x + 0.8207$ $R^2 = 0.4417$	$y = 3.3853x - 10.346$ $R^2 = 0.3879$	$y = 3.4015x - 15.557$ $R^2 = 0.4974$
Spruce	$y = 3.3964x - 22.167$ $R^2 = 0.3252$	$y = 3.2116x - 9.5252$ $R^2 = 0.3795$	$y = 3.3355x - 18.19$ $R^2 = 0.3008$
Ash, beech and oak (hardwood)	$y = 3.2507x - 18.664$ $R^2 = 0.6149$	$y = 3.3794x - 7.2297$ $R^2 = 0.5254$	$y = 3.1381x - 11.757$ $R^2 = 0.4437$
Spruce and larch (softwood)	$y = 3.1657x - 18.056$ $R^2 = 0.284$	$y = 4.0626x - 22.182$ $R^2 = 0.4944$	$y = 3.3361x - 17.638$ $R^2 = 0.3016$
All species	$y = 3.301x - 15.461$ $R^2 = 0.4048$	$y = 4.1098x - 19.796$ $R^2 = 0.5499$	$y = 3.3525x - 16.262$ $R^2 = 0.4128$

4 Conclusions

The NDT methods used in this study successfully evaluated the impact-related properties of thermally modified wood, though with significant limitations. In line with our objectives, our findings provide a clear understanding of NDT's predictive capabilities for this application. The MOED was found to be effective for predicting the

IBS. However, as hypothesized, the accuracy was species-dependent, with strong correlations observed for oak while proving less effective for larch and spruce, and no useful for assessing ash and beech. Also, this study revealed a key limitation of NDT: it failed to provide reliable estimations for maximum deflection and force during impact loading. The poor results from regression analysis of MOED, LDD, and ACE with these parameters indicate that NDT alone cannot fully characterize the dynamic fracture behavior of

the wood. This highlights a critical gap in NDT's application for impact assessment. This study provides a comprehensive view of the effectiveness of NDT for evaluating the impact-related behavior of thermally modified timber and also it shows how the efficiency of NDT is influenced by both the wood species and the specific thermal modification process. Based on these findings, we recommend further research into the use of different NDT techniques to broaden the application of NDT in this field.

Authors Contributions All authors participated in the planning, execution, and interpretation of the research presented in this manuscript. Each author contributed significantly to all aspects of the work and has reviewed and approved the final submitted version.

Funding Open access publishing supported by the institutions participating in the CzechELib Transformative Agreement. This work was supported by the MENDEL Internal Grant Agency (grant number IGA-LDF-22-IP-016).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval Not applicable.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process During the preparation of this work the authors used Grammarly and ChatGPT by OpenAI software in order to find the primary English language grammatical mistakes in the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. However, in the end, the authors decided to send the text to an expert firm to be inspected for improvement of the English language.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Sandberg, D., Kutnar, A., Karlsson, O., Jones, D.: Wood modification technologies: principles, sustainability, and the need for innovation. CRC Press (2021)
- Sandberg, D., Kutnar, A., Mantanis, G.: Wood modification technologies – a review. *iForest Biogeosci. Forestry* **10**(6), 895 (2017)
- Hill, C.A.: Wood modification: chemical, thermal and other processes. John Wiley & Sons (2007)
- Godinho, D., Araujo, S.D.O., Quilhó, T., Diamantino, T., Gominho, J.: Thermally modified wood exposed to different weathering conditions: a review. *Forests* **12**(10), 1400 (2021)
- Boonstra, M.J., Van Acker, J., Tjeerdsma, B.F., Kegel, E.V.: Strength properties of thermally modified softwoods and its relation to polymeric structural wood constituents. *Ann. For. Sci.* **64**(7), 679–690 (2007)
- Sikora, A., Kačík, F., Gaff, M., Vondrová, V., Bubeníková, T., Kubovský, I.: Impact of thermal modification on color and chemical changes of spruce and oak wood. *J. Wood Sci.* **64**, 406–416 (2018)
- Ibach, R.: Chapter 19 – specialty treatments. In: *Wood handbook – wood as an engineering material*, pp. 1–16. United States Department of Agriculture Forest Service, Madison, Wisconsin (2010)
- Kubovský, I., Kačíková, D., Kačík, F.: Structural changes of oak wood main components caused by thermal modification. *Polymers* **12**(2), 485 (2020)
- Moliński, W., Roszyk, E., Jabłoński, A., Puszyński, J., Cegiela, J.: Mechanical parameters of thermally modified ash wood determined on compression in tangential direction. *Maderas Cienc. Tecnol.* **20**(2), 267–276 (2018)
- Hossen, M.F., Hamdan, S., Rahman, M.R.: Investigation of the acoustic properties of chemically impregnated kayu malam wood used for musical instruments. *Adv. Mater. Sci. Eng.* **2018**(1), 7829613 (2018)
- Guan, C., Zhang, H., Hunt, J.F., Yan, H.: Determining shear modulus of thin wood composite materials using a cantilever beam vibration method. *Constr. Build. Mater.* **121**, 285–289 (2016)
- Ono, T., Norimoto, M.: Anisotropy of dynamic young's modulus and internal friction in wood. *Jpn. J. Appl. Phys.* **24**(8R), 960 (1985)
- Wang, L., Li, L., Qi, W., Yang, H.: Pattern recognition and size determination of internal wood defects based on wavelet neural networks. *Comput. Electron. Agric.* **69**(2), 142–148 (2009)
- Bandara, S., Rajeev, P., Gad, E., Srisantharajah, B.: Damage severity estimation of timber poles using stress wave propagation and wavelet entropy evolution. *J. Nondestruct. Eval. Diagn. Progn. Eng. Syst.* **4**(1), 011006 (2021)
- Menezzi, C.H.S., Amorim, M.R., Costa, M.A., Garcez, L.R.: Evaluation of thermally modified wood by means of stress wave and ultrasound nondestructive methods. *Mater. Sci.* **20**(1), 61–66 (2014)
- Kutnar, A., Widmann, R., Brémaud, I.: Preliminary studies for use of dynamic mechanical analysis (DMA) to verify intensity of thermal wood modifications. *Int. Wood Prod. J.* **4**(3), 158–165 (2013)
- Sun, N., Das, S., Frazier, C.E.: Dynamic mechanical analysis of dry wood: linear viscoelastic response region and effects of minor moisture changes. *Holzforschung* (2007). <https://doi.org/10.1515/HF.2007.006>
- Hassan, K.T., Horáček, P., Tippner, J.: Evaluation of stiffness and strength of Scots pine wood using resonance frequency and ultrasonic techniques. *Bioresour.* **8**(2), 1634–1645 (2013)
- Viala, R., Placet, V., Cogan, S.: Simultaneous non-destructive identification of multiple elastic and damping properties of spruce tonewood to improve grading. *J. Cult. Herit.* **42**, 108–116 (2020)
- Nop, P., Cristini, V., Zlámál, J., Vand, M.H., Šeda, V., Tippner, J.: Determination of the static bending properties of green beech and oak wood by the frequency resonance technique. *Forests* **15**(1), 150 (2024)
- Alkadri, A., Carlier, C., Wahyudi, I., Gril, J., Langbour, P., Brémaud, I.: Relationships between anatomical and vibrational properties of wavy sycamore maple. *IAWA J.* **39**(1), 63–86 (2018)

22. Fernández-Serrano, Á., Villasante, A.: Longitudinal, transverse and ultrasound vibration for the prediction of stiffness using models incorporating features in *Pinus sylvestris* timber. *Eur. J. Wood Wood Prod.* **79**(6), 1541–1550 (2021)
23. Chauhan, S., Sethy, A.: Differences in dynamic modulus of elasticity determined by three vibration methods and their relationship with static modulus of elasticity. *Maderas. Ciencia y tecnología* **18**(2), 373–382 (2016)
24. Mania, P., Skrodzka, E.: Modal parameters of resonant spruce wood (*Picea abies* L.) after thermal treatment. *J. King Saud Univ. Sci.* **32**(1), 1152–1156 (2020)
25. Magalas, L.B., Malinowski, T.: Measurement techniques of the logarithmic decrement. *Solid State Phenom.* **89**, 247–260 (2003)
26. Brémaud, I., Minato, K., Langbour, P., Thibaut, B.: Physico-chemical indicators of inter-specific variability in vibration damping of wood. *Ann. For. Sci.* **67**(7), 707 (2010)
27. Bucur, V.: The acoustics of wood (1995). CRC press. (2017)
28. Brémaud, I., Minato, K., Thibaut, B.: Mechanical damping of wood as related to species classification: a preliminary survey. In 6th Plant Biomechanics Conference PBM09 (pp. 536–542). (2009)
29. Baar, J., Tippner, J., Gryc, V.: Wood anatomy and acoustic properties of selected tropical hardwoods. *IAWA J.* **37**(1), 69–83 (2016)
30. Sedik, Y., Hamdan, S., Jusoh, I., Hasan, M.: Acoustic properties of selected tropical wood species. *J. Nondestruct. Eval.* **29**, 38–42 (2010)
31. Rujinirun, C., Phinyocheep, P., Prachyabrued, W., Laemsak, N.: Chemical treatment of wood for musical instruments. Part I: acoustically important properties of wood for the ranad (Thai traditional xylophone). *Wood Sci. Technol.* **39**, 77–85 (2005)
32. Bhujangrao, T., Froustey, C., Iriondo, E., Veiga, F., Darnis, P., Mata, F.G.: Review of intermediate strain rate testing devices. *Metals* **10**(7), 894 (2020)
33. Bruck, H.A., Evans, J.J., Peterson, M.L.: The role of mechanics in biological and biologically inspired materials. *Exp. Mech.* **42**, 361–371 (2002)
34. Gašparík, M., Gaff, M., Šafaříková, L., Vallejo, C.R., Svoboda, T.: Impact bending strength and Brinell hardness of densified hardwoods. *Bioresour.* **11**(4), 8638–8652 (2016)
35. Yamada, H., Tateyama, K., Sasaki, H., Naruke, S., Kishimoto, H., Yoshimoto, M.: Impact resistance to ballistic ejecta of wooden buildings and a simple reinforcement method using aramid fabric. *J. Volcanol. Geotherm. Res.* **359**, 37–46 (2018)
36. Polocoşer, T., Stöckel, F., Kasal, B.: Low-velocity transverse impact of small, clear spruce and pine specimens with additional energy absorbing treatments. *J. Mater. Civ. Eng.* **28**(8), 04016048 (2016)
37. Bučar, D.G., Merhar, M.: Impact and dynamic bending strength determination of Norway spruce by impact pendulum deceleration. *Bioresour.* **10**(3), 4740–4750 (2015)
38. Fortin-Smith, J., Sherwood, J., Drane, P., Kretschmann, D.: Characterization of maple and ash material properties for the finite element modeling of wood baseball bats. *Appl. Sci.* **8**(11), 2256 (2018)
39. Hassan Vand, M., Tippner, J.: Effects of moisture content on the behaviour of Scots pine heartwood and sapwood under impact. *J. Wood Sci.* **70**(1), 39 (2024)
40. Hassan Vand, M., Tippner, J., Brabec, M.: Effects of species and moisture content on the behaviour of solid wood under impact. *Eur. J. Wood Wood Prod.* **82**(1), 23–34 (2024)
41. Esteves, B., Pereira, H.: Wood modification by heat treatment: a review. *Bioresour.* **4**(1), 370–404 (2009)
42. Hughes, M., Hill, C., Pfriem, A.: The toughness of hygrothermally modified wood: COST action FP0904 2010–2014: thermo-hydro-mechanical wood behavior and processing. *Holzforschung* **69**(7), 851–862 (2015)
43. Leijten, A.J.M.: Heat treated wood and the influence on the impact bending strength. *Heron* **49**(4), 349–360 (2004)
44. Bal, B.C., Bektaş, İ.: The effects of heat treatment on some mechanical properties of juvenile wood and mature wood of *Eucalyptus grandis*. *Drying Technol.* **31**(4), 479–485 (2013)
45. Borůvka, V., Zeidler, A., Holeček, T., Dudík, R.: Elastic and strength properties of heat-treated beech and birch wood. *Forests* **9**(4), 197 (2018)
46. Rautkari, L., Honkanen, J., Hill, C.A., Ridley-Ellis, D., Hughes, M.: Mechanical and physical properties of thermally modified Scots pine wood in high pressure reactor under saturated steam at 120, 150 and 180 C. *Eur. J. Wood Prod.* **72**(1), 33–41 (2014)
47. Gaff, M., Kačík, F., Gašparík, M.: Impact of thermal modification on the chemical changes and impact bending strength of European oak and Norway spruce wood. *Compos. Struct.* **216**, 80–88 (2019)
48. Gaff, M., Kačík, F., Sandberg, D., Babiak, M., Turčani, M., Niemz, P., Hanzlík, P.: The effect of chemical changes during thermal modification of European oak and Norway spruce on elasticity properties. *Compos. Struct.* **220**, 529–538 (2019)
49. Bragov, A.M., Iuzhina, T.N., Lomunov, A.K., Igumnov, L.A., Belov, A.A., Eremeyev, V.A.: Investigation of wood properties at elevated temperature. *J. Appl. Comput. Mech.* **8**(1), 298–305 (2022)
50. Borůvka, V., Novák, D., Sedivka, P.: Comparison and analysis of radial and tangential bending of softwood and hardwood at static and dynamic loading. *Forests* **11**(8), 896 (2020)
51. Zhang, H., Sfarra, S., Sarasini, F., Fiorelli, J., Peeters, J., Avdelidis, N.P., de Lucca Sartori, D., Ibarra-Castanedo, C., Perilli, S., Mokhtari, Y., Tirillò, J.: Impact modelling and a posteriori non-destructive evaluation of homogeneous particleboards of sugarcane bagasse. *J. Nondestruct. Eval.* **37**, 1–30 (2018)
52. Hassan Vand, M., Tippner, J.: Impact bending strength of thermally modified timber. *Holzforschung* **77**(11–12), 828–837 (2023)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Paper V
**A Multimodal Vibro-Acoustic Evaluation of Wood and Wood-
Based Materials for Musical Instrument Applications**

Patrik Nop, Kryštof Loula, Jan Tippner

Journal of Sound and Vibration

Submitted

Nop, P. (corresponding author): Conceptualization, Methodology, Resources, Investigation, Formal analysis, Writing – original draft

A Multimodal Vibro-Acoustic Evaluation of Wood and Wood-Based Materials for Musical Instrument Applications

Abstract: Wood is the preferred material for musical instruments due to its vibro-acoustic properties; however, the decreasing availability of high-quality tonewood has increased interest in alternative materials. The quality evaluation of such materials is commonly based on parameters derived from a single vibration mode, although additional modes can help describe vibro-acoustic behavior more comprehensively. Dynamic elastic moduli, damping, and derived parameters were determined using the Frequency Resonance Technique, supported by Numerical Modal Analysis. Measurements were performed on solid wood and plywood specimens and were designed to evaluate the influence of sensor selection and measurement configuration. Aggregated parameters based on acoustic conversion efficiency (*ACE*) and relative acoustic conversion efficiency (*RACE*) were formulated as ratios of key material properties to combine information across multiple vibration modes (including different shapes and orders). The results demonstrate that the determination of vibro-acoustic parameters derived from damping is sensitive to measurement method. The aggregated parameters *ACE_{combi}* and *RACE_{combi}* may enable the effective classification of materials. The proposed multimodal framework offers a broader basis for vibro-acoustic material evaluation.

Keywords: Vibro-acoustic properties; Frequency Resonance Technique; Multimodal analysis; Acoustic conversion efficiency; Musical instruments

1. Introduction

Across the classes of acoustic musical instruments (idiophones, chordophones, aerophones), wood is the preferred material for their construction [1–3]. Recently, with the decline of high-quality wood, composite materials have been increasingly explored as a replacement for solid resonance wood in musical instrument manufacturing [4,5]. When evaluating these materials, it is crucial to approximate wood in as many of its unique vibro-acoustic properties as possible [6,7].

Although their importance varies among the classes of acoustic musical instruments [1,8], the key properties for the objective quality evaluation of materials have been established [1,8–12]. The fundamental indicator is the specific modulus of elasticity [6,10,13], which is the ratio between the stiffness of the material (represented by modulus of elasticity) and its density (ρ). This relationship defines the speed at which a sound wave propagates through the material [11,12]. Another parameter derived from this relationship is the radiation coefficient (K), which indicates how much sound energy is radiated by the material [1,14], or the acoustic impedance (Z), which expresses the material's resistance to sound wave propagation [1,12]. The description of vibration attenuation due to internal friction is provided by several damping parameters (loss factor, quality factor, logarithmic decrement, damping ratio, damping coefficient, etc.), whose terminology depends mainly on the determination method; however, simple conversion relationships exist among them [1,11,12,15]. Since the specific modulus of elasticity and damping are generally related [13,16,17], expressing their ratio via the acoustic conversion efficiency (*ACE*) [12,18,19], or through the relative acoustic conversion efficiency (*RACE*) [20–22], provides the most comprehensive evaluation of the material quality for musical instruments.

Previous studies evaluate vibro-acoustic parameters based on a single vibration mode. However, different mode shapes (longitudinal, bending (flexural), and torsional) reflect distinct aspects of material behavior. Considering the contribution of multiple vibration modes to the tonal characteristics of musical instruments, a multimodal approach may offer a more sensitive evaluation of material quality.

For determining vibro-acoustic properties, the Frequency Resonance Technique (FRT) is widely used, in which the natural vibration frequencies of a free-free supported specimen are measured [10,12,18,23]. The dynamic modulus of elasticity (E_{dyn}) can be determined from the frequencies of longitudinal vibration modes, and the dynamic modulus of elasticity in bending (MOE_{dyn}) from the frequencies of flexural vibration modes [24]. The shear modulus of elasticity (G_{dyn}) can be determined from the frequencies of torsional vibrations [24]. G_{dyn} is also an important parameter for musical instruments [21]; nevertheless, only a few studies address its determination together with E_{dyn} or MOE_{dyn} [14,25,26]. Even fewer studies focus on the measurement of the damping of longitudinal or bending mode together with the damping of torsional mode [21,27]. This may be due to the difficulty of identifying corresponding (torsional or bending) mode shapes [28].

During the vibration of the specimen, even when it is supported at the nodes of the desired mode and excited at its antinodes, vibrations in other modes still occur [28–30]. Based on experience, and by gradually changing the excitation point (and the positions of supports), it is possible to distinguish individual resonant modes in the frequency spectrum during the signal analysis [28]. However, reliably identifying the individual modes using a single specimen support configuration would be more efficient.

Thanks to its high accuracy, the numerical modal analysis based on the Finite Element Method (FEM) is often employed to predict the dynamic response of systems of various complexities [31,32]. Such simulations can also effectively support FRT by providing an approximate frequency range in which the vibration modes are expected to occur (thus allowing the selection of an appropriate sampling frequency for the measurements), by assisting in the selection of support and excitation point locations, and by providing insight into the sequence of mode shapes, thereby facilitating the identification process [33,34].

Thus, multiple parameters can be characterised simultaneously using a single measurement configuration [35]. However, the question arises whether this streamlining of the measurement procedure may influence the resulting parameters. If the modes have frequencies very close to each other, it may complicate their identification, and depending on the measurement method and signal analysis used, it may also affect the determination of damping parameters [29].

To describe the damping of simultaneous vibration modes, signal decomposition into individual frequency components and subsequent evaluation of the decay of their amplitudes has proven to be a suitable approach, in contrast to envelope fitting of the overall signal or determining damping from the frequency peak bandwidth (especially in the case of close frequency peaks with overlapping bases) [36].

In FRT, a microphone or a single-point axial sensor, such as a laser vibrometer or accelerometer, is commonly used [24,26,37]. Considering their inherent measurement differences and the distinction between local and global approaches, the influence of sensor type on the determination of multiple parameters should be evaluated. Considering the contribution of higher-order modes to the tonal color, determining parameters such as *ACE* or *RACE* within a multimodal approach could further refine the quality grading of materials used for musical instruments [20,36].

This study aims to introduce a multimodal approach for vibro-acoustic evaluation of material by combining longitudinal, bending, and torsional vibration modes, while accounting for the influence of sensor selection, measurement configuration, and modal identification.

2. Materials and Methods

2.1. Material

Specimens of Norway spruce (*Picea abies* (L.) H. Karst.), Norway maple (*Acer platanoides*), pine plywood (Pine_{ply}), birch plywood (Birch_{ply}), and poplar plywood (Poplar_{ply}) were prepared with dimensions of 300 × 20 × 3.1–3.8 mm (length × width × thickness). Solid wood specimens (Spruce, Maple) were crafted from clear timber with anatomical orientation as follows: length – longitudinal (L), width – radial (R), thickness – tangential (T). Specimens from commercially available three-layer joinery plywood were crafted with the outer layers oriented longitudinally and the middle layer oriented perpendicularly. The plywood specimens were randomly cut from the central part of the boards in accordance with ČSN EN 326-1 [38]. The specimens were conditioned in a climatic chamber at 20 °C and 65 % relative humidity until an equilibrium moisture content (*EMC*) of 12 % was reached. Under these conditions, the ρ of all specimens was determined as the weight-to-volume ratio. Given the assumed greater variability in the properties of solid wood compared to engineered materials, 13 specimens were prepared and tested for the Maple and Spruce groups, and 10 specimens for each plywood group. In total, 56 specimens were tested.

2.2. Numerical Modal Analysis

To predict the frequency and the order of individual modes of the tested solid wood specimens, numerical simulations were performed using ANSYS (ANSYS Inc., Canonsburg, PA, USA). Finite element models of maple, spruce, and resonant spruce specimens were built. The 20-node solid element SOLID186 was employed. The model dimensions were in accordance with those of the experimentally tested specimens. Each model consisted of a total of 240 elements and 3267 nodes. The material properties of the solid wood used are presented in Table 1.

Table 1. Material properties applied in the numerical model

Material Model	ρ (kg·m ³)	Young's moduli (GPa)			Shear moduli (GPa)			Poisson ratio		
		E _L	E _R	E _T	G _{RT}	G _{LR}	G _{LT}	ν_{LT}	ν_{LR}	ν_{RT}
Maple	590	10.00	1.52	0.87	0.29	1.22	1.10	0.50	0.46	0.82
Spruce	430	13.50	0.89	0.48	0.03	0.72	0.50	0.54	0.45	0.56
Resonant Spruce	438	10.25	1.19	0.80	0.78	1.01	0.94	0.46	0.43	−0.38

Material properties for Maple and Spruce were taken from [6]; properties for Resonant Spruce were taken from [39].

The Block-Lanczos solver was employed for the Modal analysis, yielding the modal shapes (Figure 1) and their corresponding natural frequencies (Table 2).

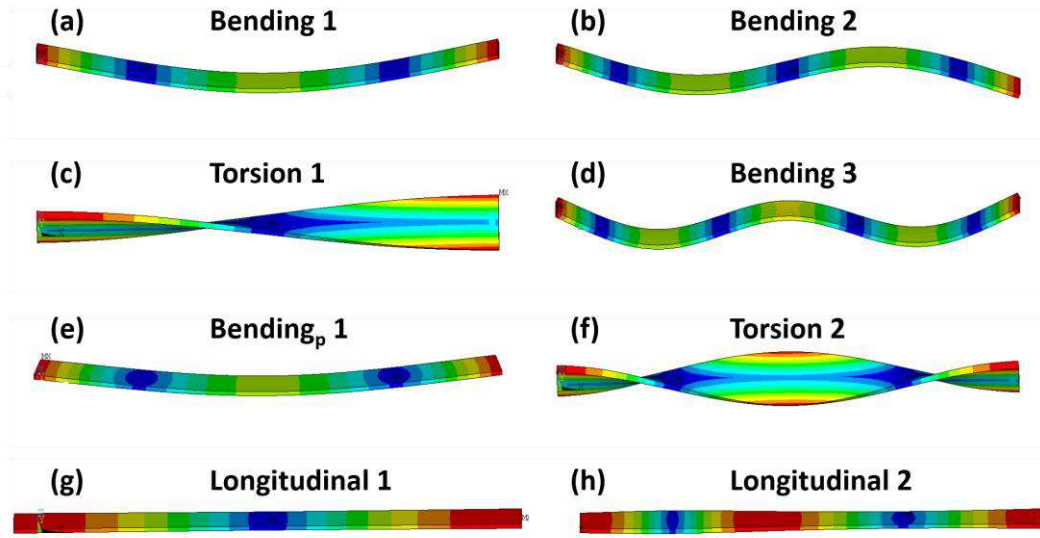


Figure 1. Mode shapes: (a)–(f) depict the first six modes obtained for the models of both spruce specimens. In the case of maple, the bending 4 and the second torsional mode (f) changed order, as the sixth and seventh modes. (e) shows the first perpendicular bending mode (Bending_p), corresponding to bending in the LR plane. The first two longitudinal modes (g) and (h) appeared in a different order for all models – mixed with higher bending and torsional modes and their combinations.

Table 2. Frequencies of individual modes – obtained from the numerical modal analysis

Material Model	$f_{B,1}$ (Hz)	$f_{B,2}$ (Hz)	$f_{T,1}$ (Hz)	$f_{B,3}$ (Hz)	$f_{Bp,1}$ (Hz)	$f_{T,2}$ (Hz)	$f_{L,1}$ (Hz)	$f_{L,2}$ (Hz)
Maple	150	412	734	807	905	1480	6859	13699
Spruce	202	555	648	1081	1208	1316	9335	18639
Resonant Spruce	175	481	765	940	1065	1548	8060	16100

Although the results differed in terms of frequencies, the frequency intervals of bending and longitudinal modes relative to the fundamental frequency of the bending mode were very similar across all materials. That is in good agreement with the intervals of bending modes provided by the Euler–Bernoulli model (2.78 : 5.45) [40]. More pronounced differences were observed in the intervals of torsional vibrations (Table 3).

Table 3. Frequency intervals (*int*) of modes – relative to the fundamental frequency $f_{B,1}$

Material Model	$int_{B,2}$ (-)	$int_{T,1}$ (-)	$int_{B,3}$ (-)	$int_{Bp,1}$ (-)	$int_{T,2}$ (-)	$int_{L,1}$ (-)	$int_{L,2}$ (-)
Maple	2.75	4.90	5.39	6.04	9.88	45.78	91.44
Spruce	2.74	3.21	5.35	5.97	6.51	46.17	92.18
Resonant Spruce	2.75	4.38	5.38	6.09	8.86	46.12	92.13

Additionally, it should be noted that the frequency intervals of the torsional modes relative to one another correspond to an octave (i.e., interval = 2), and the same applies to the frequency intervals of the longitudinal modes. For the Bending_p, the intervals of the second and third modes relative to the first Bending_p mode are as follows: Maple – 2.60 : 4.70; Spruce – 2.45 : 4.20; Resonant Spruce – 2.56 : 4.58. The discrepancy between the bending and Bending_p intervals may be attributed to differences in the anatomical orientation relative to the specimen geometry [40].

2.3. Frequency Resonance Technique

The FRT was used to determine dynamic elastic properties in accordance with ASTM E1875 [41] and ASTM E1876 [24]. Specimens were supported by polyurethane foam at the nodal points corresponding to the individual mode shapes of vibration (Figure 2).

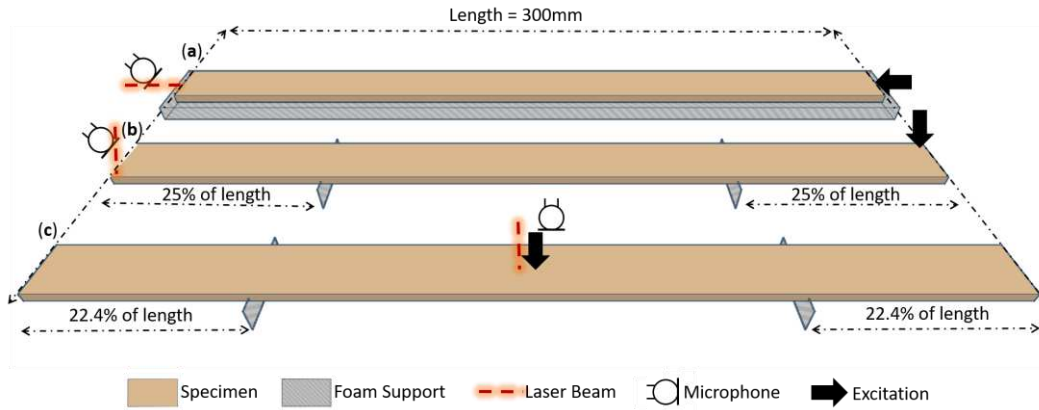


Figure 2. Different configurations used during the measurement: (a) Longitudinal configuration; (b) Torsional configuration – 2nd mode; (c) Bending configuration

Transient vibrations were excited using a plastic mallet at the antinodes of the corresponding mode shapes. For the measurement of longitudinal vibrations, the specimen was laid flat on foam and excited at the front face (Figure 1a). To measure torsional vibrations, the configuration of the second torsional mode was used, with nodes located at 25 % and 75 % of the specimen's length, and excitation applied at one corner of the specimen (Figure 1b). The configuration of the first bending mode included nodes at 22.4 % and 77.6 % of the specimen's length, with the excitation point at the center of the specimen (Figure 1c). Vibrations were sensed at the front face (longitudinal), corner (torsional), and center (bending) using a PDV-100 laser vibrometer (Polytec GmbH, Baden-Württemberg, Germany). An Audix TM-1 measurement microphone (Audix Microphones, Inc., Wilsonville, OR, USA) was positioned close to the points monitored by the vibrometer. Signals from both sensors were recorded simultaneously using a Steinberg UR22MKII external sound card (Steinberg Media Technologies GmbH, Hamburg, Germany) and Audacity software (Audacity Team, Open Source Community) at a sampling frequency (f_s) of 44.1 kHz. A minimum of seven impacts were recorded for each specimen.

2.4. Signal Processing and Spectral Analysis

All tasks related to signal processing, data analysis, and statistical evaluation were performed in MATLAB (MathWorks, Natick, MA, USA). A Fast Fourier Transform (FFT) was performed for the entire signal of specimen vibration. On its frequency spectrum, the resonance peaks were manually determined. Resonance peaks were independently determined for the vibrometer and microphone signals. Then seven impacts with the highest amplitude were selected for each specimen and measurement configuration. The impacts were identified automatically using the *findpeaks* function on the microphone signal. The same events were then extracted from the synchronised vibrometer signal to ensure that both signals were analysed from identical impacts. Each impact was represented by a signal segment composed of a short pre-impact part (2.27 ms), the full course of the impact (333.3 ms), and zero-padding at the end (743 ms). All segments were of equal length (1.08 s). A FFT was performed for each impact segment, with a frequency resolution of $\Delta f < 1$ Hz. After the FFT, the spectral curve was interpolated to obtain a 1000-times finer resolution. Based on the initial determination of frequencies from the whole signal, the automated selection (using the *findpeaks* function) was narrowed to the target frequencies, while allowing a 3% margin on both sides. This ensured that during the automated frequency selection for each impact, the desired frequencies could be refined. The resulting resonance peak frequency was the median of seven impacts.

2.5. Evaluation of Vibro-Acoustic Parameters

The frequencies of individual peaks were assigned to specific modal shapes based on their order (compared with numerical modal analysis) and by observing changes in their amplitude when changing the excitation point and the supports, according to the measurement configurations [28]. From these frequencies, the vibro-acoustic parameters were then calculated as follows:

E_{dyn} was calculated using Equation 1.

$$E_{dyn,n} = 4\rho \left(\frac{f_{L,n}}{n} \right)^2 l^2, \quad (1)$$

Where n is the index of the n th-order mode of the longitudinal vibration, $f_{L,n}$ is the corresponding frequency, and l is the length of the specimen.

MOE_{dyn} was calculated from the frequency of the n th bending mode, according to Equation 2.

$$MOE_{dyn,n} = \frac{4\pi^2 f_{B,n}^2 l^4 \rho}{\beta^4 i^2}, \quad (2)$$

Where $f_{B,n}$ is the frequency of the n th bending mode, β is a constant corresponding to the mode order ($\beta = 4.73$ for $n = 1$; $\beta = 7.853$ for $n = 2$; $\beta = 10.996$ for $n = 3$), and i is the radius of gyration, which for a specimen with rectangular cross-section is calculated using Equation 3.

$$i = \frac{t}{\sqrt{12}}, \quad (3)$$

Where t is the thickness of the specimen. G_{dyn} was determined using Equation 4.

$$G_{dyn,n} = \frac{4lmf_{T,n}^2}{wt} R, \quad (4)$$

where m is the mass of the specimen, $f_{T,n}$ is the frequency corresponding to the n th torsional mode, w is the width of the specimen, and R is a correction factor dependent on the width-to-thickness ratio, as expressed by Equation 5.

$$R = \left[\frac{1 + \left(\frac{w}{t}\right)^2}{4 - 2.521 \frac{t}{w} \left(1 - \frac{1.991}{e^{\frac{\pi w}{t}} + 1}\right)} \right] \left[1 + \frac{0.00851n^2w^2}{l^2} \right] - 0.060 \left(\frac{nw}{l}\right)^{\frac{3}{2}} \left(\frac{w}{t} - 1\right)^2, \quad (5)$$

Specific moduli were calculated according to Equation 6.

$$E_{dyn,n}' = \frac{E_{dyn,n}}{\rho}; MOE_{dyn,n}' = \frac{MOE_{dyn,n}}{\rho}; G_{dyn,n}' = \frac{G_{dyn,n}}{\rho}; \quad (6)$$

K for longitudinal (L), bending (B), and torsional (T) vibrations was calculated using Equation 7.

$$K_{L,n} = \sqrt{\frac{E_{dyn,n}}{\rho^3}}; K_{B,n} = \sqrt{\frac{MOE_{dyn,n}}{\rho^3}}; K_{T,n} = \sqrt{\frac{G_{dyn,n}}{\rho^3}}, \quad (7)$$

Z was calculated according to Equation 8.

$$Z_{L,n} = \rho \sqrt{\frac{E_{dyn,n}}{\rho}}; Z_{B,n} = \rho \sqrt{\frac{MOE_{dyn,n}}{\rho}}; Z_{T,n} = \rho \sqrt{\frac{G_{dyn,n}}{\rho}}, \quad (8)$$

Two methods were used to determine the loss factor (η) for each observed frequency peak. The first, established one – the Half-Power Method (HPM) – determines the damping from the bandwidth at 70.7% of the frequency peak amplitude (i.e., peak amplitude / $\sqrt{2}$) after FFT. At this amplitude level, the upper and lower frequencies (f_u and f_l) around the peak were used to calculate η according to Equation 9.

$$\eta = \frac{f_u - f_l}{f}, \quad (9)$$

The second method was based on monitoring the decrease in the amplitude of a frequency peak after FFT through time, while gradually shifting the analysis window through the impulse region. The initial analysis window was identical to that used in the HPM. Based on the highest analyzed frequency in the given signal (e.g., in the longitudinal configuration, this corresponds to the second longitudinal frequency), the window was then shifted by one third of the period of that frequency, and FFT was performed again to determine the amplitude of the same peaks (with a tolerance of $\pm 3\%$ frequency deviation from the previous window). In this way, the window was successively shifted until the end of the impulse signal. From the determined peak amplitudes, the values in the range from the maximum amplitude to 10% of the maximum were transformed into a linear course by applying the natural logarithm. A linear regression was then fitted, and its slope b was determined. The b represents the damping coefficient in the exponential decay of amplitude (A) over time (t) according to Equation 10.

$$A(t) = A_0 e^{-bt}, \quad (10)$$

The η from the Amplitude Decay over Time (ADT) method was calculated according to Equation 11.

$$\eta = \frac{b}{\pi f}, \quad (11)$$

The η values obtained from both methods were further used for the calculation of ACE , according to Equation 12, and for the calculation of $RACE$, according to Equation 13.

$$ACE_{L,n} = \frac{K_{L,n}}{\eta_{L,n}}; ACE_{B,n} = \frac{K_{B,n}}{\eta_{B,n}}; ACE_{T,n} = \frac{K_{T,n}}{\eta_{T,n}}, \quad (12)$$

$$RACE_{L,n} = \frac{\sqrt{\frac{E_{dyn,n}}{\rho}}}{\eta_{L,n}}; RACE_{B,n} = \frac{\sqrt{\frac{MOE_{dyn,n}}{\rho}}}{\eta_{B,n}}; RACE_{T,n} = \frac{\sqrt{\frac{G_{dyn,n}}{\rho}}}{\eta_{T,n}}, \quad (13)$$

Based on the requirement that materials for stringed-instrument soundboards should exhibit a high E/G ratio [6,35], aggregate parameters derived from ACE and $RACE$ —namely, ratios of elastic moduli, ρ and η —are introduced in Equations 14 and 16. The arithmetic means, expressing aggregation across modes (both shapes and orders), are given in Equations (15) and (17).

$$ACE_{L/T,n} = \sqrt{\frac{E_{dyn,n}}{G_{dyn,n} \cdot \rho^3}}; \quad ACE_{B/T,n} = \sqrt{\frac{MOE_{dyn,n}}{G_{dyn,n} \cdot \rho^3}} \quad (14)$$

Where $\bar{\eta}$ is the mean value of η over the set of modes considered in the calculation

$$ACE_{combi} = \frac{1}{N} \sum_{n=1}^N (ACE_{L/T,n} + ACE_{B/T,n}) \quad (15)$$

Where N is the total number of evaluated alternatives.

$$RACE_{L/T,n} = \sqrt{\frac{E_{dyn,n}}{G_{dyn,n} \cdot \rho}}; \quad RACE_{B/T,n} = \sqrt{\frac{MOE_{dyn,n}}{G_{dyn,n} \cdot \rho}} \quad (16)$$

$$RACE_{combi} = \frac{1}{N} \sum_{n=1}^N (RACE_{L/T,n} + RACE_{B/T,n}) \quad (17)$$

2.6. Data Analysis

The results obtained using the two types of sensors were compared using paired analyses within each material group. The normality of paired differences was tested, and depending on the outcome, either a paired t-test or the Wilcoxon signed-rank test was applied. Agreement between methods was further evaluated using Bland–Altman analysis, including the calculation of bias and limits of agreement. In addition to statistical significance, practical significance was also assessed.

Comparisons between material groups were performed separately for each sensor. Normality and homogeneity of variances were verified prior to testing. When the assumptions were met, one-way ANOVA with Tukey-type post-hoc analysis was used; otherwise, the Kruskal–Wallis test followed by Dunn–Sidak multiple comparisons were applied. The significance level was set at $\alpha = 0.05$.

3. Results and Discussion

The ρ determination revealed that the tested materials could be categorized into two statistically comparable groups, each containing one solid wood accompanied by plywood. The group with higher ρ consisted of Birch_{ply}, Maple, and Pine_{ply}, while the group with lower ρ consisted of Poplar_{ply} and Spruce. Birch_{ply} and Pine_{ply} showed slightly lower ρ than literature values ($650 \text{ kg} \cdot \text{m}^{-3}$ for Birch_{ply} and $630 \text{ kg} \cdot \text{m}^{-3}$ for Pine_{ply}), whereas Poplar_{ply} exhibited a noticeably lower ρ than the reported $570 \text{ kg} \cdot \text{m}^{-3}$ [40]. The ρ values of Maple and Spruce fall within literature ranges [41]. Although lower coefficient of variation (CV) was expected for plywood [42], Poplar_{ply} showed similar variability to Maple, and Pine_{ply} even higher variability (Table 4).

Table 4. Median values of ρ ($\text{kg} \cdot \text{m}^{-3}$) for each tested material, with CV (%) in parentheses

Group	ρ	CV
Birch _{ply}	613.6	(0.7)
Maple	630.3	(2.6)
Pine _{ply}	612.8	(4.4)
Poplar _{ply}	435.4	(2.6)
Spruce	444.5	(6.9)

For all specimens and both type of sensors, the first and second longitudinal modes were identified from the longitudinal configuration. The bending configuration yielded the first and third bending modes, while the torsional configuration provided the first three bending modes and the first torsional mode. The second torsional mode was not consistently identified and was therefore excluded from further analysis, along with higher and Bending_p modes).

No statistically significant differences between vibrometer and microphone data were found for the frequency of the first longitudinal mode; only Poplar_{ply} exhibited increased variability. For the frequency of the second longitudinal mode, a statistically significant systematic difference was found only for Spruce; however, high variability of the differences and wide limits of agreement across materials indicate limited reliability, likely due to low peak amplitude, especially for the microphone. All bending mode frequencies showed statistically significant differences between sensors. However, these differences were very small and practically negligible. Frequencies derived from torsional configuration were consistently higher than those from bending configuration for the first bending mode, with the largest difference observed for Poplar_{ply} (1.36 Hz). A similar result was obtained for the third bending

mode from both configurations, with statistically significant differences again, but they remained negligible. In contrast, the frequency of the first torsional mode showed limited agreement between the vibrometer and the microphone. For Poplar_{ply}, a statistically and practically significant difference was observed, and substantially increased variability in the differences was found across all materials. This disagreement is attributed not primarily to differences between the sensors but rather to inaccurate mode identification, as the torsional mode was close to certain bending, indicating that sensor choice can significantly affect torsional frequency determination.

Given the overall good agreement between the sensors and the slightly lower variability of the vibrometer data, the reported frequencies (Table 5) and the derived parameters (Tables 6–11) are based on vibrometer measurements.

Table 5. Median values of mode frequencies (Hz), with CV (%) in parentheses

Group	$f_{B,1}$	CV	$f_{B,2}$	CV	$f_{T,1}$	CV	$f_{B,3}$	CV	$f_{L,1}$	CV	$f_{L,2}$	CV
Birch _{ply}	177	(1.2)	487	(2.0)	537	(1.2)	940	(1.0)	6 946	(1.6)	13 938	(2.1)
Maple	160	(4.9)	449	(4.7)	809	(2.7)	873	(4.5)	7 303	(3.5)	14 652	(3.5)
Pine _{ply}	208	(12.1)	556	(11.7)	628	(6.2)	1020	(8.1)	6 890	(9.7)	13 513	(7.7)
Poplar _{ply}	132	(4.4)	368	(4.2)	529	(4.5)	713	(4.1)	5 386	(3.8)	10 615	(3.5)
Spruce	206	(3.2)	570	(3.4)	671	(2.6)	1103	(3.4)	9 424	(2.1)	18 871	(2.0)

As with the numerical modal analysis results, the experimentally determined frequencies, which differed across materials, were presented for further comparison as intervals for individual modes relative to the fundamental bending mode (Table 6).

Table 6. Median values of frequency intervals (–) of modes, with CV (%) in parentheses

Group	$int_{B,2}$	CV	$int_{T,1}$	CV	$int_{B,3}$	CV	$int_{L,1}$	CV	$int_{L,2}$	CV
Birch _{ply}	2.76	(1.5)	3.04	(1.9)	5.33	(0.8)	39.35	(0.8)	79.43	(1.7)
Maple	2.78	(1.9)	5.02	(4.7)	5.38	(2.1)	45.53	(1.8)	90.73	(2.5)
Pine _{ply}	2.74	(3.6)	3.07	(12.2)	5.03	(6.5)	33.34	(5.5)	64.90	(7.9)
Poplar _{ply}	2.81	(1.7)	4.14	(5.7)	5.46	(1.5)	40.79	(1.4)	81.71	(4.3)
Spruce	2.76	(0.4)	3.23	(4.5)	5.33	(0.6)	45.77	(1.9)	91.69	(2.0)

The intervals of the first longitudinal mode showed statistical comparability among Birch_{ply}–Pine_{ply}–Poplar_{ply} and Poplar_{ply}–Maple–Spruce. Intervals of the second longitudinal mode separated the materials into solid wood and plywood. For intervals of bending modes, Poplar_{ply} was comparable only with Maple, and for the third mode, Maple differed from Pine_{ply}. All other materials were mutually comparable. The bending intervals agree well with results of numerical modal analysis and with the literature values (2.76 : 5.40 [43], or 2.78 : 5.45 [40]). When complemented by results from tests on various geometries [34,44], this suggests that the bending mode intervals are independent of the material and governed by specimen geometry. The torsional intervals were statistically comparable between Birch_{ply}–Pine_{ply}–Spruce, Spruce–Poplar_{ply}, and Poplar_{ply}–Maple. Consistent with simulations, torsional peak lay between the first and second bending modes; however, in some plywood cases, the torsional peak was close to bending peaks and of lower amplitude (especially in microphone data), complicating identification. To ensure that the torsional mode can be reliably identified, it should be clearly separated from the others.

For all materials, $E_{dyn,1}$ and $E_{dyn,2}$ were statistically comparable. Both E_{dyn} were comparable to MOE_{dyn} for solid wood, but not for plywood. While E_{dyn} is typically ~ 10% higher than MOE_{dyn} [25,37,45], the differences in solid wood were <2%, whereas in plywood MOE_{dyn} was ~ 30% higher than E_{dyn} . This reflects the layered structure of plywood, where stress distribution differs between bending and longitudinal vibration [46]. Spruce exhibits the highest E_{dyn} , and statistical comparison tests did not reveal any significant differences relative to Maple. Maple is comparable to Birch_{ply}–Pine_{ply}, which are both at the limit of comparability with Poplar_{ply}, with the lowest E_{dyn} values. Compared to $MOE_{dyn,1}$, a statistically significant difference was found for $MOE_{dyn,2}$ of Poplar_{ply}, in the sense of consistently higher values; however, in practical terms, the differences were negligible for all material groups. For $MOE_{dyn,3}$ determined from both configurations, a statistically significant systematic shift of values was observed for all materials except Maple, with Birch_{ply} showing practically non-comparable results. $MOE_{dyn,1}$ from the torsional configuration exhibited a consistently small offset in values (as did the corresponding frequencies from which it was calculated); nevertheless, it was comparable for all materials with $MOE_{dyn,1}$ determined from the bending configuration. The MOE_{dyn} values were comparable across materials except Poplar_{ply} (Table 7). Spruce was at the limit of comparability with Maple. Maple exhibits the highest G_{dyn} values. Spruce is comparable to Pine_{ply}, and at the limit of comparability with Birch_{ply}. Group of plywood is comparable. The plywood MOE_{dyn} values were higher by 16% for Birch_{ply} and by 26% for Pine_{ply} compared to the stated static modulus of elasticity in bending determined on specimens with similar ρ [40]. The differences between the static and dynamic bending moduli measured on the same specimens are reported to be approximately 2–15% [26]. The results of solid wood are consistent with the range of values reported for maple and spruce with similar ρ , obtained by FRT [47].

Table 7. Median values of moduli of elasticity (GPa), with CV (%) in parentheses

Group	$E_{dyn,1}$	CV	$MOE_{dyn,1}$	CV	$G_{dyn,1}$	CV
Birch _{ply}	10.72	(3.2)	13.79	(2.6)	0.69	(2.2)

Maple	12.22	(8.5)	12.08	(11.9)	1.61	(7.9)
Pine _{ply}	10.20	(22.2)	13.36	(26.0)	0.68	(15.2)
Poplar _{ply}	4.53	(9.3)	6.01	(9.5)	0.52	(8.3)
Spruce	14.28	(9.9)	14.54	(11.0)	0.81	(6.3)

For all groups, a strong linear correlation between E_{dyn} and MOE_{dyn} was confirmed [37], but a very low linear correlation between E_{dyn} and G_{dyn} was observed (for Birch_{ply} $R^2 = 0.14$; Maple $R^2 = 0.32$; Pine_{ply} $R^2 = 0.19$; Poplar_{ply} $R^2 = 0.21$; Spruce $R^2 = 0.42$). A stronger linear correlation between G_{dyn} and MOE_{dyn} was found for the solid wood (Table 8). The inconclusive dependence in plywood is in contradiction with the strong linear correlation expected in thin wood composite materials [48].

Table 8. Linear regressions relating MOE_{dyn} to E_{dyn} and to G_{dyn}

Group	E_{dyn}	R^2	G_{dyn}	R^2
Birch _{ply}	$0.85 \cdot MOE_{dyn} - 0.90$	0.77	$-0.00 \cdot MOE_{dyn} + 0.76$	0.01
Maple	$0.67 \cdot MOE_{dyn} + 4.03$	0.88	$0.06 \cdot MOE_{dyn} + 0.90$	0.44
Pine _{ply}	$0.63 \cdot MOE_{dyn} + 2.04$	0.88	$0.02 \cdot MOE_{dyn} + 0.51$	0.26
Poplar _{ply}	$0.71 \cdot MOE_{dyn} + 0.20$	0.95	$0.03 \cdot MOE_{dyn} + 0.35$	0.15
Spruce	$0.82 \cdot MOE_{dyn} + 2.22$	0.87	$0.03 \cdot MOE_{dyn} + 0.44$	0.61

When expressing the specific moduli, the division into statistically comparable material groups turned out to be the same for E_{dyn}' as for E_{dyn} . Due to the low variability of the Birch_{ply} data, no statistically significant difference between Spruce and Birch_{ply} was detected for MOE_{dyn}' . Otherwise, Spruce formed a distinct material group with the highest values. Birch_{ply}–Pine_{ply}–Maple were statistically comparable. Pine_{ply}–Maple were both at the level of statistical comparability with Poplar_{ply}. No statistically significant difference was proved between Maple and Spruce in terms of G_{dyn}' ; same as between Spruce and Poplar_{ply}. All plywood materials were mutually comparable (Table 9).

Table 9. Median values of specific moduli of elasticity ($GPa \cdot g^{-1} \cdot cm^3$), with CV (%) in parentheses

Group	$E_{dyn,1}'$	CV	$MOE_{dyn,1}'$	CV	$G_{dyn,1}'$	CV
Birch _{ply}	17.37	(3.2)	22.35	(2.4)	1.12	(2.1)
Maple	19.20	(7.0)	19.11	(10.2)	2.57	(6.4)
Pine _{ply}	17.09	(18.9)	22.46	(23.0)	1.14	(12.4)
Poplar _{ply}	10.44	(7.5)	13.84	(7.7)	1.18	(7.7)
Spruce	31.97	(4.1)	32.37	(5.3)	1.83	(5.3)

K_L and K_B were statistically comparable for solid wood but not for plywood. Spruce exhibited the highest K_L values, clearly separated from other materials. The remaining materials were statistically comparable, except for Poplar_{ply} and Pine_{ply}. A similar pattern was observed for the K_B values. Spruce showed substantially higher values than the other groups. Among the remaining materials, Poplar_{ply} exhibited the highest values, which were not statistically comparable with Maple; the other groups with lower K_B were comparable. K_T showed comparability within Spruce–Maple–Poplar_{ply}, and Birch_{ply}–Pine_{ply}. Based on the results (Table 10), the tested Spruce was classified as resonant, according to the definition of $K > 12$ [49]. For a more detailed description of the properties of resonance wood, a requirement for a K_T may be introduced. Due to the absence of this parameter value in the literature, an indicative threshold of $K_T > 3$ is proposed based on the results of the present study; however, this issue warrants further research.

Table 10. Median values of K ($m^4 \cdot kg^{-1} \cdot s^{-1}$), with CV (%) in parentheses

Group	K_L	CV	K_B	CV	K_T	CV
Birch _{ply}	6.80	(1.8)	7.69	(1.3)	1.72	(1.3)
Maple	7.01	(3.2)	6.99	(4.2)	2.54	(3.1)
Pine _{ply}	6.75	(7.8)	7.69	(10.0)	1.77	(5.3)
Poplar _{ply}	7.33	(2.8)	8.53	(3.0)	2.52	(4.4)
Spruce	12.88	(5.5)	13.04	(5.6)	3.04	(9.1)

Z_L and Z_B were found statistically comparable for Maple, and Spruce, but not for Birch_{ply} and Poplar_{ply}. For Pine_{ply}, a statistically significant difference was found; however, due to the higher variability of the data, there was sufficient overlap between Z_L and Z_B for the differences to be at the limit of practical acceptability. The Z_L values were comparable among Maple–Birch_{ply} and Birch_{ply}–Pine_{ply}–Spruce. Poplar_{ply} was clearly isolated with the lowest values. Z_B was comparable between Birch_{ply}–Maple–Pine_{ply}, and Maple–Spruce. Poplar_{ply} again showed no comparability with the other materials. The only statistically comparable Z_T values between groups were found for Birch_{ply}–Pine_{ply} (Table 11). Compared with the available literature data for the $Z_{L,1}$ of Maple and Spruce, where the $Z_{L,1}$ value of Maple is 10.9% higher than that of Spruce, the results of the present study—showing Maple to be 9.9% higher than Spruce—are in good agreement with the reported values [41].

Table 11. Median values of Z ($kg \cdot m^{-2} \cdot s^{-1}$) $\cdot 10^{-6}$, with CV (%) in parentheses

Group	Z_L	CV	Z_B	CV	Z_T	CV
Birch _{ply}	2.57	(1.7)	2.92	(1.5)	0.65	(1.2)
Maple	2.77	(5.2)	2.76	(6.8)	1.01	(5.0)
Pine _{ply}	2.46	(13.0)	2.82	(15.0)	0.65	(9.2)
Poplar _{ply}	1.40	(5.8)	1.63	(5.9)	0.48	(4.8)
Spruce	2.52	(8.4)	2.56	(8.8)	0.60	(6.2)

Statistically significant differences in η , were observed between the microphone and vibrometer, mostly as a constant shift caused by frequency offsets; these differences were generally negligible, except as follows:

Using the HPM method to determine $\eta_{L,1}$, significant differences between the sensors were found for Birch_{ply} and Poplar_{ply}. For $\eta_{L,2}$ differences were observed across all material groups. In this case, the differences were likely due to low signal-to-noise ratio, particularly for the microphone. For η values determined using HPM for bending modes, no significant differences between the sensors were found. For $\eta_{T,1}$, high data variability was observed in all groups (CV up to 178%), along with pronounced differences between the sensors. This may have been caused by inaccurate torsional mode identification and, in particular, by the limited accuracy of the HPM method when determining η at a peak located close to another one [36]. Using the ADT method to determine $\eta_{L,1}$, significant differences between the sensors were found for Birch_{ply} and Pine_{ply}. For $\eta_{L,2}$, the sensors were not comparable across all groups except Pine_{ply}. Compared to HPM, ADT showed differences between the sensors in bending modes, namely for $\eta_{B,2}$ in Poplar_{ply} and for $\eta_{B,3}$ derived from the torsional configuration in Pine_{ply}. For the $\eta_{T,1}$ determination, a difference between the sensors was found for Spruce.

The comparison of η determination methods was performed on all data (both sensors, all groups) for the $\eta_{B,1}$ mode in bending and torsional configurations (Figure 3). A strong linear relationship was found, indicating that the experimental ADT method provides results highly consistent with the established HPM.

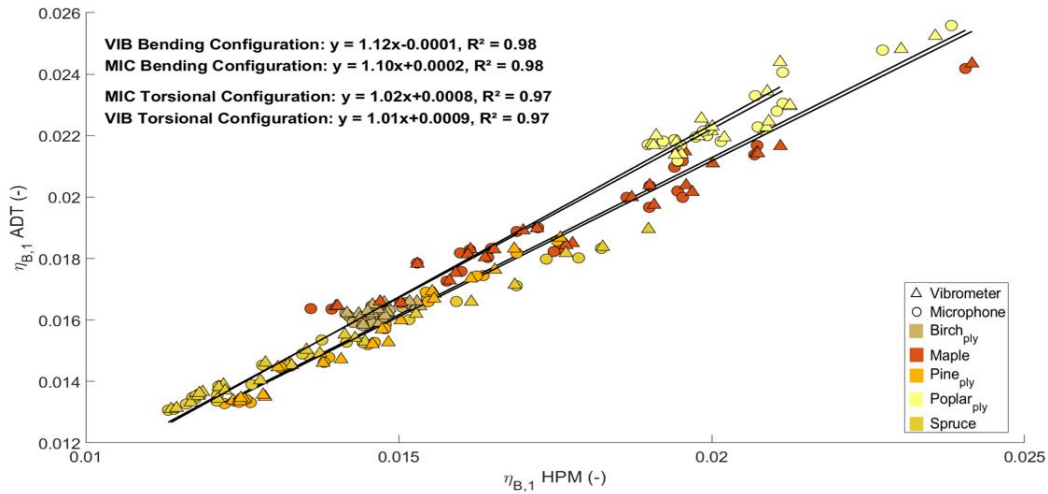


Figure 3. The comparison of ADT and HPM for the $\eta_{B,1}$ determination

Based on method comparison and consistency, and considering the higher variability of longitudinal and torsional η values obtained by HPM, ADT-derived values are presented (Table 12). Vibrometer data were selected for presentation, as they exhibited slightly lower variability, particularly in the longitudinal and torsional modes.

Table 12. Median values of η (-), with CV (%) in parentheses

Group	$\eta_{L,1}$	CV	$\eta_{L,2}$	CV	$\eta_{B,1}$	CV	$\eta_{B,2}$	CV	$\eta_{B,3}$	CV	$\eta_{T,1}$	CV
Birch _{ply}	0.0089	(7.2)	0.0170	(8.9)	0.0146	(1.9)	0.0244	(21.3)	0.0205	(8.4)	0.0281	(17.5)
Maple	0.0091	(11.4)	0.0167	(9.9)	0.0159	(6.6)	0.0293	(6.1)	0.0216	(9.3)	0.0231	(24.0)
Pine _{ply}	0.0103	(16.2)	0.0224	(17.8)	0.0142	(11.3)	0.0260	(21.3)	0.0233	(10.4)	0.0282	(16.0)
Poplar _{ply}	0.0093	(14.1)	0.0173	(26.3)	0.0195	(3.7)	0.0285	(7.4)	0.0220	(17.4)	0.0379	(8.9)
Spruce	0.0030	(46.3)	0.0141	(18.0)	0.0119	(3.8)	0.0251	(7.3)	0.0160	(4.9)	0.0268	(14.9)

The available data for $\eta_{L,1}$, determined using the HPM method for Maple ($\eta_{L,1} = 0.0175$) and Spruce ($\eta_{L,1} = 0.0115$) [41], were higher than the values obtained in the present study, and they corresponded more closely to the results for $\eta_{B,1}$. This is despite the lower presented EMC of specimens ($EMC = 8\%$) [41], which would generally be expected to reduce η [36]. The observed differences may be related to variations in specimen dimensions ($10 \times 10 \times 400$ mm), as specimen geometry is not considered in the calculation of the η coefficient; this aspect could be explored in future research.

For the longitudinal modes, there was a significant difference in damping between $\eta_{L,1}$ and $\eta_{L,2}$ for all tested materials (both sensors), with the largest difference found in Spruce, where $\eta_{L,2}$ is 372% higher than $\eta_{L,1}$. Values of $\eta_{L,1}$ and $\eta_{B,1}$ were comparable

only for microphone-measured Pine_{ply}. In all other cases, $\eta_{L,1}$ was substantially lower. For solid wood, there was a difference between $\eta_{B,1}$ from the bending configuration and the torsional configuration (for both the microphone and the vibrometer), with the torsional configuration showing an $\eta_{B,1}$ increase of 23% for Maple and 22% for Spruce. No significant difference was found for Pine_{ply}, while Birch_{ply} and Poplar_{ply} were at the limit of comparability. The increase of $\eta_{B,1}$ from the torsional configuration was 2% for Pine_{ply}, 3% for Birch_{ply}, and 7% for Poplar_{ply}. $\eta_{B,2}$ and $\eta_{B,1}$ (both configurations) were found statistically incomparable for all tested groups, with $\eta_{B,2}$ being 47% higher for Poplar_{ply} and up to 112% higher for Spruce. For $\eta_{B,3}$, no significant differences between configurations were found across groups. $\eta_{B,3}$ was not found to be comparable to $\eta_{B,1}$. $\eta_{B,3}$ and $\eta_{B,2}$ are at the limit of comparability for Pine_{ply}. Values of $\eta_{T,1}$ were not comparable with $\eta_{B,1}$ for any group, except for $\eta_{B,1}$ from the torsional configuration for Maple —vibrometer data. $\eta_{T,1}$ was very comparable with $\eta_{B,2}$, except for Maple (vibrometer), Poplar_{ply} (vibrometer), and Spruce (microphone). $\eta_{T,1}$ and $\eta_{B,3}$ from the bending configuration were comparable for Pine_{ply} (microphone), and Maple. $\eta_{T,1}$ and $\eta_{B,3}$ from the torsional configuration were comparable for Birch_{ply} (microphone), Maple, Pine_{ply} (microphone), Poplar_{ply} (microphone), and Spruce (microphone). For the bending modes, the assumption that η increases with increasing mode number [29,36] was not confirmed, because $\eta_{B,2}$ was the highest among the bending modes for all groups. For the longitudinal modes, this assumption was verified.

Across materials, Spruce exhibited the lowest $\eta_{L,1}$, significantly distinct from otherwise comparable groups. For $\eta_{L,2}$ obtained using the vibrometer, all groups were comparable except for Pine_{ply}, which was comparable only with Birch_{ply}. When measured using the microphone, Birch_{ply}–Maple–Pine_{ply}, and Pine_{ply}–Poplar_{ply} were statistically comparable. The $\eta_{B,1}$ from the bending configuration was statistically comparable for Birch_{ply}–Maple–Pine_{ply}. The $\eta_{B,1}$ determined from the torsional configuration showed comparability between Birch_{ply}–Pine_{ply}–Spruce and Maple–Poplar_{ply}. For $\eta_{B,2}$, a very similar trend was observed; however, the differences between groups were slightly smaller, leading to more groups becoming statistically comparable. The $\eta_{B,3}$ values determined from the bending configuration were comparable across all groups except Spruce. A similar pattern was observed for $\eta_{B,3}$ from the torsional configuration, although no significant difference between Spruce and Maple was proved. For $\eta_{T,1}$ based on vibrometer data, Poplar_{ply} differed from all other groups. When using microphone, all groups were comparable except for Poplar_{ply}–Spruce. Figure 4 presents a comparison of the η of individual modes based on vibrometer data across the material groups.

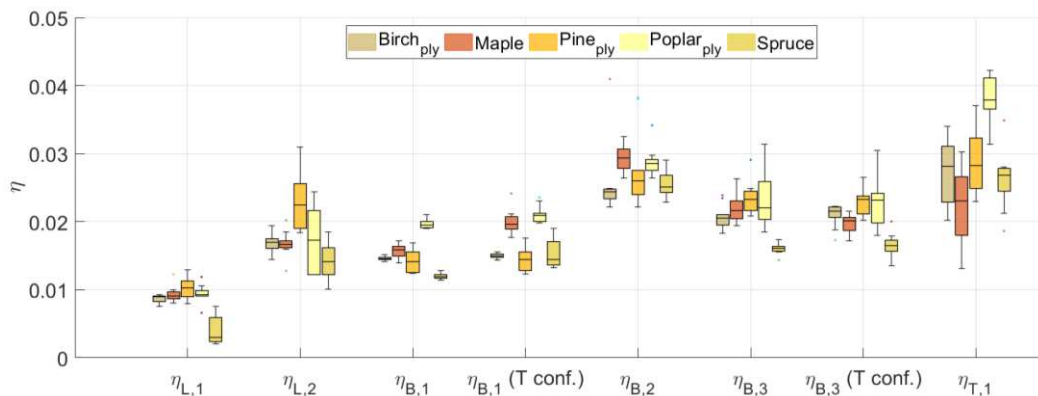


Figure 4. Modal η across material groups

The *ACE* and *RACE* parameters, due to their calculation, were significantly influenced by the results of η . The consistency of the results obtained using the microphone and the vibrometer is also derived from η , and therefore, it applies that the largest differences between the sensors were recorded for the longitudinal modes, while the most consistent determination is achieved from the bending modes. The *ACE* and *RACE* exhibited very similar trends; therefore, their evaluation is presented jointly. The results from both longitudinal modes were statistically non-comparable across groups and sensors. For the first bending mode, differences between bending and torsional configurations were found only in solid wood, consistent with η . No comparability was found between the first and second bending modes. For first and third bending modes, no significant difference was found only for Poplar_{ply}, determined by the vibrometer data. The third bending mode results were otherwise comparable across groups and sensors.

$ACE_{L,1}$ clearly separated of Spruce, which exhibited the highest values and high variability, while other materials were comparable. A similar distribution was observed for $ACE_{L,2}$; moreover, in the vibrometer data, Pine_{ply} with lowest values was not comparable with Maple–Poplar_{ply}. For bending modes, *ACE* again distinguished Spruce from other materials. The clearest separation among the remaining groups occurred for $ACE_{B,1}$ obtained from the torsional configuration, where two distinct groups emerged: Birch_{ply}–Pine_{ply} and Maple–Poplar_{ply}. For higher bending modes, differences were less pronounced and limited to individual non-comparable cases depending on the sensor. The $ACE_{T,1}$ results obtained from the vibrometer divided the materials into solid wood and plywood. When using the microphone data, Spruce was again separated from the other materials. Median *ACE* values from vibrometer data are summarised in Table 13.

Table 13. Median values of *ACE* ($\text{m}^4 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$), with CV (%) in parentheses

Group	$ACE_{L,1}$	CV	$ACE_{L,2}$	CV	$ACE_{B,1}$	CV	$ACE_{B,2}$	CV	$ACE_{B,3}$	CV	$ACE_{T,1}$	CV
Birch _{ply}	761.88	(8.9)	405.65	(9.3)	525.02	(2.9)	316.05	(13.4)	371.28	(8.2)	61.14	(18.0)
Maple	751.34	(12.0)	411.66	(11.4)	441.60	(10.3)	238.56	(8.8)	324.15	(11.4)	109.16	(27.9)
Pine _{ply}	617.94	(20.2)	279.97	(20.7)	543.14	(19.9)	261.67	(17.9)	308.17	(12.4)	60.57	(13.5)

Poplar _{ply}	790.03	(17.2)	429.55	(26.0)	440.15	(6.0)	297.77	(5.4)	389.48	(17.6)	67.48	(11.2)
Spruce	4228.94	(45.3)	957.77	(20.3)	1081.33	(5.1)	517.89	(9.2)	796.85	(9.3)	114.21	(20.3)

Evaluation of *RACE* results revealed certain differences relative to *ACE*-based group comparisons: Maple exhibited non-comparable *RACE_{L,1}* values with Pine_{ply}–Poplar_{ply}, when using microphone data. *RACE_{L,2}* additionally revealed a separation into groups: Birch_{ply}–Maple, Pine_{ply}–Poplar_{ply}. The results of *RACE* from the bending modes showed, in addition to an isolated Spruce with the highest values, also an isolated Poplar_{ply} with the lowest values. For *RACE_{B,1}*, no significant difference was proved between Poplar_{ply} and Maple. *RACE_{B,2}* indicated greater comparability among the groups, including Spruce, reducing the previously observed differences. For *RACE_{B,3}*, Spruce was again isolated, while among the remaining groups, Poplar_{ply} was comparable only with Pine_{ply} and Birch_{ply} only with Maple—thus showing a trend similar to the longitudinal modes. For *RACE_{T,1}* based on vibrometer data, Maple exhibited the highest values, non-comparable with the other groups. Spruce was at the limit of comparability with Birch_{ply}. In microphone data, Spruce was comparable with Maple, which was comparable with Birch_{ply} and Pine_{ply}. All plywood groups were mutually comparable for both sensors. An overview of the *RACE* medians obtained from vibrometer data is presented in Table 14.

Table 14. Median values of *RACE* (km·s⁻¹), with CV (%) in parentheses

Group	<i>RACE_{L,1}</i>	CV	<i>RACE_{L,2}</i>	CV	<i>RACE_{B,1}</i>	CV	<i>RACE_{B,2}</i>	CV	<i>RACE_{B,3}</i>	CV	<i>RACE_{T,1}</i>	CV
Birch _{ply}	466.33	(8.6)	249.20	(9.2)	324.05	(2.8)	193.46	(13.4)	228.03	(8.2)	37.61	(17.6)
Maple	476.92	(12.4)	259.48	(12.4)	279.41	(11.4)	151.68	(10.3)	199.10	(12.8)	69.71	(27.6)
Pine _{ply}	383.85	(21.8)	175.58	(18.4)	336.59	(22.1)	164.05	(16.6)	187.74	(12.8)	37.36	(14.2)
Poplar _{ply}	346.66	(17.3)	187.36	(26.1)	192.42	(7.3)	131.75	(4.9)	169.14	(17.5)	28.43	(11.3)
Spruce	1888.52	(43.2)	400.51	(17.6)	485.46	(5.4)	227.12	(8.0)	354.22	(4.6)	50.55	(16.4)

The results of aggregated representation using *ACE_{LT,1}* were statistically comparable for both sensors for each material, indicating a more robust representation in respect to sensors—compared to *ACE_{L,1}* and *ACE_{T,1}*. *ACE_{LT,1}* confirmed the isolated highest values of Spruce, and the mutually comparable values of the remaining materials. The *ACE_{BT,1}* exhibited comparability between data obtained using the torsional and bending testing configurations and was also statistically comparable for both sensors across materials. The *ACE_{BT,1}* results divided the materials into three distinct groups: Spruce, plywood, and Maple. When expressed as *ACE_{combi}*, Maple and Pine_{ply}–Poplar_{ply} were no longer significantly differentiated.

RACE_{LT,1} did not eliminate the inconsistency between the sensors for the Spruce and Poplar_{ply} data; however, it confirmed a greater ability to differentiate among the plywood materials, which were comparable in *ACE_{LT,1}*. The *RACE_{BT,1}* showed comparability of results derived from torsional and bending measurement configurations, but also statistical inconsistency between the sensors for Spruce data. The *RACE_{BT,1}* results distinguish the materials into three distinct groups: Spruce, Birch_{ply}–Pine_{ply}, and Maple–Poplar_{ply} (Table 15). It may therefore represent a more sensitive parameter for material assessment than *ACE_{BT,1}*. When expressed as *RACE_{combi}*, the statistical inconsistency between the sensors remained, as the division into comparable groups (Figure 5).

Table 15. Median values of *ACE* (kg^{-3/2}·m^{-9/2})·10⁻³, and *RACE* (kg^{-1/2}·m^{-3/2}), with CV (%) in parentheses

Group	<i>ACE_{LT,1}</i>	CV	<i>ACE_{BT,1}</i>	CV	<i>RACE_{LT,1}</i>	CV	<i>RACE_{BT,1}</i>	CV
Birch _{ply}	13.69	17.5	13.42	13.7	8.43	17.1	8.26	13.3
Maple	11.11	13.8	9.19	12.1	7.02	13.0	5.72	11.9
Pine _{ply}	12.82	17.7	13.01	20.6	7.74	17.8	7.73	21.4
Poplar _{ply}	13.20	7.8	12.63	7.3	5.79	7.9	5.58	7.7
Spruce	29.66	17.9	24.57	13.5	12.84	13.5	10.66	10.3

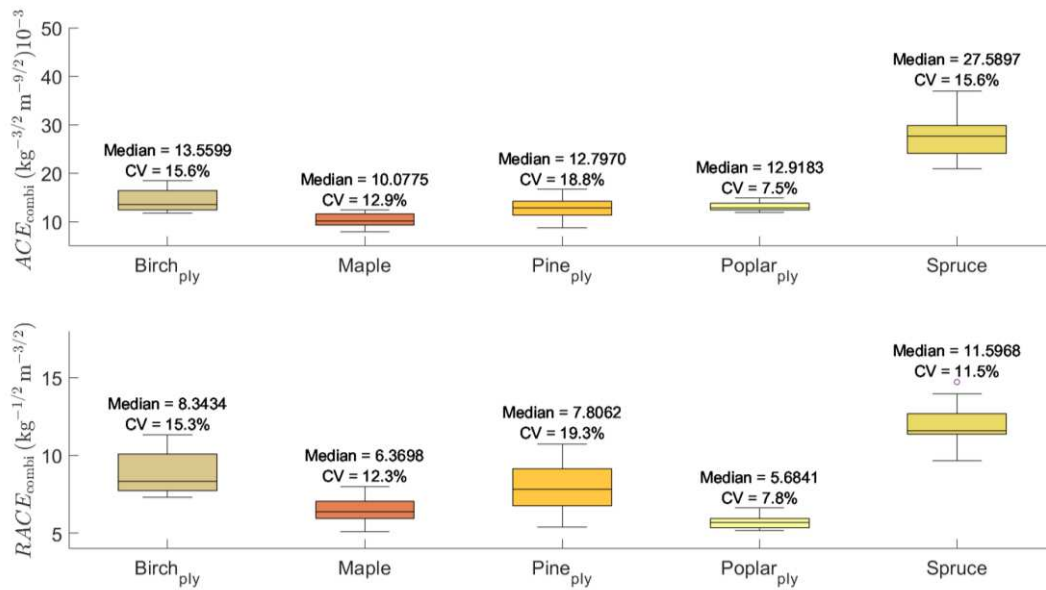


Figure 5. Comparison of material distributions using the ACE_{combi} and $RACE_{combi}$ parameters

A strong linear correlation was found between ACE_{combi} and $RACE_{combi}$ ($R^2 = 0.91$ for microphone data, and $R^2 = 0.84$ for vibrometer data), while both parameters exhibit distinct practical relevance. The use of ACE_{combi} mitigated the statistical inconsistency between the results obtained from the two sensors arising from the η estimation and enabled the classification of materials into hardwood, plywood, and softwood groups. In contrast, results expressed using $RACE_{combi}$ may be influenced by the choice of sensor; however, they demonstrate finer inter-group separation, including different plywood groups. Further improvement in the discriminative capability for evaluating material quality in terms of vibro-acoustic behavior may be achieved by increasing the number of modes used in the calculation of the aggregated parameters presented in this study, as well as by introducing potential weighting of individual modes according to their relative significance. This assumption, however, needs to be experimentally verified in future work using a larger dataset (in terms of dimensions, wood species, and material quality). At the same time, the limitations related to the objective identification of modes obtained from FRT data and their subsequent reliable analysis must be considered.

5. Conclusions

Based on the analysis of the experimental results, the following conclusions can be drawn:

- The bending modes proved to be the most stable for the determination of natural frequencies and the parameters derived from them, regardless of the measurement configuration (bending or torsional) or the instrument used (microphone or vibrometer). The bending frequencies determined using either torsional or bending configurations were mutually comparable, enabling a more efficient application of the FRT method for the simultaneous description of multiple modal shapes.
- Accurate modal characterisation requires well-separated frequencies. When modes are close, the sensor type can significantly affect torsional frequency determination. Reliable torsional analysis therefore requires specimen dimensions tailored to the material. Mode intervals can facilitate correct mode identification.
- To complement the description of the resonance wood, a threshold of $K_T > 3$ is suggested as an additional requirement.
- The parameter η is sensitive to the method of determination, already at the level of the sensor used. It also exhibits considerable variability depending on the mode shape, mode number, and the measurement configuration at which the η is evaluated.
- The aggregated parameters ACE_{combi} and $RACE_{combi}$, combining longitudinal, bending, and torsional vibration modes, were proposed. The observed inconsistencies between sensors and modes arise predominantly from the damping values used in their calculation. ACE_{combi} provided a more robust representation by mitigating these inconsistencies. If the material evaluation is simplified to $ACE_{B/T,1}$, it can be effectively determined using a single torsional measurement configuration. $RACE_{combi}$ appears to be a more sensitive parameter with respect to material differentiation and offers enhanced discriminative capability for a more detailed classification of materials.

References

- [1] U. G. K. Wegst, "Wood for sound," Am. J. Bot., vol. 93, no. 10, pp. 1439–1448, 2006, doi: 10.3732/ajb.93.10.1439.
- [2] S. Yoshikawa and C. Waltham, "Woods for Wooden Musical Instruments," in Proceedings of the International Symposium on Musical Acoustics (ISMA 2014), 2014, pp. 568–573.
- [3] X. Zhou, R. Miyauchi, and Y. Inoue, "Effect of wood attributes on the price persistence of acoustic guitars," J. Wood Sci., vol. 68, no. 1, pp. 1–15, 2022, doi: 10.1186/s10086-022-02054-1.

-
- [4] S. Brezas, M. Katsipis, K. Kaleris, H. Papadaki, D. T. G. Katerelos, N. A. Papadogiannis, M. Bakarezos, V. Dimitriou, and E. Kaselouris, “Review of manufacturing processes and vibro-acoustic assessments of composite and alternative materials for musical instruments,” *Applied Sciences*, vol. 14, no. 6, p. 2293, 2024. <https://doi.org/10.3390/app14062293>
- [5] C. C. B. Sanchez, V. R. L. Camangian, C. E. N. Camba, S. G. F. Bactong, T. A. Bonajos, J. N. Jerez, T. T. M. T. Pasion, and C. J. G. Atilano, “Eichhornia-based plywood as type 2 alternative material for musical instrument production,” research report, Taguig Science High School, Philippines, Nov. 2024. Available: ResearchGate.
- [6] V. Bucur, *Handbook of Materials for String Musical Instruments*. Cham, Switzerland: Springer, 2016. <https://doi.org/10.1007/978-3-319-32080-9>.
- [7] S. Calvano, F. Negro, F. Ruffinatto, D. Zanuttini-Frank, and R. Zanuttini, “Use and sustainability of wood in acoustic guitars: An overview based on the global market,” *Heliyon*, vol. 9, no. 4, p. e15218, 2023, doi: 10.1016/j.heliyon.2023.e15218.
- [8] I. Brémaud, “Acoustical properties of wood in string instruments soundboards and tuned idiophones: Biological and cultural diversity,” *J. Acoust. Soc. Am.*, vol. 131, no. 1, pp. 807–818, 2012, doi: 10.1121/1.3651233.
- [9] C. Besnainou, “Composite materials for musical instruments: The maturity,” *Journal of the Acoustical Society of America*, vol. 103, no. 5 (Suppl.), pp. 2872–2873, 1998. <https://doi.org/10.1121/1.421525>
- [10] B. Traoré, L. Brancheriau, P. Perré, T. Stevanovic, and P. Diouf, “Acoustic quality of vene wood (*Pterocarpus erinaceus* Poir.) for xylophone instrument manufacture in Mali,” *Ann. For. Sci.*, vol. 67, no. 8, 2010, doi: 10.1051/forest/2010054.
- [11] M. Quintavalla, “Assessing the quality of materials for stringed instrument soundboards: An approach considering orthotropic elastic and damping properties,” *Applied Acoustics*, vol. 183, p. 108521, 2022. <https://doi.org/10.1016/j.apacoust.2021.108521>
- [12] S. A. Ahmed and S. Adamopoulos, “Acoustic properties of modified wood under different humid conditions and their relevance for musical instruments,” *Appl. Acoust.*, vol. 140, pp. 92–99, Nov. 2018, doi: 10.1016/j.apacoust.2018.05.017.
- [13] I. Brémaud, K. Minato, P. Langbour, and B. Thibaut, “Physico-chemical indicators of inter-specific variability in vibration damping of wood,” *Ann. For. Sci.*, vol. 67, no. 7, pp. 707–707, 2010, doi: 10.1051/forest/2010032.
- [14] K. T. S. Hassan and J. Tippner, “Acoustic properties of neem (*Azadirachta indica*) wood,” *BioResources*, vol. 14, no. 2, pp. 2919–2930, 2019.
- [15] M. J. Casiano, “Extracting damping ratio from dynamic data and numerical solutions,” *NASA Technical Memorandum NASA/TM—2016–218227*, Sep. 2016.
- [16] T. Ono and M. Norimoto, “Anisotropy of dynamic Young’s modulus and internal friction in wood,” *Japanese Journal of Applied Physics*, vol. 24, no. 8, pp. 960–964, 1985. <https://doi.org/10.1143/jjap.24.960>
- [17] T. Ono and M. Norimoto, “Study on Young’s modulus and internal friction of wood in relation to the evaluation of wood for musical instruments,” *Japanese Journal of Applied Physics*, vol. 22, no. 4, pp. 611–614, 1983. <https://doi.org/10.1143/jjap.22.611>
- [18] M. F. Hossen, S. Hamdan, and M. R. Rahman, “Investigation of the acoustic properties of chemically impregnated Kayu Malam wood used for musical instruments,” *Advances in Materials Science and Engineering*, vol. 2018, Article ID 7829613, 7 pp., 2018. <https://doi.org/10.1155/2018/7829613>
- [19] J. Baar, J. Tippner, and V. Gryc, “Wood anatomy and acoustic properties of selected tropical hardwoods,” *IAWA J.*, vol. 37, no. 1, pp. 69–83, 2016, doi: 10.1163/22941932-20160121.
- [20] A. Straže, B. Mitkovski, J. Tippner, K. Čufar, and Gorišek, “Structural and acoustic properties of African padouk (*Pterocarpus soyauxii*) wood for xylophones,” *Eur. J. Wood Wood Prod.*, vol. 73, no. 2, pp. 235–243, Mar. 2015, doi: 10.1007/s00107-015-0878-0.
- [21] E. Obataya, T. Ono, and M. Norimoto, “Vibrational properties of wood along the grain,” *J. Mater. Sci.*, vol. 35, no. 12, pp. 2993–3001, 2000, doi: 10.1023/A:1004782827844.
- [22] T. Ray and J. Kaljun, “The influence of the acoustic properties of wood for the production of electric solid body guitars,” in *DAAAM International Scientific Book 2018*, B. Katalinic, Ed. Vienna: DAAAM International, 2018, pp. 207–218. <https://doi.org/10.2507/daaam.scibook.2018.18>

- [23] M. Roohnia, M. Kohantorabi, and A. Tajdini, "Maple wood extraction for better acoustical performance," *European Journal of Wood and Wood Products*, vol. 73, no. 1, pp. 139–142, 2015. <https://doi.org/10.1007/s00107-014-0871-z>
- [24] ASTM E1876-15, "Standard Test Method for Dynamic Young's Modulus, Shear Modulus, and Poisson's Ratio by Impulse Excitation of Vibration," ASTM International, West Conshohocken, PA, 2015. <https://doi.org/10.1520/E1876-15>
- [25] K. T. S. Hassan, P. Horáček, and J. Tippner, "Evaluation of stiffness and strength of Scots pine wood using resonance frequency and ultrasonic techniques," *BioResources*, vol. 8, no. 2, pp. 1634–1645, 2013. <https://doi.org/10.15376/biores.8.2.1634-1645>
- [26] C. L. Cho, "Comparison of three methods for determining Young's modulus of wood," *Taiwan Journal of Forest Science*, vol. 22, no. 3, pp. 297–306, 2007
- [27] H. Aizawa, E. Obataya, T. Ono, and M. Norimoto, "Acoustic converting efficiency and anisotropic nature of wood," *Wood Research: Bulletin of the Wood Research Institute, Kyoto University*, no. 85, pp. 81–83, 1998.
- [28] M. Roohnia and M. Kohantorabi, "Dynamic methods to evaluate the shear modulus of wood," *BioResources*, vol. 10, no. 3, pp. 4867–4876, 2015. <https://doi.org/10.15376/biores.10.3.4867-4876>
- [29] M. Merhar and R. Moutou Pitti, "The influence of thermal modification, moisture content, frequency, and vibration direction plane on the damping of spruce wood (*Picea abies*) as determined by the wavelet transform method," *Forests*, vol. 16, p. 1055, 2025. <https://doi.org/10.3390/f16071055>
- [30] Y. Kubojima, S. Sonoda, and H. Kato, "Practical techniques for the vibration method with additional mass: bending vibration generated by tapping cross section," *J. Wood Sci.*, vol. 64, no. 1, pp. 16–22, 2018, doi: 10.1007/s10086-017-1676-6.
- [31] R. Corradi, S. Miccoli, G. Squicciarini, and P. Fazioli, "Modal analysis of a grand piano soundboard at successive manufacturing stages," *Appl. Acoust.*, vol. 125, pp. 113–127, 2017, doi: 10.1016/j.apacoust.2017.04.010.
- [32] M. D. Stanciu, S. Năstac, V. G. Gliga, M. Campean, and V. Bucur, "Effect of wood anisotropy on eigenmodes and eigenvalues using finite element analysis: Case of violin plates," *IOP Conference Series: Materials Science and Engineering*, vol. 997, no. 1, p. 012105, 2020. <https://doi.org/10.1088/1757-899X/997/1/012105>.
- [33] R. Viala, V. Placet, and S. Cogan, "Identification of the anisotropic elastic and damping properties of complex shape composite parts using an inverse method based on finite element model updating and 3D velocity fields measurements (FEMU-3DVF): Application to bio-based composite violin soundboards," *Compos. Part A Appl. Sci. Manuf.*, vol. 106, pp. 91–103, Mar. 2018, doi: 10.1016/j.compositesa.2017.12.018.
- [34] M. Merhar, "Determination of elastic properties of beech plywood by analytical, experimental and numerical methods," *Forests*, vol. 11, no. 11, p. 1221, 2020. <https://doi.org/10.3390/f11111221>
- [35] R. Sproßmann, M. Zauer, and A. Wagenführ, "Characterization of acoustic and mechanical properties of common tropical woods used in classical guitars," *Results Phys.*, vol. 7, pp. 1737–1742, 2017, doi: 10.1016/j.rinp.2017.05.006.
- [36] M. Merhar, "Identification of Damping of Spruce Wood (*Picea abies*) under Various Levels of Moisture Content Using Time-Scale Decomposition," *Polymers (Basel)*, vol. 16, no. 10, 2024, doi: 10.3390/polym16101313.
- [37] S. Chauhan and A. Sethy, "Differences in dynamic modulus of elasticity determined by three vibration methods and their relationship with static modulus of elasticity," *Maderas Cienc. y Tecnol.*, vol. 18, no. 2, pp. 373–382, 2016, doi: 10.4067/S0718-221X2016005000034.
- [38] ČSN EN 326-1, *Desky na bázi dřeva – Odběr vzorků, řezání a kontrola – Část 1: Odběr vzorků a řezání zkušebních těles a vyjádření výsledků zkoušek*, Český normalizační institut, Praha, 2015
- [39] F. Dinulić, M. D. Stanciu, and A. Savin, "Correlation between anatomical grading and acoustic–elastic properties of resonant spruce wood used for musical instruments," *Forests*, vol. 12, no. 8, pp. 1–18, 2021, doi: 10.3390/f12081122.
- [40] J. Bretos, C. Santamaría, and J. Alonso Moral, "Finite element analysis and experimental measurements of natural eigenmodes and random responses of wooden bars used in musical instruments," *Appl. Acoust.*, vol. 56, no. 3, pp. 141–156, 1999, doi: 10.1016/S0003-682X(98)00030-9.
- [41] ASTM E1875-20, "Standard test method for dynamic Young's modulus, shear modulus, and Poisson's ratio by sonic resonance," ASTM

- International, West Conshohocken, PA, 2020. <https://doi.org/10.1520/E1875-20> 505
- [42] T. Biadała, R. Czarniecki, and D. Dukarska, “WATER RESISTANT PLYWOOD OF INCREASED ELASTICITY PRODUCED FROM EUROPEAN WOOD,” *Wood Res.*, vol. 65, no. 1, pp. 111–124, 2020, doi: doi.org/10.37763/wr.1336-4561/65.1.111124. 506 507
- [43] A. Danihelová, Z. Vidholdová, T. Gergel', L. Spišiaková Kružlicová, and M. Pástor, “Thermal modification of spruce and maple wood for special wood products,” *Polymers*, vol. 14, no. 14, p. 2813, 2022. <https://doi.org/10.3390/polym14142813> 508 509
- [44] C. Tenorio, R. Moya, and F. Muñoz, “Comparative study on physical and mechanical properties of laminated veneer lumber and plywood panels made of wood from fast-growing *Gmelina arborea* trees,” *Journal of Wood Science*, vol. 57, no. 2, pp. 134–139, 2011. <https://doi.org/10.1007/s10086-010-1149-7> 510 511 512
- [45] I. Bork, “Practical tuning of xylophone bars and resonators,” *Appl. Acoust.*, vol. 46, no. 1, pp. 103–127, 1995, doi: [10.1016/0003-682X\(95\)93953-F](https://doi.org/10.1016/0003-682X(95)93953-F). 513 514
- [46] P. R. G. Hein and L. Brancheriau, “Comparison between three-point and four-point flexural tests to determine wood strength of *Eucalyptus* specimens,” *Maderas Cienc. y Tecnol.*, vol. 20, no. 3, pp. 333–342, 2018, doi: [10.4067/S0718-221X2018005003401](https://doi.org/10.4067/S0718-221X2018005003401). 515 516
- [47] J. Kljak and M. Brezović, “Relationship between bending and tensile stress distribution in veneer plywood,” *Forest Products Journal*, vol. 57, no. 12, pp. 85–90, 2007. 517 518
- [48] A. Danihelová and M. Čulík, “Influence of ageing process on sound quality of wood for stringed musical instrument making,” in *Proceedings of Forum Acusticum 2005, Budapest, Hungary, Aug. 29–Sep. 2, 2005*, pp. 581–584. 519 520
- [49] C. Guan, H. Zhang, J. F. Hunt, and H. Yan, “Determining shear modulus of thin wood composite materials using a cantilever beam vibration method,” *Constr. Build. Mater.*, vol. 121, pp. 285–289, 2016, doi: [10.1016/j.conbuildmat.2016.06.007](https://doi.org/10.1016/j.conbuildmat.2016.06.007). 521 522
- [50] V. I. Fedyukov, E. Y. Saldaeva, M. S. Chernova, and V. Y. Chernov, “Research into dendro-acoustic properties of introduced clones’ wood as material for manufacturing musical instruments,” *South-east European Forestry*, vol. 10, no. 2, pp. 173–179, 2019, <https://doi.org/10.15177/seefor.19-18>. 523 524 525

7 Summary

This work aims to investigate the factors that influence the dynamic response of materials used for musical instruments—primarily wood and wood-based composites—and to account for these factors in the application of evaluation methods to describe their mechanical and vibro-acoustic properties. Based on the obtained findings, the approach to material assessment can be extended to materials with modified properties, alternative materials with specific internal structures, as well as to a more refined selection of high-quality materials at early stages of the manufacturing process. The key findings of this work are summarised below:

Effect of specimen dimensions

The determination of *ACE* and other vibro-acoustic parameters using FRT on boards and on smaller specimens—with preserved dimensional ratios relative to the original board—demonstrated that specimen size influenced the obtained parameters due to non-negligible structural and anatomical characteristics of wood species (such as structural irregularities) (**hypothesis 1a accepted**). However, specimen geometry, particularly its slenderness and cross-sectional shape, was shown to have a significantly greater effect on the determination of vibro-acoustic properties. Varying interactions between vibration modes, influencing the evaluated parameters (e.g., damping), can even lead to potential misidentification when closely spaced modes merge or when their order is interchanged (**hypothesis 1b accepted**) (Paper I).

Effect of material characteristics

When using FRT, it was found that above the FSP, further increases in MC did not significantly affect MOE_{dyn} or $\tan \delta$, and that the continued decrease in *ACE* is mainly caused by increasing MC, while the sample volume remains unchanged. When using DMA, $\tan \delta$ continued to increase significantly above the FSP, resulting in a steeper decrease in *ACE* (**hypothesis 2a accepted**) (Paper III). The observed significant increase in $\tan \delta$ and decrease in MOE_{dyn} due to fungal degradation acted, in contrast to the relatively small mass loss, as the dominant factors in the reduction of *ACE* ($\approx -74\%$) (**hypothesis 2b accepted**) (Paper II). Changes caused by thermal modification—whether in MOE_{dyn} , $\tan \delta$, equilibrium MC, or mass loss—led to an increase in *ACE* with increasing thermal treatment temperature; however, the significance of the *ACE* improvement is species-dependent (**hypothesis 2c accepted**) (Paper IV). Across the studies, it was demonstrated that *ACE* values were capable of distinguishing between different wood species, their modifications, and different materials (solid wood vs. plywood), while *RACE* differentiated even plywood materials at the species level (**hypothesis 2d & 2e accepted**) (Papers I, III, IV, V).

Effect of vibrational mode

When comparing vibro-acoustic parameters determined from frequencies of bending and longitudinal modes, it was shown that the determination was influenced by specimen geometry. Slender solid wood specimens (slenderness > 48) exhibited more isotropic behavior, and, for example, differences between MOE_{dyn} derived from the first bending and the first longitudinal mode shape were negligible. However, results for tested plywood (slenderness > 85) demonstrated that the material still plays a significant role in this respect. The determination based on higher modes showed statistical comparability across all tested materials. In contrast, $\tan \delta$ exhibited high variability of results and could not be interchanged between mode shapes or mode orders, which was also reflected in the non-comparability of ACE across mode shapes and mode orders (**hypothesis 3a & 3b accepted**) (Papers I, V).

Effect of the measurement technique

The choice of sensors affected the evaluation of ACE in FRT, particularly in the case of contact measurements using an accelerometer on lightweight specimens, where the mass of the accelerometer significantly influenced the dynamic response. In the case of non-contact sensors, a consistent but practically negligible difference in the determined frequencies was observed between the microphone and the vibrometer. However, differences in the amplitudes of resonance peaks of individual modes led to varying results when using selected methods for damping coefficient determination, and in the case of closely spaced modes, the choice and positioning of the sensor may affect correct mode identification, thereby leading to errors in the ACE determination. (**hypothesis 4a accepted**) (Papers I, V). The comparison of results obtained from two dynamic methods (DMA and FRT) led to the conclusion that MOE_{dyn} values determined by the two methods are not statistically comparable; however, a strong linear correlation exists between them ($R^2 = 0.84$), allowing for appropriate conversion of the results. In contrast, due to the different descriptions of behaviour above the FSP provided by the methods, the results for $\tan \delta$ did not allow for a reliable conversion between them (**hypothesis 4b accepted**) (Paper III). While the expected strong correlation between MOE_{dyn} and MOR was confirmed, the relationship between IBS and MOE_{dyn} was relatively weak. Based on these results, non-destructive testing of bending stiffness cannot be considered a reliable predictor of strength-related bending parameters in general, because the mode of failure and the rate of the loading process must be taken into account (**hypothesis 4c rejected**) (Papers II, IV).

Characteristics of the vibrational spectrum

The vibrational spectrum was shown to be governed by the geometry and material properties of the tested specimen. Specimen size and shape influence the pattern of the spectrum, affecting the number of modes and the sequence of mode shapes. Observed deviations in the intervals between individual modes were found to be associated with material properties and related factors, such as structural irregularities (**hypothesis 5a & 5b accepted**) (Papers I, V). To combine dynamic moduli and $\tan \delta$ determined from different modes (longitudinal, bending, torsional) the aggregated parameters ACE_{combi} and $RACE_{combi}$ were introduced. These parameters incorporate information on the degree of anisotropy and enable extension to include higher-order modes while accounting for the high variability of $\tan \delta$ determined from multiple modes. The results showed that both parameters differentiate between the tested materials and provide a more comprehensive representation of vibro-acoustic properties (**hypothesis 5c accepted**) (Paper V).

List of Abbreviations

ACE	Acoustic conversion efficiency
ACE_{L/T}, ACE_{B/T}	Acoustic conversion efficiency combining results from longitudinal or bending mode with torsional mode
ACE_{combi}	Acoustic conversion efficiency combining results from all investigated modes
C	Stiffness tensor
c	Speed of wave propagation
c_L, c_T	Speed of longitudinal and transverse wave propagation
CITES	Convention on International Trade in Endangered Species of Wild Fauna and Flora
DIC	Digital Image Correlation
DMA	Dynamic Mechanical Analysis
E	Young's modulus
E_{dyn}	Dynamic Young's modulus
E_L	Young's modulus in longitudinal direction
EMA	Experimental Modal Analysis
E'	Storage modulus
E''	Loss modulus
f	Frequency
F	Force
FEM	Finite Element Method
FFT	Fast Fourier Transform
FRT	Frequency Resonance Technique
FSP	Fibre saturation point
G	Shear modulus
G_{dyn}	Dynamic shear modulus
i	Radius of gyration
I	Moment of inertia
IBS	Impact bending strength

k	Constant corresponding to bending mode order
K	Radiation coefficient
l	Length
L	Longitudinal direction
l₀	Distance between supports
LDD	Logarithmic decrement of damping
m	Mass
MC	Moisture content
MOE_{dyn}	Dynamic bending modulus of elasticity
MOR	Bending modulus of rupture
n	Number
P	Correction factor
Q	Quality factor
R	Radial direction
RACE	Relative acoustic conversion efficiency
RACE_{L/T}, RACE_{B/T}	Relative acoustic conversion efficiency combining results from longitudinal or bending mode with torsional mode
RACE_{combi}	Relative acoustic conversion efficiency combining results from all investigated modes
S	Compliance tensor
t	Thickness
T	Tangential direction
tan δ	Loss tangent
ToF	Time of Flight
v	Poisson's ratio
w	Width
Z	Acoustic impedance
β	Temporal absorption coefficient
δ	Loss angle
ε	Strain
η	Loss factor

λ	Wavelength
ρ	Density
σ	Stress
σ_B	Bending stress
ζ	Damping ratio